Stochastic Analysis

LectureNotes

Phd.Student
LaTeX by Zheng.zh
SUST University

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Chapter 1

Intro

Riemann-Stieltjes Integral: If f is continuous and g is of finite variation, then $\int_a^b f(x)\ \mathrm{d}g(x)$ is well-defined, and is given by

$$\int_a^b f(x)\ \mathrm{d}g(x) := \lim_{n\to\infty} \sum_{i=0}^{n-1} f(\xi_i)\,(g(x_{i+1}) - g(x_i))$$

where $a \le x_0 < x_1 < \cdots < x_n \le b$ and $\xi_i \in [x_i, x_{i+1}]$.

Main Task: define the Itô integral

$$(H \cdot X)_t = \int_0^t H_s\ \mathrm{d}X_s.$$

The most important case is when X is Brownian motion.

Theorem 1.0.1. $\sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right)^2 \xrightarrow{a.s.} 1.$

Recall that the finite variation of a function $g : [a,b] \to \mathbb{R}$ is

$$V_g([a,b]) = \int_a^b |\mathrm{d}g(x)| = \sup_n \sum_{i=1}^{n} |g(x_{i+1}) - g(x_i)|$$

Corollary 1.0.1. $V_B([0,1]) = \infty$ a.s.

Proof for Corollary.

■.

$$\begin{aligned}(\star) \sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right)^2 &\le \sup_{1\le k\le 2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right| \cdot \sum_{k=1}^{2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right| \\ &\le V_B([0,1]) \cdot \sup_{1\le k\le 2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right|\end{aligned}$$

If $V_B([0,1]) < \infty$, then by the continuity of Brownian paths,

$$\sup_{1 \le k \le 2^n} \left| B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}} \right| \to 0,$$

and hence $(\star) \to 0$.

This contradicts $(\star) \to 1$. Therefore, with probability one,

$$V_B([0,1]) = \infty.$$

$\square$

This shows that Brownian motion is too irregular to be treated by the classical Riemann–Stieltjes theory. Therefore, to define

$$(H \cdot X)_t = \int_0^t H_s \,\mathrm{d}X_s,$$

we need a new framework, where typically H is previsible (predictable) and X is a semimartingale.

Example . Let $H_n \in \mathcal{F}_{n-1}$ and let (M_n) be a martingale.

Then $(H \cdot M)_n = \sum_{k=1}^n H_k(M_k - M_{k-1})$ is a martingale.

1.1 Set-up

Throughout, let $(\Omega, \mathcal{F}, \mathsf{P})$ be a probability space. A stochastic process $X = (X_t)_{t \ge 0}$ is a family of random variables taking values in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$:

$$X \colon [0,\infty) \times \Omega \to \mathbb{R}^d$$
$$(t,\omega) \mapsto X_t(\omega).$$

For each fixed ω, the map $t \mapsto X_t(\omega)$ is called the sample path of X corresponding to ω.

Definition 1.1.1. For two stochastic processes X and Y, we say that Y is a modification of X if

$$\forall\, t \ge 0, \quad \mathsf{P}(X_t = Y_t) = 1.$$

Definition 1.1.2. We say that X and Y are indistinguishable if

$$\mathsf{P}(X_t = Y_t,\ \forall\, t \ge 0) = 1.$$

In this case, Y is also called a version of X.

Example . Let Y be a modification of X, and suppose that both X and Y have a.s. right-continuous sample paths. Then X and Y are indistinguishable.

Definition 1.1.3. A function f is right-continuous if

$$f(x) = f(x+) = \lim_{t \downarrow x} f(t).$$

Proof. Since $\forall\ t \geq 0$, $\mathsf{P}(X_t \neq Y_t) = 0$, the two processes agree almost surely at every fixed rational time. More precisely,

$$\mathsf{P}\left(\bigcup_{q \in \mathbb{Q}^+} X_q \neq Y_q \right) \leq \sum_{q \in \mathbb{Q}^+} \mathsf{P}(X_q \neq Y_q) = 0.$$

Therefore,

$$\mathsf{P}\left(\bigcap_{q \in \mathbb{Q}^+} X_q = Y_q \right) = 1.$$

On the other hand, if two right-continuous functions f and g agree on $\mathbb{Q}^+$, then they must agree on all of $[0, \infty)$, since the rationals are dense.

Applying this observation pathwise to the sample paths of X and Y, we conclude that

$$\mathsf{P}\left(X_t = Y_t,\ \forall\ t \geq 0 \right) = 1.$$

□

Example . Let $T \sim \text{Exp}(1)$. Define

$$X_t \equiv 0, \quad \forall\ t \geq 0$$
$$Y_t = \begin{cases} 0, & t \neq T \\ 1, & t = T. \end{cases}$$

Then for every fixed $t \geq 0$,

$$\mathsf{P}(X_t \neq Y_t) = \mathsf{P}(Y_t = 1) = \mathsf{P}(t = T) = 0.$$

Hence Y is a modification of X. However, X and Y are not indistinguishable, because on the event $\{T < \infty\}$ they differ at time T.

1.2 Filtration

We now introduce filtrations, which describe the information available up to each time t.

Let $(\mathcal{F}_t)_{t \geq 0}$ be a filtration of $(\Omega, \mathcal{F})$, that is,

$$\forall\ 0 \leq s < t < \infty, \quad \mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}.$$

Example . (Natural filtration) Let $(X_t)_{t\geq 0}$ be a stochastic process. Define

$$\mathcal{F}_t^X = \sigma(X_s, 0 \leq s \leq t).$$

Then $(\mathcal{F}_t^X)$ is a filtration.

By construction, $X_t \in \mathcal{F}_t^X$ for every $t \geq 0$.

Exercise 1.7. Let X be a process whose sample paths are all RCLL. Let

$$A = \{\omega \colon X(\omega) \text{ is continuous on } [0, t_0)\}$$

for some fixed $t_0 > 0$. Show that $A \in \mathcal{F}_{t_0}^X$.

RCLL: $f(x) = f(x+)$ and $f(x-)$ exists.

Proof. Since every sample path is RCLL, continuity on $[0, t_0)$ is equivalent to continuity at every point in $(0, t_0)$. Hence

$$\begin{aligned} A &= \{\omega \colon \forall\ s \in (0, t_0),\ X(s-, \omega) = X(s, \omega)\} \\ &= \{\omega \colon \forall\ q \in (0, t_0) \cap \mathbb{Q},\ X(q-, \omega) = X(q, \omega)\} \end{aligned}$$

because for an RCLL function, possible discontinuities are jumps, and it is enough to check them on rational times.

Let $(0, t_0) \cap \mathbb{Q} = (q_m)_{m\geq 1}$. Then

$$\begin{aligned} A &= \left\{\omega \colon \forall\ q_m \text{ with } m \geq 1,\ \forall\ k \geq 1,\ \exists\ N \geq 1,\ \forall\ n \geq N, \left|X(q_m - \frac{1}{n}, \omega) - X(q_m, \omega)\right| < \frac{1}{k}\right\} \\ &= \bigcap_{m=1}^{\infty} \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left\{\omega \colon \left|X(q_m - \frac{1}{n}, \omega) - X(q_m, \omega)\right| < \frac{1}{k}\right\} \in \mathcal{F}_{t_0}^X. \end{aligned}$$

Therefore $A \in \mathcal{F}_{t_0}^X$. □

Exercise 1.8. Let X be a process whose sample paths are RCLL a.s. Then $A \notin \mathcal{F}_{t_0}^X$ is possible.

Proof. [TODO] □

Definition 1.2.1. A filtration $\{\mathcal{F}_t\}_{t\geq 0}$ is said to satisfy the usual conditions if

(a) $\mathcal{F}_t = \mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$;

(b) $\mathcal{F}_0$ contains all the $\mathsf{P}-$negligible sets in $\mathcal{F}$.

Definition 1.2.2. An event A is called P-negligible if there exists $N \in \mathcal{F}$ such that $A \subseteq N$ and $\mathsf{P}(N) = 0$.

Definition 1.2.3. We define

$$\mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s, \qquad \mathcal{F}_{t-} = \sigma\left(\bigcup_{s<t} \mathcal{F}_s\right).$$

We say that the filtration is right-continuous if $\mathcal{F}_t = \mathcal{F}_{t+}$.

Definition 1.2.4. The stochastic process X is adapted to the filtration $\{\mathcal{F}_t\}_{t\geq 0}$ if $\forall\ t \geq 0$, X_t is $\mathcal{F}_t$-measurable.

In particular, every process is adapted to its natural filtration.

Definition 1.2.5. The stochastic process X is called progressively measurable with respect to $(\mathcal{F}_t)$ if $\forall\ t \geq 0$, the map

$$(s,\omega) \mapsto X_s(\omega)$$
$$([0,t] \times \Omega, \mathcal{B}([0,t]) \otimes \mathcal{F}_t) \to (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$$

is measurable.

An adapted process need not be progressively measurable in general. The useful point here is that, under additional path regularity, adaptedness implies progressive measurability. In the usual hierarchy one has

$$\textit{previsible} \implies \textit{progressively measurable} \implies \textit{adapted.}$$

Proposition 1.2.1. If X is adapted to the filtration $(\mathcal{F}_t)$ and every sample path is right-continuous or left-continuous, then X is also progressively measurable.

Proof for Proposition.

■. It is enough to treat the right-continuous case. Fix $t > 0$. For every $n \geq 1$ and $0 \leq k \leq 2^n - 1$, define

$$X_s^{(n)}(\omega) = X_{\frac{k+1}{2^n}t}(\omega), \quad \forall\ \frac{kt}{2^n} < s \leq \frac{k+1}{2^n}t$$

Then, by right-continuity of the sample paths,

$$\lim_{n\to\infty} X_s^{(n)}(\omega) = X_s(\omega)$$

for all $(s,\omega) \in [0,t] \times \Omega$.

Therefore, it suffices to show that

$$(s,\omega) \mapsto X_s^{(n)}(\omega)$$
$$([0,t] \times \Omega, \mathcal{B}([0,t]) \otimes \mathcal{F}_t) \to (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$$

is measurable.

Indeed, for every $A \in \mathcal{B}(\mathbb{R}^d)$,

$$\begin{aligned}&\left\{(s,\omega)\colon X_s^{(n)}(\omega) \in A\right\} \\ =& \bigcup_{k=0}^{2^n-1} \left(\frac{k}{2^n}t, \frac{k+1}{2^n}t\right] \times \left\{X_{\frac{k+1}{2^n}t} \in A\right\} \in \mathcal{B}([0,t]) \otimes \mathcal{F}_t\end{aligned}$$

because X is adapted and therefore $X_{\frac{k+1}{2^n}t}$ is $\mathcal{F}_t$-measurable.

Thus $X_s^{(n)}(\omega)$ is measurable for every n. Passing to the pointwise limit, we conclude that X is progressively measurable. □

1.3 Stopping Times

We now study random times that are compatible with the filtration $(\mathcal{F}_t)_{t\geq 0}$ on $(\Omega, \mathcal{F}, \mathsf{P})$.

Definition 1.3.1. A random time T is called a stopping time with respect to $(\mathcal{F}_t)$ if

$$\forall\, t \geq 0, \quad \{T \leq t\} \in \mathcal{F}_t.$$

Definition 1.3.2. A random time T is called an optional time with respect to $(\mathcal{F}_t)$ if

$$\forall\, t \geq 0, \quad \{T < t\} \in \mathcal{F}_t.$$

Proposition 1.3.1. Every stopping time is an optional time. Moreover, if the filtration is right-continuous, then the two notions are equivalent.

Proof for Proposition.

■. "⇒" If T is a stopping time, then

$$\{T < t\} = \bigcup_{n=1}^{\infty} \left\{T \leq t - \frac{1}{n}\right\} \in \mathcal{F}_t.$$

"⇐" Now assume that T is an optional time and that the filtration is right-continuous. Then for every $m \geq 1$,

$$\{T \leq t\} = \bigcap_{n=m}^{\infty} \left\{T < t + \frac{1}{n}\right\}.$$

Since $\{T < t + \frac{1}{n}\} \in \mathcal{F}_{t+\frac{1}{n}} \subseteq \mathcal{F}_{t+\frac{1}{m}}$, we obtain

$$\{T \leq t\} \in \mathcal{F}_{t+\frac{1}{m}}.$$

Therefore

$$\{T \leq t\} \in \bigcap_{m=1}^{\infty} \mathcal{F}_{t+\frac{1}{m}} = \mathcal{F}_{t^+} = \mathcal{F}_t,$$

so T is a stopping time. $\square$

Lemma 1.3.1. If T and S are stopping times, then $T \wedge S$, $T \vee S$, $T + S$ are stopping times.

Proof. $\forall\, t \geq 0$,

1. $\{T \wedge S \leq t\} = \{T \leq t\} \cup \{S \leq t\} \in \mathcal{F}_t$.
2. $\{T \vee S \leq t\} = \{T \leq t\} \cap \{S \leq t\} \in \mathcal{F}_t$.
3. To prove $\{T + S \leq t\} \in \mathcal{F}_t$, it is enough to show that $\{T + S > t\} \in \mathcal{F}_t$.

$$\begin{aligned} &\{T + S > t\} \\ =&\{T = 0,\ S > t\} \cup \{S = 0,\ T > t\} \cup \{0 < T < t,\ T + S > t\} \cup \{S > 0,\ T \geq t\} \\ \in&\mathcal{F}_t \cup \mathcal{F}_t \cup \{0 < T < t,\ T + S > t\} \cup \mathcal{F}_t. \end{aligned}$$

Thus it remains to show that

$$\{0 < T < t,\ T + S > t\} = \bigcup_{\substack{r \in \mathbb{Q}^+ \\ 0 < r < t}} \{r < T < t,\ S > t - r\}.$$

The right-hand side belongs to $\mathcal{F}_t$.

"$\Leftarrow$ " $S > t - r > t - T$, so $S + T > t$.

"$\Rightarrow$ " Since $S + T > t$, we can choose $r \in \mathbb{Q}^+$ with $0 < r < T$ and $S + r > t$.

$\square$

Lemma 1.3.2. If $\{T_n\}_{n \geq 1}$ are stopping times, then $\sup_{n \geq 1} T_n$ is a stopping time.

Proof. $\forall\, t \geq 0$, $\{\sup_{n \geq 1} T_n \leq t\} = \bigcap_{n=1}^{\infty} \{T_n \leq t\} \in \mathcal{F}_t$. $\square$

Definition 1.3.3. The stopping σ-algebra associated with a stopping time T is

$$\mathcal{F}_T = \{A \in \mathcal{F} \mid \forall\, t \geq 0,\ A \cap \{T \leq t\} \in \mathcal{F}_t\}.$$

Proposition 1.3.2. If $T(\omega) = t_0$ for some fixed $t_0 \geq 0$, $\forall\, \omega \in \Omega$, then $\mathcal{F}_T = \mathcal{F}_{t_0}$.

Proof. Note that $\forall\, t \geq 0$,

$$\{T \leq t\} = \begin{cases} \varnothing, & t_0 > t \\ \Omega, & t_0 \leq t \end{cases}$$

"$\mathcal{F}_{t_0} \subseteq \mathcal{F}_T$"

$\forall\, A \in \mathcal{F}_{t_0}$, $\forall\, t \geq 0$,

① $t \geq t_0$, $A \cap \{T \leq t\} = A \in \mathcal{F}_{t_0} \subseteq \mathcal{F}_t$;

② $t < t_0$, $A \cap \{T \le t\} = \varnothing \in \mathcal{F}_t$.

Hence $\mathcal{F}_{t_0} \subseteq \mathcal{F}_T$.
"$\mathcal{F}_T \subseteq \mathcal{F}_{t_0}$"

$\forall\ A \in \mathcal{F}_T$, we have $A \cap \{T \le t\} \in \mathcal{F}_t$ by definition. Let $t = t_0$, then $\{T \le t_0\} = \Omega$, so

$$A \cap \{T \le t_0\} \in \mathcal{F}_{t_0} \Rightarrow\ A \cap \Omega \in \mathcal{F}_{t_0} \Rightarrow\ A \in \mathcal{F}_{t_0}.$$

Hence $\mathcal{F}_T \subseteq \mathcal{F}_{t_0}$. □

Lemma 1.3.3. $\forall$ stopping time T and S, $\forall\ A \in \mathcal{F}_S$, we have $A \cap \{S \le T\} \in \mathcal{F}_T$. In particular, if $S \le T$ for all $\omega \in \Omega$, then $\mathcal{F}_S \subseteq \mathcal{F}_T$.

Proof. First note that $T \wedge t$ is also a stopping time. Hence $T \wedge t$ is $\mathcal{F}_t$-measurable.

To prove $A \cap \{S \le T\} \in \mathcal{F}_T$, it is enough to show that for every $t \ge 0$,

$$A \cap \{S \le T\} \cap \{T \le t\} \in \mathcal{F}_t.$$

We have

$$A \cap \{S \le T\} \cap \{T \le t\} = \underbrace{A \cap \{S \le t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{T \le t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{S \wedge t \le T \wedge t\}}_{\mathcal{F}_t}.$$

Indeed, on $\{S \le t\} \cap \{T \le t\}$, the event $\{S \le T\}$ is exactly the same as $\{S \wedge t \le T \wedge t\}$. Therefore the whole intersection belongs to $\mathcal{F}_t$, and hence $A \cap \{S \le T\} \in \mathcal{F}_T$. □

Lemma 1.3.4. Let S, T be two stopping times. Then $\mathcal{F}_{T \wedge S} = \mathcal{F}_T \cap \mathcal{F}_S$ and $\{T < S\}$, $\{S < T\}$, $\{T \le S\}$, $\{S \le T\}$, $\{T = S\}$ belong to $\mathcal{F}_S \cap \mathcal{F}_T = \mathcal{F}_{S \wedge T}$.

Proof. $\mathcal{F}_{T \wedge S} \subseteq \mathcal{F}_S$ and $\mathcal{F}_{T \wedge S} \subseteq \mathcal{F}_T \Rightarrow\ \mathcal{F}_{T \wedge S} \subseteq \mathcal{F}_T \cap \mathcal{F}_S$.

To prove $\mathcal{F}_T \cap \mathcal{F}_S \subseteq \mathcal{F}_{T \wedge S}$, $\forall\ A \in \mathcal{F}_T \cap \mathcal{F}_S$, we will prove $\forall\ t \ge 0$, $A \cap \{T \wedge S \le t\} \in \mathcal{F}_t$.

$$\begin{aligned} A \cap \{T \wedge S \le t\} &= A \cap (\{T \le t\} \cup \{S \le t\}) \\ &= (A \cap \{T \le t\}) \cup (A \cap \{S \le t\}) \in \mathcal{F}_t. \end{aligned}$$

Next, $\{S \le T\} \in \mathcal{F}_T$. It remains to prove $\{S < T\} \in \mathcal{F}_T$.

Assuming this, by interchanging T and S, we get $\{T \le S\}$, $\{T > S\}$, $\{T < S\} \in \mathcal{F}_S$.

Therefore $\{S \le T\} \in \mathcal{F}_S \cap \mathcal{F}_T$.

$\{T = S\} = \{S \le T\} \cap \{T \le S\} \in \mathcal{F}_S \cap \mathcal{F}_T$.

To prove $\{S < T\} \in \mathcal{F}_T$, let $R = S \wedge T$. Then $\{S < T\} = \{R < T\}$. Since R is a stopping time, we have $R \in \mathcal{F}_R \subseteq \mathcal{F}_T$, and therefore $\{R < T\} \in \mathcal{F}_T$. □

Summary:

(i) $T \in \mathcal{F}_T$;

(ii) If $S \le T$, $\mathcal{F}_S \subseteq \mathcal{F}_T$;

(iii) $T \wedge t \in \mathcal{F}_{T\wedge t} \subseteq \mathcal{F}_t$;

(iv) $\mathcal{F}_{T\wedge S} = \mathcal{F}_T \cap \mathcal{F}_S$.

For each $\omega \in \Omega$, if $T(\omega) \in [0,\infty)$ and $X_t(\omega) \in \mathbb{R}^d$ for all $t \ge 0$, we define

$$X_T(\omega) = X_{T(\omega)}(\omega).$$

Proposition 1.3.3. Let $\{X_t\}_{t\ge 0}$ be a progressively measurable process, and let T be a stopping time. Then X_T, defined on $\{T < \infty\}$, is $\mathcal{F}_T$-measurable, and the stopped process $\{X_{T\wedge t}\}_{t\ge 0}$ is progressively measurable.

Proof. Let $B \in \mathcal{B}(\mathbb{R}^d)$. By definition of $\mathcal{F}_T$,

$$\{X_T \in B\} \in \mathcal{F}_T \quad \Longleftrightarrow \quad \forall\, t \ge 0,\ \{X_T \in B\} \cap \{T \le t\} \in \mathcal{F}_t.$$

On the event $\{T \le t\}$ we have $X_T = X_{T\wedge t}$, so

$$\{X_T \in B\} \cap \{T \le t\} = \{X_{T\wedge t} \in B\} \cap \{T \le t\}.$$

Therefore, it is enough to prove that $X_{T\wedge t}$ is $\mathcal{F}_t$-measurable for every $t \ge 0$.

Observe that the map

$$(s,\omega) \mapsto (T(\omega) \wedge s, \omega)$$

is $\mathcal{B}([0,t]) \otimes \mathcal{F}_t$-measurable.

By assumption, $(s,\omega) \mapsto X_s(\omega)$ is $\mathcal{B}([0,t]) \otimes \mathcal{F}_t$-measurable.

Hence the composition

$$(s,\omega) \mapsto X_{T(\omega)\wedge s}(\omega)$$

is also $\mathcal{B}([0,t]) \otimes \mathcal{F}_t$-measurable. In particular, evaluating at $s = t$ shows that $X_{T\wedge t}$ is $\mathcal{F}_t$-measurable.

This also proves that the stopped process $(X_{T\wedge t})_{t\ge 0}$ is progressively measurable. □

1.4 Cts-time martingale

We now turn to continuous-time martingales and some of their basic properties.

Definition 1.4.1. On $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\ge 0}, \mathbb{P})$, a martingale is an adapted stochastic process $(X_t)_{t\ge 0}$ such that

$$\begin{cases} (i)\ \forall\, t \ge 0,\ \mathbb{E}|X_t| < \infty \\ (ii)\ \forall\, t \ge s \ge 0,\ \mathbb{E}(X_t \mid \mathcal{F}_s) = X_s. \end{cases}$$

If $\mathbb{E}(X_t \mid \mathcal{F}_s) \ge X_s$, we call (X_t) a submartingale.

If $\mathbb{E}(X_t \mid \mathcal{F}_s) \le X_s$, we call (X_t) a supermartingale.

Proposition 1.4.1. Let $\varphi : \mathbb{R} \to \mathbb{R}$ be a convex (concave up) function. If $(X_t)_{t\geq 0}$ is a martingale, then $(\varphi(X_t))_{t\geq 0}$ is a submartingale.

Examples. $|X_t|$, X_t^2, $|X_t|^p$ for all $p \geq 1$, and $(X_t)^+$ are all submartingales.
We assume $\mathbb{E}\,|\varphi(X_t)| < \infty$, $\forall\ t \geq 0$.

Proof. Jensen's inequality. □

Define

$$\begin{aligned} T_b^{(0)} &= -\infty, \\ T_a^{(1)} &= \inf\{t \geq (T_b^{(0)})^+ \colon X_t \leq a\}, \\ T_b^{(1)} &= \inf\{t \geq T_a^{(1)} \colon X_t \geq b\}. \end{aligned}$$

$\forall\ l \geq 2$,

$$\begin{aligned} T_a^{(l)} &= \inf\{t \geq T_b^{(l-1)} \colon X_t \leq a\}, \\ T_b^{l} &= \inf\{t \geq T_a^{l} \colon X_t \geq b\} \end{aligned}$$

Define

$$U_{ab}^X[0,n] = \max\{k \geq 0 \colon T_b^{(k)} \leq n\}$$

to be the number of upcrossings of the interval $[a,b]$ on $[0,n]$.

Theorem 1.4.1 (Continuous-time Doob's upcrossing inequality)**.** Let $(X_t)_{t\geq 0}$ be a right-continuous submartingale. For all $T > 0$,

$$\mathbb{E}\left(U_{ab}^X[0,T]\right) \leq \frac{\mathbb{E}(X_T - a)^+}{b-a}$$

Proof. For each $n \geq 1$, let $D_n = \mathbb{Z}/2^n = \{0, \frac{\pm 1}{2^n}, \frac{\pm 2}{2^n}, \cdots\}$.

Consider the upcrossings of X on $[0,T] \cap D_n$. The sampled process $\left\{\frac{X_j}{2^n}\right\}_{j=0}^{\infty}$ is a discrete-time submartingale, so

$$U_{ab}^X\left([0,T] \cap D_n\right) \leq U_{ab}^X[0,T], \qquad U_{ab}^X\left([0,T] \cap D_n\right) \uparrow U_{ab}^X([0,T]).$$

Applying the discrete-time upcrossing inequality, we obtain

$$\mathbb{E}\left(U_{ab}^X\left([0,T] \cap D_n\right)\right) \leq \frac{1}{b-a}\mathbb{E}(X_{T_n} - a)^+,$$

where $T_n = \frac{[T\cdot 2^n]}{2^n}$. Letting $n \to \infty$, we get

$$\mathbb{E}\left(U_{ab}^X([0,T])\right) \leq \frac{1}{b-a}\mathbb{E}(X_T - a)^+.$$

□

Proposition 1.4.2. If $(X_t)_{t\geq 0}$ is a right-continuous submartingale and $\sup_{t\geq 0} \mathbb{E}X_t^+ < \infty$, then there exists $X \in L^1$ such that $X_t \xrightarrow{a.s.} X$ as $t \to \infty$.

Proof.
$$\begin{aligned}\mathbb{E}\left(U_{ab}^X([0,\infty))\right) &= \lim_{n\to\infty}\mathbb{E}\left(U_{ab}^X([0,n])\right)\\ &\le \limsup_{n\to\infty}\frac{1}{b-a}\mathbb{E}(X_n-a)^+\\ &\le \limsup_{n\to\infty}\frac{1}{b-a}\left(\mathbb{E}(X_n^+)+|a|\right)<\infty.\end{aligned}$$

Hence for every $a<b$, we have $U_{ab}^X[0,\infty)<\infty$ a.s.

Therefore, with probability one, $U_{ab}^X[0,\infty)<\infty$ for all rational $a<b$.

Hence the event
$$\left\{\liminf_{t\to\infty} X_t \le a \le b < \limsup_{t\to\infty} X_t\right\}$$
cannot occur.

Therefore,
$$\liminf_{t\to\infty} X_t = \limsup_{t\to\infty} X_t$$
almost surely. □

Definition 1.4.2. A family of random variables $(X_t)_{t\ge 0}$ is uniformly integrable (U.I.) if
$$\lim_{M\to\infty}\sup_{t\ge 0}\mathbb{E}\left(|X_t|\,1_{|X_t|>M}\right)=0.$$

Remark. If $\sup_{t\ge 0}\mathbb{E}\,|X_t|^p<\infty$ for some $p>1$, then U.I.

Theorem 1.4.2. If $X_n \xrightarrow{P} X$, then the following are equivalent:
$$\begin{cases} X_n \xrightarrow{L^1} X,\\ \mathbb{E}\,|X_n| \to \mathbb{E}\,|X|\,,\\ \{X_n\} \text{ is U.I..}\end{cases}$$

Proposition 1.4.3. Let $Z\in L^1$. Then
$$\{\mathbb{E}(Z\mid\mathcal{G})\colon \mathcal{G} \text{ is a sub-}\sigma\text{-field of } \mathcal{F}\}$$
is uniformly integrable.

In particular, $\{\mathbb{E}(Z\mid\mathcal{F}_t)\}_{t\ge 0}$ is uniformly integrable, and it is a martingale by the tower property.

Proposition 1.4.4. A right-continuous martingale $(X_t)_{t\ge 0}$ is uniformly integrable if and only if there exists $X_\infty\in L^1$ such that
$$X_t=\mathbb{E}(X_\infty\mid\mathcal{F}_t).$$

Proof. "⇒" Uniform integrability implies $\sup_{t\ge 0}\mathbb{E}\,|X_t|<\infty$.

Hence the martingale convergence theorem yields some X_∞ such that $X_t \xrightarrow{a.s.} X_\infty$ and

$$\mathbb{E}\,|X_\infty| = \mathbb{E}\left(\lim_{t\to\infty} |X_t|\right) \le \liminf_{t\to\infty} \mathbb{E}\,|X_t| < \infty.$$

To prove that $X_t = \mathbb{E}(X_\infty \mid \mathcal{F}_t)$ for every $t \ge 0$, it is enough to show that

$$\mathbb{E}(X_t \mathbf{1}_A) = \mathbb{E}(X_\infty \mathbf{1}_A), \qquad \forall\ A \in \mathcal{F}_t.$$

For any $s \ge 0$, the martingale property gives

$$\mathbb{E}(X_{t+s} \mid \mathcal{F}_t) = X_t,$$

and hence

$$\mathbb{E}(X_{t+s}\mathbf{1}_A) = \mathbb{E}(X_t\mathbf{1}_A).$$

On the other hand,

$$|\mathbb{E}(X_{t+s}\mathbf{1}_A) - \mathbb{E}(X_\infty\mathbf{1}_A)| \le \mathbb{E}\,|X_{t+s} - X_\infty| \to 0.$$

Letting $s \to \infty$, we obtain

$$\mathbb{E}(X_t\mathbf{1}_A) = \mathbb{E}(X_\infty\mathbf{1}_A).$$

"$\Leftarrow$" If there exists $X_\infty \in L^1$ such that $X_t = \mathbb{E}(X_\infty \mid \mathcal{F}_t)$, then $\{X_t\}_{t\ge 0}$ is uniformly integrable by the previous proposition. □

Theorem 1.4.3 (Optional sampling Theorem). Let $(X_t)_{t\ge 0}$ be a right-continuous martingale. Let $S \le T$ be two stopping times. Suppose that either

① $\{X_t\}$ is uniformly integrable;

② there exists a constant $K > 0$ such that $S, T \le K$.

Then

$$\mathbb{E}(X_T \mid \mathcal{F}_S) = X_S.$$

In particular,

$$\mathbb{E}(X_T) = \mathbb{E}(X_S) = \mathbb{E}(X_0).$$

Proof. In case ①, let $Z = X_\infty$. In case ②, let $Z = X_K$.

In either case, we have $X_t = \mathbb{E}(Z \mid \mathcal{F}_t)$: for all $t \ge 0$ in case ①, and for all $t \le K$ in case ②.

It therefore suffices to prove that $X_T = \mathbb{E}(Z \mid \mathcal{F}_T)$. Then similarly $X_S = \mathbb{E}(Z \mid \mathcal{F}_S)$.

Hence

$$\begin{aligned}\mathbb{E}(X_T \mid \mathcal{F}_S) &= \mathbb{E}\left(\mathbb{E}(Z \mid \mathcal{F}_T) \mid \mathcal{F}_S\right)\\ &= \mathbb{E}(Z \mid \mathcal{F}_S)\\ &= X_S.\end{aligned}$$

First suppose that T takes values in some countable set $\{t_1, t_2, \dots\}$. Then for every $A \in \mathcal{F}_T$,

$$\begin{aligned}
\mathbb{E}(X_T 1_A) &= \sum_{n=1}^{\infty} \mathbb{E}(X_{t_n} 1_{\{T=t_n\}} 1_A) \\
&= \sum_{n=1}^{\infty} \mathbb{E}\left(X_{t_n} 1_{A\cap\{T=t_n\}}\right) \\
&= \sum_{n=1}^{\infty} \mathbb{E}\left(\mathbb{E}(Z \mid \mathcal{F}_{t_n}) 1_{A\cap\{T=t_n\}}\right) \\
&= \sum_{n=1}^{\infty} \mathbb{E}\left(Z 1_{A\cap\{T=t_n\}}\right) \\
&= \mathbb{E}(Z 1_A).
\end{aligned}$$

because $A \cap \{T = t_n\} \in \mathcal{F}_{t_n}$ and $\mathbb{E}(Z \mid \mathcal{F}_{t_n}) = X_{t_n}$.

Now consider a general stopping time T. Define $T_k = \frac{[2^k T]+1}{2^k}$.

For every $A \in \mathcal{F}_T \subseteq \mathcal{F}_{T_k}$, the first step gives

$$\mathbb{E}(Z 1_A) = \mathbb{E}(X_{T_k} 1_A).$$

Letting $k \to \infty$, we conclude that

$$\mathbb{E}(Z 1_A) = \mathbb{E}(X_T 1_A).$$

Indeed, $X_{T_k} \xrightarrow{a.s.} X_T$ by right-continuity, and $\{X_{T_k}\}$ is uniformly integrable because $X_{T_k} = \mathbb{E}(Z \mid \mathcal{F}_{T_k})$. Hence $X_{T_k} \xrightarrow{L^1} X_T$. □

Theorem 1.4.4 (Doob's Maximal Inequality)**.** Let $(X_t)_{t\geq 0}$ be a right-continuous submartingale. Then for every $\lambda > 0$,

$$\mathsf{P}(\sup_{0\leq s\leq t} X_s > \lambda) \leq \frac{1}{\lambda}\mathbb{E}X_t^+,$$

$$\mathsf{P}(\inf_{0\leq s\leq t} X_s < -\lambda) \leq \frac{1}{\lambda}(\mathbb{E}X_t^+ - \mathbb{E}X_0).$$

Proof. For $\lambda > 0$, let

$$T = \inf\{t \geq 0 : X_t > \lambda\}.$$

Then

$$\mathsf{P}(\sup_{0\leq s\leq t} X_s > \lambda) = \mathsf{P}(T \leq t).$$

Consider the stopped process $\{X_{t\wedge T}, t \geq 0\}$. Then

$$\begin{aligned}\lambda \mathsf{P}(T \leq t) = \lambda \mathbb{E}(1_{T\leq t}) &\leq \mathbb{E}(X_T 1_{T\leq t}) \\ &= \mathbb{E}(X_{T\wedge t} 1_{T\leq t}) \\ &= \mathbb{E}(X^+_{T\wedge t} 1_{T\leq t}) \\ &= \mathbb{E}(X^+_{T\wedge t}) \leq \mathbb{E}(X^+_t).\end{aligned}$$

Next let

$$S = \inf\{t \geq 0 : X_t < -\lambda\}.$$

Then

$$\mathsf{P}(\inf_{0\leq s\leq t} X_s < -\lambda) = \mathsf{P}(S \leq t).$$

Therefore

$$\lambda \mathsf{P}(S \leq t) = \lambda \mathbb{E}(1_{S\leq t}).$$

Because $X_S \leq -\lambda$, $\lambda \leq -X_S$, so

$$\lambda \mathsf{P}(S \leq t) \leq \mathbb{E}(-X_S 1_{S\leq t}) = -\mathbb{E}(X_{S\wedge t} 1_{S\leq t}).$$

Applying optional stopping to the submartingale, we obtain

$$\begin{aligned}\mathbb{E}X_0 \leq \mathbb{E}X_{t\wedge S} &= \mathbb{E}(X_{t\wedge S} 1_{S\leq t}) + \mathbb{E}(X_{t\wedge S} 1_{S>t}) \\ &\leq -\lambda \mathsf{P}(S \leq t) + \mathbb{E}(X^+_{t\wedge S}) \\ &\leq -\lambda \mathsf{P}(S \leq t) + \mathbb{E}(X^+_t).\end{aligned}$$

□

Theorem 1.4.5. $\forall\, p > 1$, $\mathbb{E}\left(\sup_{s\leq t} |X_s|\right)^p \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}\,|X_t|^p$.

Proof. Let $Y = \sup_{0\leq s\leq t} |X_s|$.

$$\begin{aligned}\lambda \mathsf{P}(Y \geq \lambda) &\leq \mathbb{E}(|X_{T\wedge t}|\, 1_{T\leq t}) \\ &= \mathbb{E}\left(|X_{T\wedge t}|\right) - \mathbb{E}\left(|X_{T\wedge t}|\, 1_{T>t}\right) \\ &\leq \mathbb{E}\left(|X_t|\right) - \mathbb{E}\left(|X_t|\, 1_{T>t}\right) \\ &= \mathbb{E}\,|X_t| - \mathbb{E}\left(|X_t|\, 1_{Y<\lambda}\right) \\ &= \mathbb{E}\left(|X_t|\, 1_{Y\geq\lambda}\right).\end{aligned}$$

Hence $\mathsf{P}(Y \geq \lambda) \leq \frac{1}{\lambda}\mathbb{E}\left(|X_t| \, 1_{Y \geq \lambda}\right)$.

$$\begin{aligned}
\mathbb{E}Y^p &= \int_0^\infty p\lambda^{p-1}\mathsf{P}(Y \geq \lambda)\ d\lambda \\
&\leq \int_0^\infty p\lambda^{p-2}\mathbb{E}\left(|X_t| \, 1_{Y \geq \lambda}\right)\ d\lambda \\
&= \mathbb{E}\left(|X_t| \int_0^\infty p\lambda^{p-2} 1_{Y \geq \lambda}\ d\lambda\right) \\
&= \mathbb{E}\left(|X_t| \int_0^Y p\lambda^{p-2}\ d\lambda\right) \\
&= \mathbb{E}\left(|X_t| \frac{p}{p-1} Y^{p-1}\right) \\
&\leq \frac{p}{p-1}\left(\mathbb{E}\,|X_t|^p\right)^{\frac{1}{p}}\left(\mathbb{E}(Y^{p-1})^q\right)^{\frac{1}{q}} \\
&= \frac{p}{p-1}\left(\mathbb{E}\,|X_t|^p\right)^{\frac{1}{p}}\left(\mathbb{E}Y^p\right)^{\frac{1}{q}}.
\end{aligned}$$

Therefore

$$\left(\mathbb{E}Y^p\right)^{\frac{1}{p}} \leq \frac{p}{p-1}\left(\mathbb{E}\,|X_t|^p\right)^{\frac{1}{p}}.$$

To justify the argument rigorously, consider $Y \wedge k$ for $k > 0$. Then

$$\begin{aligned}
\mathbb{E}\,(Y \wedge k)^p &= \int_0^\infty p\lambda^{p-1}\mathsf{P}(Y \wedge k \geq \lambda)\ d\lambda \\
&= \int_0^k p\lambda^{p-1}\mathsf{P}(Y \wedge k \geq \lambda)\ d\lambda + \underbrace{\int_k^\infty p\lambda^{p-1}\mathsf{P}(Y \wedge k \geq \lambda)\ d\lambda}_{=0} \\
&= \int_0^k p\lambda^{p-1}\mathsf{P}(Y \wedge k \geq \lambda)\ d\lambda \\
&= \int_0^k p\lambda^{p-1}\mathsf{P}(Y \geq \lambda)\ d\lambda \\
&\leq \mathbb{E}\left(|X_t| \int_0^k p\lambda^{p-1} 1_{Y \geq \lambda}\ d\lambda\right) \\
&= \mathbb{E}\left(|X_t| \int_0^{k \wedge Y} p\lambda^{p-2}\ d\lambda\right) \\
&= \mathbb{E}\left[|X_t| \frac{p}{p-1}(Y \wedge k)^{p-1}\right].
\end{aligned}$$

$\implies \mathbb{E}(Y \wedge k)^p \leq \frac{p}{p-1}\left(\mathbb{E}\,|X_t|^p\right)^{\frac{1}{p}}\left(\mathbb{E}(Y \wedge k)^p\right)^{\frac{1}{q}}$.
$\implies \left(\mathbb{E}(Y \wedge k)^p\right)^{\frac{1}{p}} \leq \frac{p}{p-1}\left(\mathbb{E}\,|X_t|^p\right)^{\frac{1}{p}}$.
Letting $k \to \infty$, we obtain

$$\left(\mathbb{E}Y^p\right)^{\frac{1}{p}} \leq \frac{p}{p-1}\left(\mathbb{E}\,|X_t|^p\right)^{\frac{1}{p}}.$$

□

1.4.1 The Doob-Meyer Decomposition

We now turn to the continuous-time analogue of the discrete Doob decomposition.

Recall the discrete-time Doob decomposition

$$X_n = M_n + A_n,$$

where X_n is a submartingale, M_n is a martingale, and A_n is a predictable increasing process.

$$\begin{aligned}
&\mathbb{E}\left(M_{n+1} \mid \mathcal{F}_n\right) = M_n \\
\Longrightarrow\ &\mathbb{E}(X_{n+1} - A_{n+1} \mid \mathcal{F}_n) = X_n - A_n \\
\Longrightarrow\ &\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) - A_{n+1} = X_n - A_n \\
\Longrightarrow\ &A_{n+1} - A_n = \mathbb{E}(X_{n+1} \mid \mathcal{F}_n) - X_n \geq 0.
\end{aligned}$$

$$\begin{cases} A_n = \sum_{k=0}^{n-1} \left(\mathbb{E}(X_{k+1} \mid \mathcal{F}_k) - X_k\right) \\ A_0 = 0. \end{cases}$$

For a right-continuous submartingale $(X_t)_{t\geq 0}$, we seek a decomposition of the form

$$X_t = M_t + A_t,$$

where M_t is a martingale and A_t is an increasing and “predictable” process.

So we first clarify what properties the increasing part should satisfy.

Definition 1.4.3. An adapted process A is called increasing if, almost surely,

(a) $A_0 = 0$;

(b) $t \mapsto A_t$ is nondecreasing and right-continuous;

(c) $\mathbb{E}(A_t) < \infty$, $\forall\, t \geq 0$.

We say that A is integrable if

$$\mathbb{E}(A_\infty) = \mathbb{E}(\lim_{t\to\infty} A_t) < \infty.$$

Definition 1.4.4. An increasing process A is called natural if for every bounded, right-continuous martingale $(M_t)_{t\geq 0}$ and every $t > 0$,

$$\mathbb{E}\int_0^t M_s \, dA_s = \mathbb{E}\int_0^t M_{s-} \, dA_s.$$

Remark. If $s \mapsto A_s$ is a.s. continuous, then A is natural.

Definition 1.4.5. An increasing sequence $(A_n, \mathcal{F}_n)$ is called natural if for every bounded martingale $\{M_n, \mathcal{F}_n\}$,

$$\mathbb{E}(M_n A_n) = \mathbb{E}\left(\sum_{k=1}^{n} M_{k-1}(A_k - A_{k-1})\right).$$

Proposition 1.4.5. In discrete time, predictability is equivalent to naturality.

Proof. "$\Rightarrow$" Assume that A is predictable, i.e. $A_n \in \mathcal{F}_{n-1}$. We show that A is natural.

Indeed,

$$\begin{aligned}
\mathbb{E}(M_n A_n) &= \mathbb{E}\left(\sum_{k=1}^{n} M_k A_k - \sum_{k=1}^{n} M_{k-1} A_{k-1}\right) \\
&= \sum_{k=1}^{n} \mathbb{E}(M_k A_k) - \sum_{k=1}^{n} \mathbb{E}(M_{k-1} A_{k-1}) \\
&\iff \sum_{k=2}^{n} \mathbb{E}(M_{k-1} A_{k-1}) + \mathbb{E}(M_n A_n) = \sum_{k=1}^{n} \mathbb{E}(M_{k-1} A_k) \\
&\iff \sum_{k=1}^{n} \mathbb{E}(M_k A_k) = \sum_{k=1}^{n} \mathbb{E}(M_{k-1} A_k) \\
&\iff \sum_{k=1}^{n} \mathbb{E}\left[(M_k - M_{k-1}) A_k\right] = 0.
\end{aligned}$$

Since $A_n \in \mathcal{F}_{n-1}$,

$$\mathbb{E}\left(A_n (M_n - M_{n-1})\right) = \mathbb{E}\left[A_n \underbrace{\mathbb{E}(M_n - M_{n-1} \mid \mathcal{F}_{n-1})}_{=0}\right] = 0.$$

Therefore A is natural.

"$\Leftarrow$" Now assume that A is natural. Then for every bounded martingale (M_n),

$$\mathbb{E}\left[A_n (M_n - M_{n-1})\right] = 0, \quad \forall\, n \geq 1.$$

Our goal is to prove that

$$A_n = \mathbb{E}(A_n \mid \mathcal{F}_{n-1}) \quad \text{a.s.},$$

which is equivalent to $A_n \in \mathcal{F}_{n-1}$.

$$\begin{aligned}
\mathbb{E}\left[M_n \left(A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})\right)\right] &= \mathbb{E}(M_n A_n) - \mathbb{E}\left[M_n \, \mathbb{E}(A_n \mid \mathcal{F}_{n-1})\right] \\
&= \mathbb{E}(M_n A_n) - \mathbb{E}\left[\mathbb{E}(A_n \mid \mathcal{F}_{n-1}) \, \mathbb{E}(M_n \mid \mathcal{F}_{n-1})\right] \\
&= \mathbb{E}(M_n A_n) - \mathbb{E}\left[\mathbb{E}(A_n \mid \mathcal{F}_{n-1}) M_{n-1}\right] \\
&= \mathbb{E}(M_n A_n) - \mathbb{E}(A_n M_{n-1}) \\
&= \mathbb{E}\left[A_n (M_n - M_{n-1})\right] = 0.
\end{aligned}$$

Fix $n \geq 1$, and let

$$X = \operatorname{sgn}\left(A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})\right)$$

Then $X\left(A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})\right) = |A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})|$.

Define

$$M_k = \begin{cases} \mathbb{E}(X \mid \mathcal{F}_k), & 0 \le k \le n-1 \\ X, & k = n \end{cases}$$

Then (M_k) is a bounded martingale.

$$\begin{aligned} 0 &= \mathbb{E}\left[M_n(A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1}))\right] \\ &= \mathbb{E}\left|A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})\right| \end{aligned}$$

Therefore $A_n = \mathbb{E}(A_n \mid \mathcal{F}_{n-1})$ a.s.

Hence $A_n \in \mathcal{F}_{n-1}$, so A is predictable. □

The previous proposition gives the discrete-time characterization of natural increasing processes. We now explain why the continuous-time definition is the correct analogue.

$$\begin{aligned} \mathbb{E}(M_n A_n) &= \sum_{k=1}^{n} \mathbb{E}\left(M_{k-1}(A_k - A_{k-1})\right) \\ \Longleftrightarrow \mathbb{E}(M_t A_t) &= \mathbb{E}\left(\int_0^t M_{s-}\, dA_s\right) = \mathbb{E}\left(\int_0^t M_s\, dA_s\right). \end{aligned}$$

Indeed, the discrete sum should be viewed as a Riemann–Stieltjes approximation of the corresponding continuous-time integral.

$$\int_0^t M_s\, dA_s = \lim_{\|\Pi\| \to 0} \int_0^t M_s^{\Pi}\, dA_s.$$

where $\Pi = \{t_0, t_1, \cdots, t_n\} \subseteq [0, t]$ with $0 = t_0 < t_1 < \cdots < t_n = t$ and $\|\Pi\| = \max_{1 \le k \le n}(t_k - t_{k-1})$. By right-continuity,

$$M_s^{\Pi} = \sum_{k=1}^{n} M_{t_k} \mathbb{1}_{(t_{k-1}, t_k]}(s) \xrightarrow{a.s.} M_s.$$

For each Π,

$$\begin{aligned} \mathbb{E}\int_0^t M_s^{\Pi}\, dA_s &= \mathbb{E}\sum_{k=1}^{n} M_{t_k}(A_{t_k} - A_{t_{k-1}}) \\ &= \sum_{k=1}^{n} \mathbb{E}(M_{t_k} A_{t_k}) - \sum_{k=1}^{n} \mathbb{E}(M_{t_k} A_{t_{k-1}}) \\ &= \mathbb{E}(M_{t_n} A_{t_n}) + \sum_{k=1}^{n-1} \mathbb{E}(M_{t_k} A_{t_k}) - \sum_{k=2}^{n} \mathbb{E}(M_{t_k} A_{t_{k-1}}) \\ &= \mathbb{E}(M_t A_t) + \sum_{k=1}^{n-1} \mathbb{E}(M_{t_k} A_{t_k}) - \sum_{k=1}^{n-1} \mathbb{E}(M_{t_{k+1}} A_{t_k}) \\ &= \mathbb{E}(M_t A_t) - \underbrace{\sum_{k=1}^{n-1} \mathbb{E}\left(A_{t_k}(M_{t_{k+1}} - M_{t_k})\right)}_{=0}. \end{aligned}$$

Thus the continuous-time identity is exactly the limit of the discrete-time naturality relation.

Definition 1.4.6. Let

$$S = \{\text{stopping time } T \text{ with } T < \infty \text{ a.s.}\}.$$

We say that a right-continuous process $\{X_t\}_{t\geq 0}$ is of class D if $\{X_T\}_{T\in S}$ is uniformly integrable. In particular, this implies that $\{X_t\}_{t\geq 0}$ itself is uniformly integrable.

Fix $a > 0$, and let

$$S_a = \{\text{stopping time } T \text{ with } T \leq a \text{ a.s.}\}.$$

We say that $\{X_t\}_{t\geq 0}$ is of class DL if $\{X_T\}_{T\in S_a}$ is uniformly integrable for every $a > 0$.

Theorem 1.4.6 (Doob-Meyer Decomposition)**.** If $\{X_t\}$ is of class DL, then $X_t = M_t + A_t$, where $\{M_t\}$ is a martingale and $\{A_t\}$ is increasing.

If (A_t) is natural, then (M_t, A_t) is unique.

Proof. Assume that

$$X_t = M_t + A_t = M_t' + A_t',$$

where (M_t) and (M_t') are martingales, and (A_t) and (A_t') are natural increasing processes.

Let

$$B_t = M_t - M_t' = A_t' - A_t.$$

Then (B_t) is a martingale, and also a process of bounded variation. Moreover, it is natural.

To prove uniqueness, let $(\xi_t)_{t\geq 0}$ be any bounded right-continuous martingale. Then

$$\begin{aligned}
\mathbb{E}\big[\xi_t(A_t - A_t')\big] &= \mathbb{E}\int_0^t \xi_{s-}(dA_s - dA_s') \\
&= \mathbb{E}\left[\int_0^t \xi_{s-}\, dB_s\right] \\
&= \lim_{n\to\infty} \mathbb{E}\sum_{j=1}^{m_n} \xi_{t_{j-1}^{(n)}}\left[B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}}\right],
\end{aligned}$$

where $\Pi_n = \{t_0^{(n)}, \cdots, t_{m_n}^{(n)}\}$ is a partition of $[0, t]$.

For each j,

$$\mathbb{E}\left[\xi_{t_{j-1}^{(n)}}\big(B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}}\big)\right] = \mathbb{E}\left[\xi_{t_{j-1}^{(n)}}\underbrace{\mathbb{E}\big(B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}} \mid \mathcal{F}_{t_{j-1}^{(n)}}\big)}_{=0}\right].$$

Hence $\mathbb{E}\big[\xi_t(A_t - A_t')\big] = 0$.

Now let ξ be any bounded $\mathcal{F}_t$-measurable random variable. Applying the preceding identity with the martingale

$$\xi_s = \mathbb{E}(\xi \mid \mathcal{F}_s), \qquad 0 \leq s \leq t,$$

we get

$$\mathbb{E}\left[\xi(A_t - A'_t)\right] = 0.$$

Taking $\xi = \operatorname{sgn}(A_t - A'_t)$, we obtain

$$\mathbb{E}\left|A_t - A'_t\right| = 0.$$

Hence $A_t = A'_t$ a.s. for every $t > 0$. In particular, this holds for all $q \in \mathbb{Q}_+$. By right-continuity, we conclude that $A_t = A'_t$ for all $t \geq 0$ a.s.

Existence: Fix $a > 0$.

It suffices to construct the decomposition on $[0, a]$.

Consider

$$Y_t \triangleq X_t - \mathbb{E}(X_a \mid \mathcal{F}_t), \qquad 0 \leq t \leq a.$$

Then $(Y_t)_{0\leq t\leq a}$ is a submartingale, because

$$\begin{aligned}\mathbb{E}(Y_t \mid \mathcal{F}_s) &= \mathbb{E}(X_t \mid \mathcal{F}_s) - \mathbb{E}(X_a \mid \mathcal{F}_s) \\ &\geq X_s - \mathbb{E}(X_a \mid \mathcal{F}_s) = Y_s, \qquad 0 \leq s \leq t \leq a.\end{aligned}$$

Let $\Pi_n = \{t_0^{(n)}, \cdots, t_{2^n}^{(n)}\}$ with $t_j^{(n)} = \frac{j}{2^n}a$, $0 \leq j \leq 2^n$. For each n, the discrete-time Doob decomposition gives

$$Y_{t_j^{(n)}} = M_{t_j^{(n)}}^{(n)} + A_{t_j^{(n)}}^{(n)}, \qquad 0 \leq j \leq 2^n,$$

where

$$A_{t_j^{(n)}}^{(n)} = \sum_{k=0}^{j-1} \mathbb{E}\left(Y_{t_{k+1}^{(n)}} - Y_{t_k^{(n)}} \middle| \mathcal{F}_{t_k^{(n)}}\right).$$

Since

$$\begin{cases} Y_0 = X_0 - \mathbb{E}(X_a \mid \mathcal{F}_0) \leq 0, \\ Y_a = X_a - \mathbb{E}(X_a \mid \mathcal{F}_a) = 0, \end{cases}$$

we have

$$0 = Y_a = M_a^{(n)} + A_a^{(n)}.$$

Hence

$$M_{t_j^{(n)}}^{(n)} = \mathbb{E}(M_a^{(n)} \mid \mathcal{F}_{t_j^{(n)}}) = -\mathbb{E}(A_a^{(n)} \mid \mathcal{F}_{t_j^{(n)}}),$$

and therefore

$$Y_{t_j^{(n)}} = A_{t_j^{(n)}}^{(n)} - \mathbb{E}(A_a^{(n)} \mid \mathcal{F}_{t_j^{(n)}}).$$

We will show that $\{A_a^{(n)}\}_{n\geq 1}$ is uniformly integrable. Then, by the Dunford-Pettis compactness criterion, there exists an integrable random variable A_a such that

$$A_a^{(n_k)} \rightharpoonup A_a \qquad \text{weakly in } L^1$$

along some subsequence. Equivalently, for every bounded random variable ξ,

$$\lim_{k\to\infty} \mathbb{E}(\xi A_a^{(n_k)}) = \mathbb{E}(\xi A_a).$$

Define

$$A_t = Y_t + \mathbb{E}(A_a \mid \mathcal{F}_t), \qquad 0 \le t \le a.$$

By Problem 4.11, the convergence $\mathbb{E}(\xi A_a^{(n_k)}) \to \mathbb{E}(\xi A_a)$ for every bounded random variable ξ implies

$$\mathbb{E}(\xi \mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t)) \to \mathbb{E}(\xi \mathbb{E}(A_a \mid \mathcal{F}_t)),$$

that is, $\mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t)$ converges weakly in L^1 to $\mathbb{E}(A_a \mid \mathcal{F}_t)$.

Fix a dyadic time t. Then, for all sufficiently large k, we have

$$\begin{cases} A_t = Y_t + \mathbb{E}(A_a \mid \mathcal{F}_t), \\ A_t^{(n_k)} = Y_t + \mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t) \end{cases}$$

because t belongs to the dyadic partition Π_{n_k} once k is large enough. Therefore

$$A_t^{(n_k)} - A_t = \mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t) - \mathbb{E}(A_a \mid \mathcal{F}_t).$$

Since $\mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t)$ converges weakly in L^1 to $\mathbb{E}(A_a \mid \mathcal{F}_t)$, it follows that

$$\mathbb{E}(\xi A_t^{(n_k)}) \to \mathbb{E}(\xi A_t)$$

for every bounded random variable ξ.

For any dyadic times $0 \le s < t \le a$,

$$\mathbb{E}(\xi(A_t - A_s)) = \lim_{k\to\infty} \mathbb{E}(\xi(A_t^{(n_k)} - A_s^{(n_k)})) \ge 0$$

whenever $\xi \ge 0$.

Taking $\xi = \mathbf{1}_{A_s > A_t}$, we get

$$\mathbb{E}\left[(A_t - A_s)\mathbf{1}_{A_s > A_t}\right] \ge 0.$$

Therefore $A_t - A_s \ge 0$ a.s.

Since

$$A_0^{(n)} = 0 = Y_0 + \mathbb{E}(A_a^{(n)} \mid \mathcal{F}_0),$$

passing to the limit gives

$$A_0 = Y_0 + \mathbb{E}(A_a \mid \mathcal{F}_0) = 0.$$

Moreover, for any dyadic time t,

$$X_t - \mathbb{E}(X_a \mid \mathcal{F}_t) = Y_t = A_t^{(n_k)} - \mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t)$$

for all sufficiently large k.

Passing to the limit along the subsequence gives

$$Y_t = A_t - \mathbb{E}(A_a \mid \mathcal{F}_t) = X_t - \mathbb{E}(X_a \mid \mathcal{F}_t).$$

Therefore

$$X_t = A_t + \mathbb{E}(X_a - A_a \mid \mathcal{F}_t)$$

for every dyadic time $t \in [0, a]$. By right-continuity, the same identity then holds for all $t \in [0, a]$.

Define

$$M_t = \mathbb{E}(X_a - A_a \mid \mathcal{F}_t), \qquad 0 \le t \le a.$$

Then $(M_t)_{0 \le t \le a}$ is a martingale and

$$X_t = A_t + M_t.$$

Finally, we prove that (A_t) is natural. It is enough to show that for every bounded right-continuous martingale (ξ_t),

$$\mathbb{E} \int_0^a \xi_s \, dA_s = \mathbb{E} \int_0^a \xi_{s-} \, dA_s.$$

We first note that

$$\mathbb{E} \int_0^a \xi_s \, dA_s = \mathbb{E}(\xi_a A_a) = \lim_{n \to \infty} \mathbb{E}(\xi_a A_a^{(n)}).$$

On the other hand,

$$\mathbb{E} \int_0^a \xi_{s-} \, dA_s = \lim_{n \to \infty} \mathbb{E} \sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} (A_{t_j^{(n)}} - A_{t_{j-1}^{(n)}}).$$

Since $\{A^{(n)}_{t_j^{(n)}}\}_{0 \le j \le 2^n}$ is predictable, and $t \mapsto \mathbb{E}(A_a \mid \mathcal{F}_t)$ is a martingale,

$$\begin{aligned}
\mathbb{E}(\xi_a A_a^{(n)}) &= \mathbb{E}\left(\sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} (A^{(n)}_{t_j^{(n)}} - A^{(n)}_{t_{j-1}^{(n)}}) \right) \\
&= \mathbb{E}\left(\sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} (Y_{t_j^{(n)}} - Y_{t_{j-1}^{(n)}}) \right) \\
&= \mathbb{E}\left(\sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} \Big[(A_{t_j^{(n)}} - A_{t_{j-1}^{(n)}}) - \big(\mathbb{E}(A_a \mid \mathcal{F}_{t_j^{(n)}}) - \mathbb{E}(A_a \mid \mathcal{F}_{t_{j-1}^{(n)}}) \big) \Big] \right) \\
&= \mathbb{E}\left(\sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} (A_{t_j^{(n)}} - A_{t_{j-1}^{(n)}}) \right).
\end{aligned}$$

Here the last equality follows because $t \mapsto \mathbb{E}(A_a \mid \mathcal{F}_t)$ is a martingale. Letting $n \to \infty$, we obtain

$$\mathbb{E}\int_0^a \xi_s \, dA_s = \mathbb{E}\int_0^a \xi_{s-} \, dA_s.$$

Therefore (A_t) is natural. □

1.5 Square Integrable

Definition 1.5.1. Let $(X_t)_{t\geq 0}$ be a right-continuous martingale. We say that X is square-integrable if

$$\mathbb{E}X_t^2 < \infty, \qquad \forall\, t \geq 0.$$

Throughout this section, we assume $X_0 = 0$ and write $X \in \mathcal{M}_2$.

Then $Y_t = X_t^2$ is a submartingale. Recall that a process is of class DL if, for every $a > 0$, the family $\{Y_T : T \leq a \text{ is a stopping time}\}$ is uniformly integrable. By the Doob–Meyer decomposition,

$$Y_t = X_t^2 = M_t + A_t, \qquad t \geq 0.$$

If $X \in \mathcal{M}_2^c$ is continuous, then we may choose M_t and A_t to be continuous as well.

Definition 1.5.2. For $X \in \mathcal{M}_2$, the quadratic variation of X is defined by $\langle X \rangle_t \triangleq A_t$, that is, $\langle X \rangle_t$ is the unique adapted natural increasing process such that $\langle X \rangle_0 = 0$ and

$$X_t^2 - \langle X \rangle_t$$

is a martingale.

Remark. The notation $\langle X \rangle_t$ is introduced here through the Doob–Meyer decomposition of the submartingale X_t^2.

If $X, Y \in \mathcal{M}_2$, then

$$\begin{cases} (X_t + Y_t)^2 - \langle X + Y \rangle_t \text{ is a martingale,} \\ (X_t - Y_t)^2 - \langle X - Y \rangle_t \text{ is a martingale.} \end{cases}$$

Subtracting the second relation from the first, we see that

$$4X_tY_t - \big(\langle X + Y \rangle_t - \langle X - Y \rangle_t\big)$$

is a martingale.

Definition 1.5.3. For $X, Y \in \mathcal{M}_2$, we define the cross variation by

$$\langle X, Y \rangle_t \triangleq \frac{1}{4}\left(\langle X + Y \rangle_t - \langle X - Y \rangle_t\right),$$

so that

$$X_tY_t - \langle X, Y\rangle_t$$

is a martingale.

Remark. $\langle X, X\rangle_t = \langle X\rangle_t$.

Definition 1.5.4. Let $\Pi = \{t_0, \cdots, t_m\}$ be a partition of $[0,t]$. The p-th variation of X over Π is defined by

$$V_t^{(p)}(\Pi) = \sum_{k=1}^{m} \left|X_{t_k} - X_{t_{k-1}}\right|^p.$$

When $p=2$, we call $V_t^{(2)}(\Pi)$ the quadratic variation along the partition Π.

Theorem 1.5.1. Let $X \in \mathcal{M}_2^c$. For any partition Π of $[0,t]$, we have

$$V_t^{(2)}(\Pi) \xrightarrow{P} \langle X\rangle_t$$

as $\|\Pi\| \to 0$.

Proof. Let $\Pi = \{t_0, \ldots, t_m\}$ and write

$$\Delta_k X = X_{t_k} - X_{t_{k-1}}, \qquad \Delta_k\langle X\rangle = \langle X\rangle_{t_k} - \langle X\rangle_{t_{k-1}}.$$

Then

$$V_t^{(2)}(\Pi) - \langle X\rangle_t = \sum_{k=1}^{m} \left((\Delta_k X)^2 - \Delta_k\langle X\rangle\right).$$

Hence

$$\begin{aligned}
&\mathbb{E}\left(V_t^{(2)}(\Pi) - \langle X\rangle_t\right)^2 \\
=&\mathbb{E}\left[\sum_{k=1}^{m}\left((\Delta_k X)^2 - \Delta_k\langle X\rangle\right)\right]^2 \\
=&\sum_{k=1}^{m}\mathbb{E}\left[\left((\Delta_k X)^2 - \Delta_k\langle X\rangle\right)^2\right] \\
&+2\sum_{1\le j<k\le m}\mathbb{E}\left[\left((\Delta_k X)^2 - \Delta_k\langle X\rangle\right)\left((\Delta_j X)^2 - \Delta_j\langle X\rangle\right)\right].
\end{aligned}$$

The cross terms vanish. Indeed, for $j<k$, the random variable

$$(\Delta_j X)^2 - \Delta_j\langle X\rangle$$

is $\mathcal{F}_{t_{k-1}}$-measurable, while

$$\mathbb{E}\left((\Delta_k X)^2 - \Delta_k\langle X\rangle \middle| \mathcal{F}_{t_{k-1}}\right) = 0$$

because $X_s^2 - \langle X \rangle_s$ is a martingale. Therefore

$$\mathbb{E}\left[\left((\Delta_k X)^2 - \Delta_k \langle X \rangle\right)\left((\Delta_j X)^2 - \Delta_j \langle X \rangle\right)\right] = 0.$$

Consequently,

$$\begin{aligned}
\mathbb{E}\left(V_t^{(2)}(\Pi) - \langle X \rangle_t\right)^2 &= \sum_{k=1}^m \mathbb{E}\left[\left((\Delta_k X)^2 - \Delta_k \langle X \rangle\right)^2\right] \\
&\le 2\sum_{k=1}^m \mathbb{E}(\Delta_k X)^4 + 2\sum_{k=1}^m \mathbb{E}(\Delta_k \langle X \rangle)^2 \\
&= I + II \\
&= 2\mathbb{E}V_t^{(4)}(\Pi) + II
\end{aligned}$$

We first consider the bounded case. Assume that

$$|X_s| \le K, \qquad \langle X \rangle_s \le K, \qquad 0 \le s \le t.$$

Lemma 1.5.1. Let $X \in \mathcal{M}_2$ satisfy $|X_s| \le K < \infty$ for all $0 \le s \le t$. Then

$$\mathbb{E}\left[V_t^{(2)}(\Pi)^2\right] \le 6K^4.$$

Assuming this lemma, we obtain

$$V_t^{(4)}(\Pi) \le V_t^{(2)}(\Pi)\, m_t^2(X; \|\Pi\|),$$

where

$$m_t(X, \delta) = \sup\{|X_r - X_s| : |s - r| \le \delta,\ 0 \le r < s \le t\}.$$

Hence, by Cauchy–Schwarz,

$$\begin{aligned}
\mathbb{E}\left[V_t^{(4)}(\Pi)\right] &\le \mathbb{E}\left[V_t^{(2)}(\Pi)\, m_t^2(X, \|\Pi\|)\right] \\
&\le \left(\mathbb{E}\left[V_t^{(2)}(\Pi)^2\right]\right)^{1/2} \left(\mathbb{E}m_t^4(X, \|\Pi\|)\right)^{1/2} \\
&\le (6K^4)^{1/2} \left(\mathbb{E}m_t^4(X, \|\Pi\|)\right)^{1/2}.
\end{aligned}$$

Since X is continuous and bounded on $[0, t]$, bounded convergence yields

$$\mathbb{E}m_t^4(X, \|\Pi\|) \to 0$$

as $\|\Pi\| \to 0$. Thus $I \to 0$.

For II, note that $\langle X\rangle$ is continuous and increasing, so

$$\sum_{k=1}^{m}(\Delta_k\langle X\rangle)^2 \le \sup_{1\le k\le m}|\Delta_k\langle X\rangle|\ \langle X\rangle_t.$$

Since $\langle X\rangle$ is uniformly continuous on $[0,t]$ and bounded by K, it follows that

$$\mathbb{E}\left[\sup_{1\le k\le m}|\Delta_k\langle X\rangle|\ \langle X\rangle_t\right]\to 0$$

as $\|\Pi\|\to 0$. Hence $II\to 0$ as well.

Therefore, in the bounded case,

$$\mathbb{E}\left[\left(V_t^{(2)}(\Pi)-\langle X\rangle_t\right)^2\right]\to 0$$

as $\|\Pi\|\to 0$, and hence

$$V_t^{(2)}(\Pi)\xrightarrow{P}\langle X\rangle_t.$$

For a general $X\in\mathcal{M}_2^c$, we use localization. For $n\ge 1$, define

$$T_n=\inf\{t\ge 0:|X_t|\ge n \text{ or } \langle X\rangle_t\ge n\}.$$

Then $T_n\uparrow\infty$ a.s., and

$$X_t^{(n)}=X_{t\wedge T_n}$$

is a bounded martingale. Moreover,

$$X_{t\wedge T_n}^2-\langle X\rangle_{t\wedge T_n}$$

is a martingale, so by uniqueness of the quadratic variation,

$$\langle X^{(n)}\rangle_t=\langle X\rangle_{t\wedge T_n}.$$

Hence

$$\begin{cases}\left|X_s^{(n)}\right|\le n, \\ \langle X^{(n)}\rangle_s\le n,\end{cases}\qquad 0\le s\le t,$$

and the bounded case gives

$$\mathbb{E}\left[\left(V_{t\wedge T_n}^{(2)}(X)-\langle X\rangle_{t\wedge T_n}\right)^2\right]\to 0$$

as $\|\Pi\|\to 0$.

Now fix $\varepsilon>0$ and $\eta>0$. Choose n so large that

$$\mathsf{P}(T_n<t)\le\frac{\eta}{2}.$$

Then

$$\mathsf{P}\left(\left|V_t^{(2)}(X) - \langle X\rangle_t\right| > \varepsilon\right) \le \mathsf{P}(T_n < t) + \mathsf{P}\left(T_n \ge t,\ \left|V_{t\wedge T_n}^{(2)}(X) - \langle X\rangle_{t\wedge T_n}\right| > \varepsilon\right)$$
$$\le \frac{\eta}{2} + \frac{1}{\varepsilon^2}\mathbb{E}\left[\left(V_{t\wedge T_n}^{(2)}(X) - \langle X\rangle_{t\wedge T_n}\right)^2\right].$$

Taking $\|\Pi\|$ small enough, the second term is at most $\eta/2$. Therefore

$$\mathsf{P}\left(\left|V_t^{(2)}(X) - \langle X\rangle_t\right| > \varepsilon\right) \le \eta.$$

This proves that

$$V_t^{(2)}(\Pi) \xrightarrow{P} \langle X\rangle_t.$$

□

Corollary 1.5.1. If $V_t^{(2)}(\Pi) \xrightarrow{P} \langle X\rangle_t$, then

$$\begin{cases} \forall\ q > 2, \quad V_t^{(q)}(\Pi) \xrightarrow{P} 0, \\ \forall\ q < 2, \quad V_t^{(q)}(\Pi) \xrightarrow{P} \infty \end{cases}$$

on $\{\langle X\rangle_t > 0\}$.

We use the following notation:

$$\mathcal{M}_2 = \{X : \mathbb{E}X_t^2 < \infty,\ \forall\ t \ge 0\},$$

$$\mathcal{M}^2 = \left\{X : \sup_{t\ge 0} \mathbb{E}X_t^2 < \infty\right\},$$

and

$$\mathcal{M}_0^2 = \{X \in \mathcal{M}^2 : X_0 = 0\}.$$

1.6 L^2-Theory

The space $\mathcal{M}^2$ is naturally identified with $L^2(\mathcal{F}_\infty)$ through the correspondence

$$X \longmapsto X_\infty.$$

Indeed, for every $X \in \mathcal{M}^2$, we have

$$\begin{cases} X_\infty = \lim_{t\to\infty} X_t \quad \text{a.s.,} \\ X_t \xrightarrow{L^2} X_\infty. \end{cases}$$

Conversely, if $X_\infty \in L^2(\mathcal{F}_\infty)$, then

$$X_t \triangleq \mathbb{E}\left(X_\infty | \mathcal{F}_t\right)$$

defines a martingale in $\mathcal{M}^2$. Moreover,

$$\mathbb{E}\left(X_t^2\right) = \mathbb{E}\left[\mathbb{E}\left(X_\infty|\mathcal{F}_t\right)^2\right] \le \mathbb{E}\left[\mathbb{E}\left(X_\infty^2\middle|\mathcal{F}_t\right)\right] = \mathbb{E}[X_\infty^2].$$

Thus, for $X, Y \in \mathcal{M}^2$, it is natural to define

$$\langle X, Y\rangle \triangleq \mathbb{E}\left[X_\infty Y_\infty\right].$$

Remark. Here $\langle X, Y\rangle$ denotes the inner product on the Hilbert space $\mathcal{M}^2$. It is different from the time-dependent bracket process $\langle X\rangle_t$ or $\langle X, Y\rangle_t$ introduced in the previous section.

Theorem 1.6.1 (Doob's L^2 inequality)**.**

$$\mathbb{E}\left[\sup_{0\le s\le t}|X_s|^2\right] \le 4\mathbb{E}\left[X_t^2\right].$$

Letting $t \to \infty$, we obtain

$$\mathbb{E}\left[\sup_{s\ge 0}|X_s|^2\right] \le 4\mathbb{E}[X_\infty^2].$$

The space $\mathcal{M}^2$ is complete. Indeed, if $\{M^{(n)}\}$ is a Cauchy sequence in $\mathcal{M}^2$, then $\{M_\infty^{(n)}\}$ is Cauchy in $L^2(\mathcal{F}_\infty)$. Hence there exists $M_\infty \in L^2(\mathcal{F}_\infty)$ such that

$$M_\infty^{(n)} \xrightarrow{L^2} M_\infty.$$

Let

$$M_t = \mathbb{E}(M_\infty \mid \mathcal{F}_t).$$

Then

$$\sup_{t\ge 0}\left|M_t^{(n)} - M_t\right| \xrightarrow{L^2} 0.$$

Proof. By Doob's L^2 inequality,

$$\begin{aligned}
\mathbb{E}\left[\sup_{t\ge 0}\left|M_t^{(n)} - M_t\right|^2\right] &= \mathbb{E}\left[\sup_{t\ge 0}\left|\mathbb{E}\left(M_\infty^{(n)} - M_\infty\middle|\mathcal{F}_t\right)\right|^2\right] \\
&\le 4\mathbb{E}\left[\left(M_\infty^{(n)} - M_\infty\right)^2\right] \to 0.
\end{aligned}$$

□

Corollary 1.6.1. The space $c\mathcal{M}^2$ of continuous martingales in $\mathcal{M}^2$ is closed in the topology of $\mathcal{M}^2$.

Proof. Let $M^{(n)} \to M$ in $\mathcal{M}^2$, with each $M^{(n)} \in c\mathcal{M}^2$. Then

$$\sup_{t\ge 0}\left|M_t^{(n)} - M_t\right| \xrightarrow{L^2} 0.$$

Hence, after passing to a subsequence if necessary,

$$\sup_{t\geq 0}\left|M_t^{(n_k)} - M_t\right| \xrightarrow{a.s.} 0.$$

Therefore $M^{(n_k)}$ converges uniformly almost surely to M, and since each $M^{(n_k)}$ has continuous sample paths, the limit process M is also continuous. □

Definition 1.6.1 (Weakly orthogonal). Two martingales $M, N \in \mathcal{M}_0^2$ are said to be weakly orthogonal if

$$\mathbb{E}[M_\infty N_\infty] = 0.$$

Definition 1.6.2 (Strongly orthogonal). Two martingales $M, N \in \mathcal{M}_0^2$ are said to be strongly orthogonal if, for every stopping time T,

$$\mathbb{E}[M_T N_T] = 0.$$

Lemma 1.6.1. Two martingales are strongly orthogonal if and only if MN is uniformly integrable.

Lemma 1.6.2. Let M be a progressively measurable process such that for every stopping time T,

$$\begin{cases} \mathbb{E}\left|M_T\right| < \infty, \\ \mathbb{E}M_T = 0. \end{cases}$$

Then M is uniformly integrable.

Definition 1.6.3 (Stable subspaces of $\mathcal{M}_0^2$). A subspace $\mathcal{N}_0 \subseteq \mathcal{M}_0^2$ is called stable if

① $\mathcal{N}_0$ is a closed subspace;

② for every $N \in \mathcal{N}_0$ and every stopping time T, the stopped process N^T, defined by $N_t^T = N_{t\wedge T}$, also belongs to $\mathcal{N}_0$.

Example . $c\mathcal{M}_0^2$ is a stable subspace.

Theorem 1.6.2. Let $\mathcal{N}_0$ be a stable subspace of $\mathcal{M}_0^2$. Let $\mathcal{N}_0^\perp$ be the set of $Z \in \mathcal{M}_0^2$ which are weakly orthogonal to every element of $\mathcal{N}_0$, i.e.

$$\mathcal{N}_0^\perp = \{Z \in \mathcal{M}_0^2 \colon \langle Z, N\rangle = 0,\ \forall\ N \in \mathcal{N}_0\}.$$

Then $\mathcal{N}_0^\perp$ is a stable subspace of $\mathcal{M}_0^2$. Moreover, if $Z \in \mathcal{N}_0^\perp$ and $N \in \mathcal{N}_0$, then Z and N are strongly orthogonal.

Finally, every $M \in \mathcal{M}_0^2$ admits a decomposition

$$M = N + Z$$

with $N \in \mathcal{N}_0$ and $Z \in \mathcal{N}_0^\perp$.

Proof. Since $\mathcal{N}_0$ is a closed subspace of the Hilbert space $\mathcal{M}_0^2$, its orthogonal complement $\mathcal{N}_0^\perp$ is also a closed subspace, and every $M \in \mathcal{M}_0^2$ admits an orthogonal decomposition

$$M = N + Z$$

with $N \in \mathcal{N}_0$ and $Z \in \mathcal{N}_0^\perp$.

It remains to prove that $\mathcal{N}_0^\perp$ is stable and that orthogonality is in fact strong. Let $Z \in \mathcal{N}_0^\perp$, $N \in \mathcal{N}_0$, and let T be a stopping time. Then

$$\begin{aligned}
\mathbb{E}[Z_\infty^T N_\infty] &= \mathbb{E}[Z_T N_\infty] \\
&= \mathbb{E}\left[Z_T \mathbb{E}\left(N_\infty | \mathcal{F}_T\right)\right] \\
&= \mathbb{E}[Z_T N_T] \\
&= \mathbb{E}\left[\mathbb{E}\left(Z_\infty | \mathcal{F}_T\right) N_T\right] \\
&= \mathbb{E}[Z_\infty N_T] \\
&= \mathbb{E}[Z_\infty N_\infty^T].
\end{aligned}$$

Since $\mathcal{N}_0$ is stable, we have $N^T \in \mathcal{N}_0$, and hence

$$\mathbb{E}[Z_\infty N_\infty^T] = 0.$$

Therefore

$$\mathbb{E}[Z_\infty^T N_\infty] = 0$$

for every $N \in \mathcal{N}_0$, so $Z^T \in \mathcal{N}_0^\perp$. This proves that $\mathcal{N}_0^\perp$ is stable.

In particular, since $Z^T \in \mathcal{N}_0^\perp$, taking the same $N \in \mathcal{N}_0$ in the preceding identity yields

$$\mathbb{E}[Z_T N_T] = \mathbb{E}[Z_\infty^T N_\infty] = 0.$$

Thus Z and N are strongly orthogonal. $\square$

Definition 1.6.4. Define

$$d\mathcal{M}_0^2 \triangleq (c\mathcal{M}_0^2)^\perp.$$

Then every $M \in \mathcal{M}_0^2$ can be written as

$$M = M^c + M^d,$$

where $M^c \in c\mathcal{M}_0^2$, $M^d \in d\mathcal{M}_0^2$, and M^c and M^d are strongly orthogonal. In particular, $M^c M^d$ is uniformly integrable.

For a càdlàg function f, define

$$\Delta f(t) = f(t) - f(t-).$$

We now extend the quadratic variation from the continuous case to a general martingale in $\mathcal{M}_0^2$. In the continuous case, the bracket $\langle M\rangle$ is characterized by the fact that $M^2-\langle M\rangle$ is a martingale. When jumps are present, however, the increasing process must also record the jump contribution. The next theorem shows that there is a unique increasing process $[M]$ which plays this role and satisfies $\Delta[M]=(\Delta M)^2$.

Theorem 1.6.3 (Meyer)**.** Let $M\in\mathcal{M}_0^2$. Then there exists a unique increasing process $[M]$, null at 0, such that

① $M^2-[M]$ is uniformly integrable;

② $\Delta[M]=(\Delta M)^2$.

Proof. Write

$$M=M^c+M^d,$$

where $M^c\in c\mathcal{M}_0^2$ and $M^d\in d\mathcal{M}_0^2$. Then

$$M_t^2=(M_t^c+M_t^d)^2=(M_t^c)^2+2M_t^cM_t^d+(M_t^d)^2.$$

The theorem is obtained by combining the continuous quadratic variation of M^c, the purely discontinuous part M^d, and the strong orthogonality between M^c and M^d. □

For $M,N\in\mathcal{M}_0^2$, define

$$[M,N]\triangleq\frac{1}{4}\left([M+N]-[M-N]\right).$$

Theorem 1.6.4 (Kunita & Watanabe)**.** The process $[M,N]$ is the unique finite-variation process, null at 0, such that

① $MN-[M,N]$ is uniformly integrable;

② $\Delta[M,N]=\Delta M\cdot\Delta N$.

1.7 Itô Integral

Let $M\in\mathcal{M}_0^2$. We begin with the simplest class of integrands.

If $H_t=Z1_{(S,T]}(t)$, where $Z\in\mathcal{F}_S$ is bounded and $S\le T$ are stopping times, define

$$\int_0^t H_s\,\mathrm{d}M_s\equiv(H\cdot M)_t\equiv Z\big(M_{T\wedge t}-M_{S\wedge t}\big).$$

We now show that $(H\cdot M)$ belongs to $\mathcal{M}_0^2$. As a first step, we prove that $(H\cdot M)$ is a martingale. By Lemma 1.6.2, it suffices to verify that for every stopping time R,

$$\begin{cases}\mathbb{E}\,|(H\cdot M)_R|<\infty,\\ \mathbb{E}(H\cdot M)_R=0.\end{cases}$$

Proof. Let R be any stopping time. Since Z is bounded, say $|Z| \le K$, we have

$$\mathbb{E}\,|(H \cdot M)_R| \le K\mathbb{E}\,|M_{T\wedge R}| + K\mathbb{E}\,|M_{S\wedge R}|\,.$$

Because $M \in \mathcal{M}_0^2$, the martingale M is uniformly integrable, and

$$M_{T\wedge R} = \mathbb{E}(M_\infty \mid \mathcal{F}_{T\wedge R}).$$

Hence

$$\mathbb{E}\,|M_{T\wedge R}| \le \mathbb{E}\left[\mathbb{E}(|M_\infty| \mid \mathcal{F}_{T\wedge R})\right] = \mathbb{E}\,|M_\infty|\,,$$

and similarly for $M_{S\wedge R}$. Therefore

$$\mathbb{E}\,|(H \cdot M)_R| < \infty.$$

It remains to show that $\mathbb{E}(H \cdot M)_R = 0$. By linearity, it is enough to treat the case $Z = 1_A$ with $A \in \mathcal{F}_S$. Then

$$\mathbb{E}(H \cdot M)_R = \mathbb{E}\left[1_A(M_{T\wedge R} - M_{S\wedge R})\right].$$

Define the stopping times

$$S_A \equiv \begin{cases} S, & \text{on } A, \\ \infty, & \text{on } A^c. \end{cases} \qquad T_A \equiv \begin{cases} T, & \text{on } A, \\ \infty, & \text{on } A^c. \end{cases}$$

Then

$$1_A M_{S\wedge R} = M_{S_A\wedge R}, \qquad 1_A M_{T\wedge R} = M_{T_A\wedge R}.$$

Hence the expectation above is the difference of expectations of stopped values of the martingale M, and therefore equals 0 by optional sampling. Thus $(H \cdot M)$ is a martingale. $\square$

Lemma 1.7.1. If $Z \in b\mathcal{F}_S$ and M is a uniformly integrable martingale, then $H \cdot M$ is a uniformly integrable martingale.

More generally, consider a finite linear combination of such elementary processes:

$$H_t = \sum_{i=1}^{n} Z_i 1_{(S_i,T_i]}(t),$$

where each $Z_i \in b\mathcal{F}_{S_i}$.

We define

$$(H \cdot M)_t \triangleq \sum_{i=1}^{n} Z_i\big(M_{T_i\wedge t} - M_{S_i\wedge t}\big).$$

Corollary 1.7.1. If $Z_i \in b\mathcal{F}_{S_i}$ and M is U.I. martingale, then $(H \cdot M)_t$ is U.I. martingale.

We next introduce the predictable σ-algebra, which is the natural measurability class for Itô integrands.

Definition 1.7.1 (Previsible process)**.** The previsible σ-algebra $\mathcal{P}$ on $(0,\infty)\times\Omega$ is defined to be the smallest σ-algebra for which every adapted L-process is $\mathcal{P}$-measurable.

An L-process is a process $(X_t)_{t>0}$ such that $X_t = X_{t-}$ and X_{t+} exists.

We say that a process $(X_t)_{t>0}$ is previsible if it is $\mathcal{P}$-measurable.

Lemma 1.7.2. Suppose S and T are stopping times with $S \leq T \leq \infty$. Let $Z \in b\mathcal{F}_S$ and $H = Z1_{(S,T]}$. Then H is previsible.

Proof. Fix $t > 0$ and let B be a Borel set. We may assume that $0 \notin B$. Then

$$\{H_t \in B\} = \{Z \in B\} \cap \{S < t\} \cap \{t \leq T\}.$$

Moreover,

$$\{S < t\} = \bigcup_{n=1}^{\infty} \left\{S \leq t - \frac{1}{n}\right\}.$$

Since $\{Z \in B\} \in \mathcal{F}_S$, we have

$$\{Z \in B\} \cap \left\{S \leq t - \frac{1}{n}\right\} \in \mathcal{F}_{t-\frac{1}{n}} \subseteq \mathcal{F}_t.$$

Also, $\{t \leq T\} \in \mathcal{F}_t$. Therefore $\{H_t \in B\} \in \mathcal{F}_t$, so H is adapted. Since H is clearly an L-process, it follows from the definition of $\mathcal{P}$ that H is previsible. □

Definition 1.7.2. Let

$$b\varepsilon = \left\{\sum_{i=1}^{n} Z_i 1_{(S_i,T_i]} : n \geq 1,\ Z_i \in b\mathcal{F}_{S_i},\ S_1 \leq T_1 \leq S_2 \leq \cdots\right\}.$$

Claim. $\sigma(b\varepsilon) = \mathcal{P}$.

Proof. "$\subseteq$" The inclusion $\sigma(b\varepsilon) \subseteq \mathcal{P}$ follows from the previous lemma.

"$\supseteq$" For any bounded left-continuous process,

$$H_t = \lim_{k\uparrow\infty} \lim_{n\uparrow\infty} \sum_{i=2}^{n_k} H_{\frac{i-1}{n}} 1_{(\frac{i-1}{n},\frac{i}{n}]}(t).$$

Since H is adapted, we get $H_{\frac{i-1}{n}} \in b\mathcal{F}_{\frac{i-1}{n}}$.

Hence

$$\sum_{i=2}^{n_k} H_{\frac{i-1}{n}} 1_{(\frac{i-1}{n},\frac{i}{n}]}(t) \in b\varepsilon.$$

$\implies H_t \in \sigma(b\varepsilon)$.

□

Proof. □

Lemma 1.7.3. Let $H \in b\varepsilon$ and $M \in \mathcal{M}_0^2$. Then $H \cdot M \in \mathcal{M}_0^2$, and if

$$H = \sum_{i=1}^{n} Z_{i-1} 1_{(T_{i-1}, T_i]},$$

then

$$\mathbb{E}(H \cdot M)_\infty^2 = \sum_{i=1}^{n} \mathbb{E}\left[Z_{i-1}^2 (M_{T_i} - M_{T_{i-1}})^2\right].$$

Proof.

$$(H \cdot M)_t = \sum_{i=1}^{n} Z_{i-1}\left(M_{T_i \wedge t} - M_{T_{i-1} \wedge t}\right).$$

By the previous corollary 1.7.1, $(H \cdot M)$ is a U.I. martingale.

Hence $(H \cdot M)_t \to (H \cdot M)_\infty$ a.s. and in L^1, where

$$(H \cdot M)_\infty = \sum_{i=1}^{n} Z_{i-1} \Delta_i, \qquad \Delta_i = M_{T_i} - M_{T_{i-1}}.$$

Therefore

$$\mathbb{E}\left[(H \cdot M)_\infty^2\right] = \mathbb{E} \sum_{i=1}^{n} Z_{i-1}^2 \Delta_i^2 + 2\mathbb{E}\left(\sum_{i<j} Z_{i-1} \Delta_i Z_{j-1} \Delta_j\right).$$

For $i < j$, the random variable $Z_{i-1} \Delta_i Z_{j-1}$ is $\mathcal{F}_{T_{j-1}}$-measurable, so

$$\mathbb{E}\left[Z_{i-1} \Delta_i Z_{j-1} \Delta_j\right] = \mathbb{E}\left[Z_{i-1} \Delta_i Z_{j-1} \mathbb{E}\left(\Delta_j \middle| \mathcal{F}_{T_{j-1}}\right)\right] = 0.$$

Thus all cross terms vanish.

By Doob's L^2 inequality,

$$\mathbb{E}\left[\sup_{t \geq 0} (H \cdot M)_t^2\right] \leq 4\mathbb{E}\left[(H \cdot M)_\infty^2\right] < \infty.$$

Therefore $H \cdot M \in \mathcal{M}_0^2$. □

Recall from Meyer's theorem that for $M \in \mathcal{M}_0^2$ there exists a quadratic variation $[M]$ such that $M^2 - [M]$ is a uniformly integrable martingale. Applying this to the previous lemma 1.7.3, we get

$$\mathbb{E}(H \cdot M)_\infty^2 = \sum_{i=1}^{n} \mathbb{E}\left[Z_{i-1}^2 \left(M_{T_i} - M_{T_{i-1}}\right)^2\right].$$

Moreover,

$$\begin{aligned} \mathbb{E}\left[\left(M_{T_i} - M_{T_{i-1}}\right)^2 \middle| \mathcal{F}_{T_{i-1}}\right] &= \mathbb{E}\left[M_{T_i}^2 - M_{T_{i-1}}^2 \middle| \mathcal{F}_{T_{i-1}}\right] \\ &= \mathbb{E}\left[[M]_{T_i} - [M]_{T_{i-1}} \middle| \mathcal{F}_{T_{i-1}}\right], \end{aligned}$$

because $M^2 - [M]$ is a martingale and $\mathbb{E}(M_{T_i} \mid \mathcal{F}_{T_{i-1}}) = M_{T_{i-1}}$. Therefore

$$\mathbb{E}(H \cdot M)_\infty^2 = \sum_{i=1}^n \mathbb{E}\left[Z_{i-1}^2\big([M]_{T_i} - [M]_{T_{i-1}}\big)\right] = \mathbb{E}\left[\int_0^\infty H_s^2 \,\mathrm{d}[M]_s\right].$$

This shows that, for simple predictable integrands, the map

$$H \mapsto (H \cdot M)_\infty$$

is an isometry from $(L^2(M) \cap b\varepsilon, \|\cdot\|_M)$ into $L^2(\mathcal{F}_\infty)$ equipped with the usual L^2 norm.

For fixed $M \in \mathcal{M}_0^2$, define

$$\|H\|_M = \left(\mathbb{E}\int_0^\infty H_s^2 \,\mathrm{d}[M]_s\right)^{1/2}.$$

Let

$$L^2(M) = \{\text{previsible } H : \|H\|_M < \infty\}.$$

Then the map

$$\begin{aligned} I : L^2(M) \cap b\varepsilon &\to L^2(\mathcal{F}_\infty), \\ H &\mapsto (H \cdot M)_\infty \end{aligned}$$

is an isometry, since

$$\mathbb{E}\left[\int_0^\infty H_s^2 \,\mathrm{d}[M]_s\right] = \mathbb{E}(H \cdot M)_\infty^2.$$

Hence I extends uniquely from simple predictable processes to all of $L^2(M)$.

Definition 1.7.3 (Itó Integral)**.** Let $M \in \mathcal{M}_0^2$ and $H \in L^2(M)$. Then Itó Integral $(H \cdot M)_t$ is the image of H under extension of $I : L^2(M) \to L^2(\mathcal{F}_\infty)$.

Localization.

Definition 1.7.4. Let T be a stopping time and let H be any process. Define

$$H(0,T](t) = \begin{cases} H_t, & 0 < t \le T, \\ 0, & t > T. \end{cases}$$

Remark. If H is previsible, then so is $H(0,T]$.

Definition 1.7.5. Let T be a stopping time and let X be any process. Define

$$X^T(t) = \begin{cases} X(t), & 0 \le t \le T, \\ X(T), & t > T. \end{cases}$$

For an elementary integrand, the localization identity can be checked directly. Indeed,

$$(H \cdot M)_t^T = (H(0,T] \cdot M^T)_t,$$

and

$$\left[Z_i \left(M_{T_i \wedge t} - M_{T_{i-1} \wedge t}\right)\right]^T = \left(Z_i 1_{(T_{i-1} \wedge T,\, T_i \wedge T]} \cdot M^T\right)_t = Z_i \left(M_{T_i \wedge T \wedge t} - M_{T_{i-1} \wedge T \wedge t}\right).$$

Theorem 1.7.1. Fix $M \in \mathcal{M}_0^2$. For any $H \in L^2(M)$ and any stopping time T:

(i) $(H \cdot M)^T = H(0,T] \cdot M = H \cdot M^T$.

(ii) $(H \cdot M)_t^2 - \int_0^t H_s^2 \, \mathrm{d}[M]_s$ is a uniformly integrable martingale. In particular,

$$[H \cdot M]_t = \int_0^t H_s^2 \, \mathrm{d}[M]_s.$$

(iii) If $H, K \in b\mathcal{P}$, then

$$(HK) \cdot M = H \cdot (K \cdot M) = K \cdot (H \cdot M).$$

(iv) $\Delta(H \cdot M) = H \Delta M$.

(v) If M is continuous, then $H \cdot M$ is continuous.

Proof. We first prove the identities for elementary processes and then extend them by density.

For (i), let $H = Z1_{(U,V]}$. Then

$$(H \cdot M)_t = Z(M_{V \wedge t} - M_{U \wedge t}),$$

so

$$(H \cdot M)_t^T = (H \cdot M)_{t \wedge T} = Z(M_{V \wedge T \wedge t} - M_{U \wedge T \wedge t}).$$

On the other hand,

$$H(0,T] = Z1_{(U \wedge T, V \wedge T]},$$

and therefore

$$(H(0,T] \cdot M)_t = Z(M_{V \wedge T \wedge t} - M_{U \wedge T \wedge t}).$$

Similarly,

$$(H \cdot M^T)_t = Z(M^T_{V \wedge t} - M^T_{U \wedge t}) = Z(M_{V \wedge T \wedge t} - M_{U \wedge T \wedge t}).$$

Hence (i) holds for elementary processes, and then for all $H \in L^2(M)$ by continuity of the integral map.

For (ii), set

$$N_t = (H \cdot M)_t^2 - \int_0^t H_s^2 \, \mathrm{d}[M]_s.$$

Since

$$\mathbb{E}\left[\sup_{t\geq 0}(H\cdot M)_t^2\right] \leq 4\mathbb{E}\left[\int_0^\infty H_s^2\,\mathrm{d}[M]_s\right] < \infty,$$

the family $\{N_t : t \geq 0\}$ is uniformly integrable. It therefore suffices to show that $\mathbb{E}N_T = 0$ for every stopping time T.

By part (i),

$$(H\cdot M)_T = (H(0,T]\cdot M)_\infty.$$

Applying the isometry to the integrand $H(0,T]$, we obtain

$$\begin{aligned}\mathbb{E}(H\cdot M)_T^2 &= \mathbb{E}(H(0,T]\cdot M)_\infty^2 \\ &= \mathbb{E}\int_0^\infty H(0,T]_s^2\,\mathrm{d}[M]_s \\ &= \mathbb{E}\int_0^T H_s^2\,\mathrm{d}[M]_s.\end{aligned}$$

Hence $\mathbb{E}N_T = 0$, and (ii) follows.

For (iii) First take $H = Z_1\mathbf{1}_{(S_1,T_1]}$ and $K = Z_2\mathbf{1}_{(S_2,T_2]}$. Then

$$HK = Z_1Z_2\,\mathbf{1}_{(S,T]},$$

where

$$S = S_1\vee S_2, \qquad T = (T_1\wedge T_2)\vee(S_1\vee S_2).$$

Hence

$$((HK)\cdot M)_\infty = Z_1Z_2(M_T - M_S).$$

On the other hand,

$$\begin{aligned}(H\cdot(K\cdot M))_\infty &= Z_1\Big((K\cdot M)_{T_1} - (K\cdot M)_{S_1}\Big) \\ &= Z_1\Big((K\cdot M^{T_1})_\infty - (K\cdot M^{S_1})_\infty\Big) \\ &= Z_1Z_2\Big[(M_{T_2\wedge T_1} - M_{S_2\wedge T_1}) - (M_{T_2\wedge S_1} - M_{S_2\wedge S_1})\Big] \\ &= Z_1Z_2(M_T - M_S).\end{aligned}$$

Therefore, for every $H, K \in b\varepsilon$,

$$H\cdot(K\cdot M) = (HK)\cdot M.$$

The identity

$$K\cdot(H\cdot M) = (HK)\cdot M$$

is proved in the same way.

Now let $K \in b\mathcal{P}$. Choose $(K^n)_{n\geq 1} \subset b\varepsilon$ such that

$$K^n \xrightarrow{L^2(M)} K.$$

Fix $H \in b\varepsilon$. By the isometry,

$$\begin{aligned}
\mathbb{E}\left[(H\cdot(K\cdot M) - H\cdot(K^n\cdot M))_\infty^2\right] &= \mathbb{E}\left[\big(H\cdot\big((K-K^n)\cdot M\big)\big)_\infty^2\right] \\
&= \mathbb{E}\left[\int_0^\infty H_s^2 \,\mathrm{d}[(K-K^n)\cdot M]_s\right] \\
&= \mathbb{E}\left[\int_0^\infty H_s^2 (K_s - K_s^n)^2 \,\mathrm{d}[M]_s\right] \\
&\leq \|H\|_\infty^2 \,\mathbb{E}\left[\int_0^\infty (K_s - K_s^n)^2 \,\mathrm{d}[M]_s\right] \to 0.
\end{aligned}$$

Hence

$$H\cdot(K^n\cdot M) \xrightarrow{L^2(\mathcal{F}_\infty)} H\cdot(K\cdot M).$$

On the other hand, for each n we already know that

$$H\cdot(K^n\cdot M) = (HK^n)\cdot M.$$

Since $H \in b\varepsilon$, we also have

$$\mathbb{E}\left[\int_0^\infty (H_sK_s^n - H_sK_s)^2 \,\mathrm{d}[M]_s\right] \leq \|H\|_\infty^2 \,\mathbb{E}\left[\int_0^\infty (K_s^n - K_s)^2 \,\mathrm{d}[M]_s\right] \to 0,$$

so $HK^n \xrightarrow{L^2(M)} HK$. Applying the isometry once again,

$$(HK^n)\cdot M \xrightarrow{L^2(\mathcal{F}_\infty)} (HK)\cdot M.$$

By uniqueness of the $L^2(\mathcal{F}_\infty)$ limit,

$$H\cdot(K\cdot M) = (HK)\cdot M.$$

(iv) First let $H = Z\mathbb{1}_{(S,T]}$. Then

$$(H\cdot M)_t = Z(M_{t\wedge T} - M_{t\wedge S}),$$

so

$$\Delta(H\cdot M)_t = Z\,\mathbb{1}_{(S,T]}(t)\,\Delta M_t = H_t\Delta M_t.$$

Hence the identity holds for all $H \in b\varepsilon$.

Now let $H \in L^2(M)$, and choose $H^n \in b\varepsilon$ such that

$$H^n \xrightarrow{L^2(M)} H.$$

By the isometry,

$$H^n \cdot M \xrightarrow{L^2(\mathcal{F}_\infty)} H \cdot M,$$

and hence, after passing to a subsequence,

$$\sup_{t \geq 0} |(H^{n_k} \cdot M)_t - (H \cdot M)_t| \xrightarrow{a.s.} 0.$$

Also,

$$\int_0^\infty (H_s^{n_k} - H_s)^2 \, \mathrm{d}[M]_s \xrightarrow{a.s.} 0$$

along a further subsequence.

Fix $t \geq 0$. Since

$$\Delta(H^{n_k} \cdot M)_t = H_t^{n_k} \Delta M_t,$$

we distinguish two cases.

Case 1. If $\Delta M_t = 0$, then $\Delta(H^{n_k} \cdot M)_t = 0$ for every k, and therefore

$$\Delta(H \cdot M)_t = 0 = H_t \Delta M_t.$$

Case 2. If $\Delta M_t \neq 0$, then $\Delta[M]_t = (\Delta M_t)^2 > 0$. Hence

$$\int_0^\infty (H_s^{n_k} - H_s)^2 \, \mathrm{d}[M]_s \geq (H_t^{n_k} - H_t)^2 \Delta[M]_t,$$

and so $H_t^{n_k} \to H_t$. Letting $k \to \infty$ in

$$\Delta(H^{n_k} \cdot M)_t = H_t^{n_k} \Delta M_t,$$

we obtain

$$\Delta(H \cdot M)_t = H_t \Delta M_t.$$

□

Theorem 1.7.2 (Kunita-Watanabe)**.** Let $M \in \mathcal{M}_0^2$ and $H \in L^2(M)$. Then $\forall \, N \in \mathcal{M}_0^2$,

$$\mathbb{E}\left[(H \cdot M)_\infty \cdot N_\infty\right] = \mathbb{E}\left[(H \cdot [M, N])_\infty\right].$$

Moreover,

$$[H \cdot M, N] = H \cdot [M, N] \overset{def}{=\!=} \int H_s \, \mathrm{d}[M, N]_s.$$

Theorem 1.7.3. Let $M, N \in \mathcal{M}_0^2$. Then

$$\left(\int_0^\infty |H_s K_s| \,\mathrm{d}[M,N]_s\right)^2 \le \int_0^\infty H_s^2 \,\mathrm{d}[M]_s \cdot \int_0^\infty K_s^2 \,\mathrm{d}[N]_s. \quad \text{a.s.}$$

Proof. $|H_s K_s| = H_s K_s \cdot \operatorname{sgn}(H_s K_s)$.

$|\mathrm{d}[M,N]_s| = \mathrm{d}[M,N]_s \cdot \operatorname{sgn}([M,N]_s)$.

$$\begin{aligned} LHS &= \int_0^\infty H_s K_s \cdot \operatorname{sgn}(H_s K_s) \cdot \operatorname{sgn}([M,N]_s) \,\mathrm{d}[M,N]_s \\ &= \int_0^\infty \tilde{H}_s K_s \,\mathrm{d}[M,N]_s \end{aligned}$$

$$RHS = \int_0^\infty \tilde{H}_s \,\mathrm{d}[M]_s \cdot \int_0^\infty K_s^2 \,\mathrm{d}[N]_s$$

$$\Leftrightarrow \left(\int_0^\infty \tilde{H}_s K_s \,\mathrm{d}[M,N]_s\right)^2 \le \int_0^\infty \tilde{H}_s \,\mathrm{d}[M]_s \int_0^\infty K_s^2 \,\mathrm{d}[N]_s.$$

$$\begin{aligned} & H_n = \sum_{i=1}^n U_i (t_i, t_{i+1}] \\ \Leftrightarrow & \\ & K_n \equiv \sum_{i=1}^n V_i (t_i, t_{i+1}]. \end{aligned}$$

$$\Leftrightarrow \left(\sum_{i=1}^n U_i V_i \left([M,N]_{t_{i+1}} - [M,N]_{t_i}\right)\right)^2 \le \sum_{i=1}^2 U_i^2 ([M]_{t_{i+1}} - [M]_{t_i}) \cdot \sum_{i=1}^n V_i^2 ([N]_{t_{i+1}} - [N]_{t_i}).$$

It suffices to prove

$$\left|[M,N]_{t_{i+1}} - [M,N]_{t_i}\right| \le \left([M]_{t_{i+1}} - [M]_{t_i}\right)^{\frac{1}{2}} \left([N]_{t_{i+1}} - [N]_{t_i}\right)^{\frac{1}{2}}.$$

like $(\sum_i a_i b_i)^2 \le \sum_i a_i^2 \sum_i b_i^2$.

To prove this, note that $\forall\ \lambda \in \mathbb{Q}$, $[M + \lambda N]$ is increasing and so

$$[M + \lambda N]_{t_{i+1}} - [M + \lambda N]_{t_i} \ge 0 \quad \text{a.s..}$$

Also

$$[M + \lambda N]_t = [M]_t + 2\lambda [M,N]_t + \lambda^2 [N]_t.$$

because $(M + \lambda N)_t^2 - [M + \lambda N]_t$ is U.I. and by the uniqueness (?)

$$M_t^2 + 2\lambda M_t N_t + \lambda^2 N_t^2 - ([M]_t + 2\lambda [M,N]_t + \lambda^2 [N]_t).$$

Hence

$$\left([M]_{t_{i+1}} - [M]_{t_i}\right) + 2\lambda \left([M,N]_{t_{i+1}} - [M,N]_{t_i}\right) + \lambda^2 \left([N]_{t_{i+1}} - [N]_{t_i}\right) \ge 0$$

$$\Leftrightarrow b^2 - 4ac \le 0.$$

So

$$\left([M,N]_{t_{i+1}} - [M,N]_{t_i}\right)^2 \le \left([M]_{t_{i+1}} - [M]_{t_i}\right)\left([N]_{t_{i+1}} - [N]_{t_i}\right).$$

$$\mathbb{E}\left[(H \cdot M)_\infty N_\infty\right] = \mathbb{E}\left[(H \cdot [M,N])_\infty\right].$$

If $H = Z(S,T]$,

$$\begin{cases} LHS = \mathbb{E}\left[Z(M_T - M_S) \cdot N_\infty\right] \\ RHS = \mathbb{E}\left[Z([M,N]_T - [M,N]_S)\right] \end{cases}$$

$$\begin{aligned} LHS &= \mathbb{E}\left[\mathbb{E}(N_\infty \mid \mathcal{F}_T) \cdot Z(M_T - M_S)\right] \\ &= \mathbb{E}\left[Z(M_T N_T - M_S N_T)\right] \\ &= \mathbb{E}[Z M_T N_T] - \mathbb{E}\left[\mathbb{E}(N_T \mid \mathcal{F}_S) \cdot Z M_S\right] \\ &= \mathbb{E}\left[Z M_T N_T\right] - \mathbb{E}[Z M_S N_S]. \end{aligned}$$

Because $M_t N_t - [M,N]_t$ is U.I.

$$\mathbb{E}[M_T N_T] = \mathbb{E}[[M,N]_T] \implies \mathbb{E}[Z(M_T N_T - [M,N]_T)] = 0.$$

And

$$\begin{aligned} LHS &= \mathbb{E}\left[Z(M_T N_T - M_S N_S)\right] \\ &= \mathbb{E}\left[Z \mathbb{E}\left[M_T N_T - M_S N_S \mid \mathcal{F}_S\right]\right] \\ &= \mathbb{E}\left[Z \mathbb{E}[[M,N]_T - [M,N]_S \mid \mathcal{F}_S]\right] \\ &= \mathbb{E}[Z([M,N]_T - [M,N]_S)]. \end{aligned}$$

Hence $\forall\ H \in b\varepsilon$

$$\mathbb{E}[(H \cdot M)_\infty \cdot N_\infty] = \mathbb{E}\left[(H \cdot [M,N])_\infty\right].$$

If $H_n \xrightarrow{L^2(M)} H$, i.e.

$$\mathbb{E}\left(\int_0^\infty (H_n(s) - H_s)^2 \,\mathrm{d}[M]_s\right) \to 0$$

$\implies (H_n \cdot M)_\infty \xrightarrow{L^2} (H \cdot M)_\infty$ by

$$\mathbb{E}[(H_n \cdot M - H \cdot M)^2_\infty] = \mathbb{E}\left[\int_0^\infty (H_n - H) \,\mathrm{d}[M]_s\right] \to 0.$$

Hence $\mathbb{E}[(H_n \cdot M)_\infty \cdot N_\infty] \to \mathbb{E}[(H \cdot M)_\infty N_\infty]$.

Next,

$$\begin{aligned} \mathbb{E}\left[|(H_n \cdot [M,N])_\infty - (H \cdot [M,N])_\infty|\right] &= \mathbb{E}\left|((H_n - H) \cdot [M,N])_\infty\right| \\ &\le \left(\mathbb{E}\left(\int_0^\infty (H_n - H) \,\mathrm{d}[M]_s\right)\right)^{\frac{1}{2}} (\mathbb{E}[N]_\infty)^{\frac{1}{2}} \to 0 \end{aligned}$$

by using

$$\left(\int_0^\infty |H_s| \, \mathrm{d}[M,N]_s\right) \le \left(\int_0^\infty H_s^2 \, \mathrm{d}[M]_s\right)^{\frac{1}{2}} \left(\int_0^\infty 1 \, \mathrm{d}[N]_s\right)^{\frac{1}{2}}.$$

Hence $\mathbb{E}[(H_n \cdot [M,N])_\infty] \to \mathbb{E}[(H \cdot [M,N])_\infty]$. □

Remark. proof of $[H \cdot M, N] = H \cdot [M,N] \Leftrightarrow$ It suffices to. prove $(H \cdot M)_t N_t - \int_0^\infty H_s \, \mathrm{d}[M,N]_s$ is a U.I. martingale.

$\Leftrightarrow \forall$ stopping time R

$$\mathbb{E}[(H \cdot M)_R \cdot N_R] = \mathbb{E}\left[\int_0^R H_s \, \mathrm{d}[M,N]_s\right].$$

$$\begin{aligned} LHS &= \mathbb{E}\left[(H \cdot M)_\infty^R \cdot N_\infty^R\right] \\ &= \mathbb{E}\left[(H^R \cdot M)_\infty \cdot N_\infty^R\right] \\ &= \mathbb{E}[(H^R \cdot [M, N^R])_\infty] \\ &= \mathbb{E}\left[\int_0^\infty H^R(s) \, \mathrm{d}[M, N^R]_s\right] \\ &= \mathbb{E}\left[\int_0^R H(s) \, \mathrm{d}[M,N]_s\right] = RHS. \end{aligned}$$

Definition 1.7.6 (local martingale)**.** $\mathcal{M}_{loc} = \{\text{local martingale}\}$.

$\exists$ stopping times $T_n \uparrow \infty$ s.t. M^{T_n} is a martingale.

Definition 1.7.7 (continuous local martingale)**.** $c\mathcal{M}_{loc} = \{\text{continuous local martingale}\}$.

Quadratic variation of $c\mathcal{M}_{loc}$.

If $M \in c\mathcal{M}_{loc}$, then $\exists$ $[M]$ increasing process such that $M^2 - [M] \in c\mathcal{M}_{loc}$.

Lemma 1.7.4. If $M \in c\mathcal{M}_{loc}$. Fix $n \ge 1$, define

$$T_n = \inf\{t \ge 0 \colon |M_t| \ge n\}.$$

Then M^{T_n} is a bounded martingale.

Proof. By definition, $\exists$ stopping $(S_k) \uparrow \infty$ s.t. M^{S_k} is a martingale.

$\implies M^{S_k \wedge T_n}$ is also a martingale.

So $\forall$ $s < t$,

$$\mathbb{E}(M_t^{S_k \wedge T_n} \mid \mathcal{F}_s) = M_s^{S_k \wedge T_n}.$$

Let $k \to \infty$ to get $\mathbb{E}(M_t^{T_n} \mid \mathcal{F}_s) = M_s^{T_n}$.

pf of Quadratic variation:

(Existence.) $\forall$ $n \ge 1$, we let

$$T_n = \inf\{t \ge 0 \colon |M_t| \ge n\}.$$

Then define $\forall\, t < T_n$,

$$[M]_t = [M^{T_n}]_t.$$

We need to check that $[M^{T_n}]_t$ is well-defined.

$$[M^{T_2}]^{T_1} = [M^{T_2}]_{t\wedge T_1}$$

Because $[M]^T = [M^T]$,

$$[M^{T_2}]^{T_1} = [M^{T_2\wedge T_1}] = [M^{T_1}].$$

By $T_n \uparrow \infty$, we have $[M]_t$ is defined for all $t \geq 0$.

(uniqueness.) If $\exists\, A, A'$ increasing such that $N = M^2 - A$ and $N' = M^2 - A' \in c\mathcal{M}_{loc}$. Then $N - N' = A' - A \in c\mathcal{M}_{loc}$ is of finite variation.

Lemma 1.7.5. If $M \in c\mathcal{M}_{loc}$ and M is of finite variation, then $M_t \equiv 0$.

Proof. Let $V_M(t) =$ total variation of M on $[0,t]$. Define $T_n = \inf\{t \geq 0\colon V_M(t) > n\}$.

Then M^{T_n} is a bounded martingale and is of finite variation. Next, define

$$A_t^n \equiv \sum_{k=1}^{2^n} \left(M(t_k^n) - M(t_{k-1}^n)\right)^2$$

with $t_k^n = \frac{k}{2^n}t$.

So

$$A_t^n \leq \sup_{1\leq k\leq 2^n} \left|M(t_k^n) - M(t_{k-1}^n)\right| \cdot V_M(t) \xrightarrow{a.s.} 0$$

by continuity of M.

Thus $A_t^n \to 0$, $\forall\, t \geq 0$.

We will prove $A_t^n \to A_t$ such that $M_t^2 - A_t$ is martingale.

$\implies A_t = 0$ and $\mathbb{E}M_t^2 = \mathbb{E}M_0^2 = 0$.

$\implies M_t = 0$ a.s. $\forall\, t \geq 0$.

By a.s. continuity, $M_t \equiv 0$, a.s. □

□

1.7.1 Itô Integral of local martingale

Theorem 1.7.4. Let $M \in c\mathcal{M}_{loc}$ with $M_0 = 0$. Let $H \in \mathcal{P}$ such that W.P.1

$$\int_0^t H_s^2 \,\mathrm{d}[M]_s < \infty$$

for all $t > 0$. Then $H \cdot M$ is well-defined such that $H \cdot M \in c\mathcal{M}_{0,loc}$.

Proof. Let $U_n = \inf\{t\colon |M_t| > n\}$, $V_n = \inf\{t\colon \int_0^t H_s^2 \,\mathrm{d}[M]_s > n\}$.

(Recall $L^2(M) \implies \mathbb{E}\int_0^\infty H_s^2 \,\mathrm{d}[M]_s < \infty$).

$\implies T_n = U_n \wedge V_n$. In this way, $M^{T_n} \in c\mathcal{M}_0^2$ and $H^{T_n} \in L^2(M^{T_n})$.
So $H^{T_n} \cdot M^{T_n}$ is defined by previous L^2 theory.
$\implies (H \cdot M)^{T_n} = H^{T_n} \cdot M^{T_n}$ $(t < T_n)$ is a L^2-bounded martingale.
$\implies H \cdot M \in c\mathcal{M}_{0,loc}$. □

Definition 1.7.8 (semimartingale)**.** We say X is a semimartingale if

$$X_t = X_0 + M_t + A_t.$$

M_t is local mart, A_t is finite variation.

Definition 1.7.9. Let X be a continuous semimartingale. Then $\exists$ unique

$$\begin{cases} M \in c\mathcal{M}_{0,loc} \\ A \in cFV_0 \end{cases}$$

such that $X_t = X_0 + M_t + A_t$, $\forall\ t \geq 0$.

Proof. If $\exists$ M' and A' such that

$$X = X_0 + M' + A' = X_0 + M + A.$$

Then $M - M' = A' - A \in c\mathcal{M}_{0,loc} \cap cFV = \{0\}$. □

We write $M = X^{cm}$ and

$$[X, X] = [M, M] = [X^{cm}, X^{cm}].$$

Similarly, $[X, Y] = [X^{cm}, Y^{cm}]$.
For any $H \in lb\mathcal{P}$ define

$$H \cdot X = H \cdot M + H \cdot A.$$

Obviously $H \cdot M \in c\mathcal{M}_{0,loc}$ and $H \cdot A \in FV$.
And

$$\begin{aligned} [H \cdot X, K \cdot Y] &= [H \cdot X^{cm}, K \cdot Y^{cm}] \\ &= H \cdot [X^{cm}, K \cdot Y^{cm}] \\ &= H \cdot K[X^{cm}, Y^{cm}] \\ &= (HK) \cdot [X^{cm}, Y^{cm}] = (HK) \cdot [X, Y]. \end{aligned}$$

1.7.2 Itô's Formula

Theorem 1.7.5 (Itô's Formula)**.** Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be C^2. Let

$$X_t \equiv (X_t^1, \cdots, X_t^n)$$

be a continuous semimartingale.

Then

$$f(x_t) - f(x_0) = \int_0^t \nabla f(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t \sum_{i,j} D_{ij} f(X_s)\,\mathrm{d}[X^i, X^j]_s$$

where $\nabla f(X_s) = (D_1 f(X_s), \cdots, D_n f(X_s))$. $D_i f(x) = \frac{\partial}{\partial x_i} f(x)$, $D_{ij} = D_i D_j = D_j D_i$.

Theorem 1.7.6 (Integration by Parts)**.** Let X, Y be continuous semimartingales. Then

$$X_t Y_t - X_0 Y_0 = \int_0^t X_s\,\mathrm{d}Y_s + \int_0^t Y_s\,\mathrm{d}X_s + [X,Y]_t.$$

Proof. Assume IBP holds.

Let

$$\mathcal{A} = \left\{ f \in C^2(\mathbb{R}^n) \colon\ \mathrm{d}f(X_t) = \sum_i D_i f(X_t)\,\mathrm{d}X_t + \frac{1}{2}\sum_{i,j} D_{ij} f(X_t)\,\mathrm{d}[X^i, X^j]_t \right\}.$$

(i). $f, g \in \mathcal{A} \implies f + g \in \mathcal{A}$. (ii). $f, g \in \mathcal{A} \implies fg \in \mathcal{A}$.

$$\begin{aligned}
\mathrm{d}(fg(X_t)) =& \,\mathrm{d}(f(X_t)\cdot g(X_t)) \\
=& f(X_t)\cdot \mathrm{d}g(X_t) + g(X_t)\,\mathrm{d}f(X_t) + \mathrm{d}[f(X_t), g(X_t)]_t \\
=& f(X_t)\left(\sum_i D_i g(X_t)\,\mathrm{d}X_t + \frac{1}{2}\sum_{i,j} D_{ij} g(X_t)\,\mathrm{d}[X^i, X^j]_t \right) \\
& + g(X_t)\left(\sum_i D_i f(X_t)\,\mathrm{d}X_t + \frac{1}{2}\sum_{i,j} D_{ij} f(X_t)\,\mathrm{d}[X^i, X^j]_t \right) \\
& + \mathrm{d}\left[\int_0^t \sum_i D_i f(X_t)\,\mathrm{d}X_t, \int_0^t \sum_j D_j g(X_t)\,\mathrm{d}X_t^j \right] \\
=& \,\mathrm{d}X_t \left(\sum_i f D_i g(X_t) + \sum_i g D_i f(X_t) \right) \\
& + \frac{1}{2}\sum_{i,j} \left(g D_{ij} f(X_t) + f D_{ij} g(X_t) + 2 D_i f D_j g(X_t) \right) \mathrm{d}[X^i, X^j]_t \\
=& \sum_i D_i(fg)(X_t)\,\mathrm{d}X_t + \frac{1}{2}\sum_{i,j} D_{ij}(fg)(X_t)\,\mathrm{d}[X^i, X^j]_t \\
\implies & \quad fg \in \mathcal{A}.
\end{aligned}$$

(iii). $f(x) = x \in \mathcal{A} \implies \forall$ polynomial $\in \mathcal{A}$. $\square$

Lemma 1.7.6. $\forall\, f \in C^2(\mathbb{R}^d)$, $\exists$ polynomials $\{P_n(x)\}_{n\geq 1}$ such that

$$\begin{cases} P_n(x) \to f(x) \\ D_i P_n(x) \to D_i f(x) \\ D_{ij} P_n(x) \to D_{ij} f(x) \end{cases}$$

uniformly on compacts.

Define $U_N = \inf\{t \geq 0 \colon\, |X_t| + [M]_t > N\}$ for each $N > 0$.

Then $U_N \uparrow \infty$. So it suffices to prove Itô formula holds for $\forall\, t \leq U_N$.

Note that

$$P_n(X_t) = P_n(X_0) + \int_0^t \sum_i D_i P_n(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t \sum_{i,j} D_{ij} P_n(X_s)\,\mathrm{d}[X^i, X^j]_s$$

for $0 \leq t \leq U_N$.

By letting $n \to \infty$, we will prove the limits are

$$f(X_t) = f(X_0) + \int_0^t \sum_i D_i f(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t \sum_{ij} D_{ij} f(X_s)\,\mathrm{d}[X^i, X^j]_s.$$

$$\begin{aligned} I_{1,n} &= \int_0^t \sum_i D_i P_n(X_s)\,\mathrm{d}X_s \\ &= \int_0^t \sum_i D_i P_n(X_s)\,\mathrm{d}M_s + \int_0^t \sum_i D_i P_n(X_s)\,\mathrm{d}A_s. \end{aligned}$$

Notice that

$$\left| \int_0^t \sum_i D_i P_n(X_s)\,\mathrm{d}A_s - \int_0^t \sum_i D_i f(X_s)\,\mathrm{d}A_s \right| \leq d \cdot \sup_i \underbrace{\|D_i P_n - D_i f\|}_{\text{max on compacts}} \cdot A_t \xrightarrow{n\to\infty} 0 \text{ a.s.}$$

Next,

$$\begin{aligned} &\mathbb{E} \sup_{s \leq t} \left| \int_0^t \sum_i \left(D_i P_n(X_s) - D_i f(X_s)\right) \mathrm{d}M_s \right|^2 \\ \leq &\mathbb{E} \int_0^t \left(\sum_i D_i P_n(X_s) - D_i f(X_s) \right)^2 \mathrm{d}[M]_s \\ \leq &d \cdot \sup_{1 \leq i \leq d} \|D_i P_n - D_i f\|^2 \cdot \underbrace{\mathbb{E}[M]_{t\wedge U_N}}_{N} \xrightarrow{n\to\infty} 0. \end{aligned}$$

By taking subsequence, $\int_0^t \sum_i D_i P_{n_k}(X_s)\,\mathrm{d}M_s \xleftarrow{\text{a.s.}} \int_0^t \sum_i D_i f(X_s)\,\mathrm{d}M_s$.

$$I_{2,n} = \frac{1}{2}\int_0^t \sum_{i,j} D_{ij} P_n(X_s)\,\mathrm{d}[X^i, X^j]_s.$$

Notice that

$$\left| I_{2,n} - \frac{1}{2}\int_0^t \sum_{i,j} D_{ij} f(X_s)\,\mathrm{d}[X^i, X^j]_s \right| \le C \sum_{i,j} \|D_{ij} P_n - D_{ij} f\| \cdot \underbrace{\sum_{i,j} [X^i, X^j]_t}_{\le C_N} \xrightarrow{n\to\infty} 0.$$

Lemma 1.7.7. Let M be a bounded continuous martingale with $M_0 = 0$. Let V be continuous FV_0

$$\begin{aligned} M_t^2 &= \int_0^t 2M_s\,\mathrm{d}M_s + [M]_t, \\ M_t V_t &= \int_0^t M_s\,\mathrm{d}V_s + \int_0^t V_s\,\mathrm{d}M_s. \end{aligned}$$

Proof.

$$\begin{aligned} M_t^2 &= \left[\sum_{k=1}^{2^n} M(t_k^n) - M(t_{k-1}^n)\right]^2 \\ &= \sum_{k=1}^{2^n} \left(M(t_k^n) - M(t_{k-1}^n)\right)^2 + 2\sum_{k=1}^{2^n} \left[M(t_k^n)M(t_{k-1}^n) - M(t_{k-1}^n)^2\right] \\ &= M(t_{2^n}^n)^2 - M(t_0^n)^2 = M(t)^2 \end{aligned}$$

where $t_k^n = \frac{k}{2^n} t$.

Note that $\sum_{k=1}^{2^n} \left(M(t_k^n) - M(t_{k-1}^n)\right)^2 \to [M]_t$ and

$$2\sum_{k=1}^{2^n} \left[M(t_k^n)M(t_{k-1}^n) - M(t_{k-1}^n)^2\right] = 2\sum_{k=1}^{2^n} M(t_{k-1}^n)(M(t_k^n) - M(t_{k-1}^n)) \to 2\int_0^t M_s\,\mathrm{d}M_s.$$

$$\begin{aligned} M_t V_t &= M_{t_{2^n}^n} V_{t_{2^n}^n} - M_{t_0^n} V_{t_0^n} \\ &= \sum_{k=1}^{2^n} \left[M(t_k^n)V(t_k^n) - M(t_{k-1}^n)V(t_{k-1}^n)\right] \\ &= \sum_{k=1}^{2^n} M(t_k^n)(V(t_k^n) - V(t_{k-1}^n)) + \sum_{k=1}^{n} V(t_{k-1}^n)\left(M(t_k^n) - M(t_{k-1}^n)\right) \\ &\to \int_0^t M_s\,\mathrm{d}V_s + \int_0^t V_s\,\mathrm{d}M_s. \end{aligned}$$

□

Theorem 1.7.7 (Lévy Thm). Let $\{B_t^i\}_{t\ge 0} \in c\mathcal{M}_{0,loc}$ for $i = 1, 2, \cdots, n$. If $\forall\, i, j$, we have

$$[B^i, B^j]_t = \begin{cases} t, & i = j, \\ 0, & i \ne j. \end{cases}$$

Then $B_t = (B_t^1, \cdots, B_t^n)$ is a Brownian motion on $\mathbb{R}^n$ with $\mathcal{N}(0, I_n)$.

Proof. $\forall\ s < t$, we will prove

$$\mathbb{E}\left[e^{i\theta(B_t - B_s)}\middle|\mathcal{F}_s\right] = e^{-\frac{1}{2}(t-s)|\theta|^2}$$

for all $\theta \in \mathbb{R}^n$.

Define $f(x,t) = e^{i\theta x + \frac{1}{2}|\theta|^2 t}$, $\mathbb{R}^{n+1} \to \mathbb{C}$.

Apply Itô's formula to $f(B_t, t) \equiv f(X_t)$.

$X_t = (B_t^1, B_t^2, \cdots, B_t^n, t)$.

$$\begin{aligned} f(X_t) =& f(x_0) + \int_0^t \sum_{i=1}^n (D_i f(X_s \,\mathrm{d}B_s^i)) + \int_0^1 D_{n+1} f(X_s) ds \\ &+ \frac{1}{2}\int_0^t \sum_{1 \le i,j \le n} D_{ij} f(X_s)\,\mathrm{d}[B^i, B^j]_s \\ &+ \frac{1}{2}\int_0^t \sum_{1 \le k \le n} D_{k,n+1} f(X_s)\,\mathrm{d}[B^k, s]_s \end{aligned}$$

$$\begin{aligned} \implies f(B_t, t) =& f(B_0, 0) + \int_0^t \sum_i D_i f(B_s, s)\,\mathrm{d}B_s^i \\ &+ \int_0^t \left(e^{i\theta B_s + \frac{1}{2}|\theta|^2 s} \cdot \frac{1}{2}|\theta|^2\right)\mathrm{d}s \\ &+ \int_0^t \frac{1}{2}\left(\sum_{j=1}^n e^{i\theta B_s + \frac{1}{2}|\theta|^2 s}(i\theta_j)^2\right)\mathrm{d}s. \end{aligned}$$

We conclude that $f(B_t, t) = f(B_0, 0) + c\mathcal{M}_{0,loc}$

$$\implies e\underbrace{{}^{i\theta B_t + \frac{1}{2}|\theta|^2 t}}_{M_t^\theta \in c\mathcal{M}_{0,loc}} = 1 + c\mathcal{M}_{0,loc}$$

Since $\left|M_t^\theta\right| = e^{\frac{1}{2}|\theta|^2 t} < \infty$ and

$$\mathbb{E}(M_t^{T_n} \mid \mathcal{F}_s) = M_s^{T_n}$$

where T_n reduces M.

We may take $n \to \infty$ to get $\mathbb{E}(M_t \mid \mathcal{F}_s) = M_s$.

$$\implies \mathbb{E}(e^{i\theta B_t + \frac{1}{2}|\theta|^2 t} \mid \mathcal{F}_s) = e^{i\theta B_s + \frac{1}{2}|\theta|^2 s}.$$

$$\implies \mathbb{E}\left(e^{i\theta(B_t - B_s)} \mid \mathcal{F}_s\right) = e^{\frac{1}{2}|\theta|^2(t-s)}.$$

$$\implies B_t - B_s \sim \mathcal{N}(0, t-s)$$

Q independent of $\mathcal{F}_s$. $\square$

Theorem 1.7.8 (Dubins and Schwarz)**.** let $M \in c\mathcal{M}_{0,loc}$ such that $[M]_t \uparrow \infty$ as $t \to \infty$. Then $M_t = B([M]_t)$ where B is some Brownian motion.

Let $\tau_t = \inf\{u \geq 0\colon [M]_u > t\}$. Set $\mathcal{G}_t \equiv \mathcal{F}(\tau_t)$. Then $B_t \equiv M(\tau_t)$ is a B.M. with respect to $\mathcal{G}_t$.

Proof. (i) $M(\tau_t) \in \mathcal{F}_{\tau_t} \implies B_t \in \mathcal{G}_t$.

(ii) $t \mapsto B_t$ is continuous.

$$\begin{cases} \tau_{a-} = b, \\ \tau_{a+} = c. \end{cases}$$

We only need to prove $M(b) = M(c)$.

Assume M is bounded $\forall\ q \in \mathbb{Q}_+$, define

$$S_q \equiv \inf\{t > q\colon [M]_t > [M]_q\}.$$

We will show $M_t \equiv M_q$ for all $t \in [q, S_q)$

Note $S_q \geq q$, by OST,

$$\mathbb{E}\left(M_{S_q}^2 - [M]_{S_q} \mid \mathcal{F}_q\right) = M_q^2 - [M]_q.$$

Because $[M]_{S_q} = [M]_q$,

$$\implies \mathbb{E}[M_{S_q}^2 - M_q^2 \mid \mathcal{F}_q] = 0.$$

$$\implies \mathbb{E}((M_{S_q} - M_q)^2 \mid \mathcal{F}_q) = 0.$$

$$\implies \mathbb{E}(M_{S_q} - M_q)^2 = 0.$$

$$\implies M_{S_q} = M_q.$$

(iii) B_t and $B_t^2 - t \in c\mathcal{M}_{0,loc} \implies [B]_t = t \implies$ B.M.

Recall $B_t = M(\tau_t)$. Define

$$T_n = \inf\{t \geq 0\colon |M_t| > n\},$$
$$U_n = [M]_{T_n}.$$

We will check $\tau_{t\wedge U_n} = T_n \wedge \tau_t$.

$$t \wedge U_n \leq t \implies \tau_{t\wedge U_n} \leq \tau_t.$$

$$\tau_{t\wedge U_n} = \inf\{u\colon [M]_u > t \wedge U_n = t \wedge [M]_{T_n}\}.$$

Claim: $\tau_t \wedge T_n = \tau_{t\wedge U_n}$

case 1. $t \leq U_n$ we have

$$\tau_{t\wedge U_n} = \inf\{u\colon [M]_u > t\} = \tau_t,$$

and

$$t \le U_n = [M]_{T_n} \implies T_n \ge \tau_t \implies \tau_t \wedge T_n = \tau_t = \tau_{t\wedge U_n}.$$

case 2. $t > U_n$ we have

$$\tau_{t\wedge U_n} = \tau_{U_n} = \inf\{u\colon [M]_u > U_n\}$$

and

$$t > U_n > [M]_{T_n} \implies T_n < \tau_t \implies \tau_t \wedge T_n = T_n.$$

It suffices to prove $T_n = \tau_{[M]_{T_n}} = \tau_{U_n}$.

One hand, $\tau_{U_n} \ge T_n$ can be proved by def of τ_{U_n}.

On the other hand, $T_n = \inf\{t\colon |M_t| > n\}$. Take $\forall\, \varepsilon > 0$, $u_\varepsilon \in (T_n, T_n + \varepsilon)$ s.t. $|M_{u_\varepsilon}| > n$.

Hence $M_{u_\varepsilon} \ne M_{T_n}$. Suppose $[M]_{u_\varepsilon} = [M]_{T_n}$, then M is constant by the property of continuous local martingale. Contradiction. So $[M]_{u_\varepsilon} > [M]_{T_n} = U_n \implies \tau_{U_n} \le u_\varepsilon$. Let $\varepsilon \downarrow 0$ to get $\tau_{U_n} \le T_n$.

Then $B_{t\wedge U_n} = M(\tau_{t\wedge U_n}) = M(T_n \wedge \tau_t)$, $B_t^{U_n} = M_{\tau_t}^{T_n}$.

By OST

$$\mathbb{E}\left(B_t^{U_n} \middle| \mathcal{F}_{\tau_s}\right) = \mathbb{E}\left(M_{\tau_t}^{T_n} \middle| \mathcal{F}_{\tau_s}\right) = M_{\tau_s}^{T_n} = B_s^{U_n}.$$

Then B^{U_n} is a martingale w.r.t. $\mathcal{G}_t = \mathcal{F}_{\tau_t}$ + (2) U_n is $\mathcal{G}_t$-stopping time $\implies B \in c\mathcal{M}_{0,loc}$

pf of (2)

$$\{[M]_{T_n} \le t\} = \{U_n \le t\} \in \mathcal{F}_{\tau_t}.$$

$$\{T_n \le \tau_t\} \in \mathcal{F}_{\tau_t}.$$

$\left(M^2 - [M]\right)^{T_n}$ is martingale. Then

$$\mathbb{E}\left[(M_{\tau_t}^{T_n})^2 - [M]_{\tau_t}^{T_n} \middle| \mathcal{F}_{\tau_s}\right] = \left(M_{\tau_s}^{T_n}\right)^2 - [M]_{\tau_s}^{T_n}$$

$$\implies \mathbb{E}\left[(B_t^{U_n})^2 - t \wedge U_n \middle| \mathcal{F}_s\right] = (B_s^{U_n})^2 - s \wedge U_n.$$

$\implies (B_t^2 - t)^{U_n}$ is a martingale $\implies B_t^2 - t \in \mathcal{M}_{loc} \implies [B]_t = t.$ □

If $[M]_\infty < \infty$, we get $\tau_t = \infty$, $\forall\, t \ge [M]_\infty$.

$$B_t = M(\infty), \quad \forall\, t \ge [M]_\infty.$$

So this def does not give a B.M.

Define

$$B_t = \begin{cases} M(\tau_t), & t < [M]_\infty, \\ M(\infty) + W(t) - W([M]_\infty), & t \ge [M]_\infty, \end{cases}$$

where W is a standard B.M. independent of M.

Or

$$B_t \begin{cases} M(\tau_t), & t < [M]_\infty \\ M(\infty) + W(t - [M]_\infty), & t \geq [M]_\infty. \end{cases}$$

Then $B([M]_t) = M_t$.

Lemma 1.7.8. On $\{[M]_\infty < \infty\}$, the limit $M_\infty \equiv \lim_{t\uparrow\infty} M_t$ exists a.s.

Proof. Define $S_K = \inf\{t \geq 0 \colon [M]_t > K\}$. Then on $\{[M]_\infty < \infty\}$, $S_K = \infty$, $\forall\ K > [M]_\infty$

M^{S_K} is a L^2-bounded martingale.

$\implies \lim_{t\uparrow\infty} M_t^{S_K}$ exists a.s.

Let $K \uparrow \infty$ to get $\lim_{t\uparrow\infty} M_t$ exists a.s.

By an enlargement of the prob. space $(\Omega.\mathcal{F}, \mathsf{P})$

$$\begin{aligned} \implies \tilde{\Omega} &= \Omega \times \Omega^* \\ \tilde{\mathcal{F}}_t &= \mathcal{F}_t \times \mathcal{F}_t^* \\ \tilde{\mathsf{P}} &= \mathsf{P} \times \mathsf{P}^* \end{aligned}$$

where on $(\Omega^*, \mathcal{F}^*, \mathcal{F}_t^*, \mathsf{P}^*)$ we have W^* is B.M. independent of M. □

Theorem 1.7.9 (Dubins-Schwarz (general version))**.** Let $M \in c\mathcal{M}_{0,loc}$. Then $\exists$ B.M. B s.t. $M_t = B([M]_t)$ for all $t \geq 0$.

Theorem 1.7.10 (Doléans' Thm)**.** Let X be a semimartingale with $X_0 = 0$. Suppose $Z_0 \in \mathcal{F}_0$. Then $\exists$ a unique continuous semimartingale Z s.t.

$$Z_t = Z_0 + \int_0^t Z_s \, \mathrm{d}X_s.$$

The solution $Z_t = Z_0 e^{X_t - \frac{1}{2}[X]_t}$.

Notation: $\mathcal{E}(X_t) = e^{X_t - \frac{1}{2}[X]_t}$ is called the exponential semimartingale of X.

Proof. $\mathcal{E}(X_t) = e^{X_t - \frac{1}{2}[X]_t} = f(X_t, [X]_t)$ with $f(x,y) = e^{x - \frac{1}{2}y}$.

By apply Itô's formula,

$$\mathrm{d}f(X_t, [X]_t) = \partial_x f(X_t, [X]_t)\, \mathrm{d}X_t + \partial_y f(X_t, [X]_t)\, \mathrm{d}[X]_t + \frac{1}{2}\left(\partial_{xx} f(X_t, [X]_t)\, \mathrm{d}[X]_t\right).$$

Because

$$\begin{cases} \mathrm{d}X_t \, \mathrm{d}X_t = \mathrm{d}[X]_t \\ \mathrm{d}X_t \, \mathrm{d}[X]_t = 0 \\ \mathrm{d}[X]_t \, \mathrm{d}[X]_t = 0. \end{cases}$$

Hence

$$\mathrm{d}\mathcal{E}(X_t) = \mathcal{E}(X_t)\, \mathrm{d}X_t.$$

So $\mathcal{E}(X)$ is a solution to

$$\mathrm{d}Z_t = Z_t\,\mathrm{d}X_t.$$

(Uniqueness) Assume Z is another solution. We will prove

$$Z_t = Z_0\mathcal{E}(X_t).$$

We now have $\mathrm{d}Z_t = Z_t\,\mathrm{d}X_t$.
Define

$$Y_t = e^{-X_t+\frac{1}{2}[X]_t} = (\mathcal{E}(X)_t)^{-1}.$$

It suffices to prove $\mathrm{d}(Z_tY_t) = 0$ which implies

$$Z_tY_t = Z_0Y_0 = Z_0, \quad \forall\ t.$$

We have

$$\mathrm{d}(Z_tY_t) = Z_t\,\mathrm{d}Y_t + Y_t\,\mathrm{d}Z_t + \mathrm{d}[Y,Z]_t.$$

Notice that by Itô's formula,

$$\mathrm{d}Y_t = (-1)Y_t\,\mathrm{d}X_t + \frac{1}{2}Y_t\,\mathrm{d}[X]_t + \frac{1}{2}(-1)^2Y_t\,\mathrm{d}[X]_t = -Y_t\,\mathrm{d}X_t + Y_t\,\mathrm{d}[X]_t.$$

$$\implies \mathrm{d}(Z_tY_t) = Z_t(-Y_t\,\mathrm{d}X_t + Y_t\,\mathrm{d}[X]_t) + Y_t(Z_t\,\mathrm{d}X_t) + Y_t(Z_t\,\mathrm{d}X_t) + (-Y_tZ_t)\,\mathrm{d}[X]_t = 0.$$

□

Let $M \in c\mathcal{M}_{0,loc}$. Then $\mathcal{E}(\theta M)_t = e^{\theta M_t - \frac{1}{2}\theta^2[M]_t}$ is in $c\mathcal{M}_{loc}$.
By Fatou's lemma, we get

$$\begin{aligned}
\mathbb{E}(\mathcal{E}(\theta M)_t \mid \mathcal{F}_s) &= \mathbb{E}(\lim_{n\to\infty} \mathcal{E}(\theta M)_{t\wedge T_n} \mid \mathcal{F}_s) \\
&\le \liminf_{n\to\infty} \mathbb{E}(\mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s) \\
&= \liminf_{n\to\infty} \mathcal{E}(\theta M)_s^{T_n} \\
&= \mathcal{E}(\theta M)_s.
\end{aligned}$$

Hence $\mathcal{E}(X_t)$ is a supermartingale and $\mathbb{E}(\mathcal{E}(M)_t) \le 1$.

Theorem 1.7.11. Suppose $\forall\ t > 0$, $\exists\ K_t > 0$ s.t.

$$[M]_t < K_t \quad \text{a.s.}$$

If for some $t > 0$, $\exists\ K_t > 0$ s.t.

$$[M]_t < K_t \quad \text{a.s.}$$

Then for this $t > 0$,

$$\mathsf{P}\left(\max_{s\le t} M_s > y\right) \le e^{-\frac{y^2}{2K_t}}, \quad \forall\ y > 0.$$

and $\mathcal{E}(\theta M)$ is a true martingale.

Proof. By using $\mathcal{E}(\theta M)$ is a supermartingale, by Doob's inequality,

$$\begin{aligned}
\mathsf{P}\left(\max_{s\le t} M_s > y\right) &= \mathsf{P}\left(\max_{s\le t} \mathcal{E}(\theta M)_s > e^{\theta y - \frac{1}{2}\theta^2 K_t}\right) \\
&\le \frac{\mathbb{E}(\mathcal{E}(\theta M)_t)}{e^{\theta y - \frac{1}{2}\theta^2 K_t}} \\
&\le e^{-\theta y + \frac{1}{2}\theta^2 K_t}.
\end{aligned}$$

Pick $\theta = \frac{y}{K_t}$ to get $e^{-\frac{y^2}{2K_t}}$.

Next, define $Z_t = \max_{s\le t} M_s$. Then $\forall\ \theta > 0$,

$$\mathbb{E}e^{\theta Z_t} = \int_{\mathbb{R}} e^{\theta y}\mathsf{P}(Z_t \in \mathrm{d}y).$$

Notice that $\forall\ z \in \mathbb{R}$,

$$e^{\theta Z} = 1 + \int_0^Z \theta e^{\theta y}\,\mathrm{d}y.$$

$$\begin{aligned}
\implies \mathbb{E}(e^{\theta Z_t}) &= 1 + \mathbb{E}\left(\int_0^{Z_t} \theta e^{\theta y}\,\mathrm{d}y\right) \\
&= 1 + \mathbb{E}\left(\int_0^{\infty} 1_{y<Z_t}\theta e^{\theta y}\,\mathrm{d}y\right) \\
&= 1 + \int_0^{\infty} \theta e^{\theta y}\,\mathrm{d}y\mathsf{P}(Z_t > y) \\
&\le 1 + \int_0^{\infty} \theta e^{\theta y} e^{-\frac{y^2}{2K_t}}\,\mathrm{d}y \\
&< \infty.
\end{aligned}$$

By $\mathcal{E}(\theta M)_t$ is a $\mathcal{M}_{loc}$, we get $\exists\ T_n \uparrow \infty$ s.t. $\mathcal{E}(\theta M)_t^{T_n}$ is a martingale.

$$\implies \forall\ s < t, \quad \mathbb{E}[\mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s] = \mathcal{E}(\theta M)_s^{T_n}.$$

By DCT for $\mathcal{E}(\theta M)_t^{T_n} \le Z_t$ and $\mathbb{E}Z_t < \infty$, we get

$$\lim_{n\to\infty} \mathbb{E}(\mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s) = \mathbb{E}(\mathcal{E}(\theta M)_t \mid \mathcal{F}_s).$$

□

Corollary 1.7.2. Let B be a B.M. in $\mathbb{R}^n$ and let H be a bounded previsible process. Then

$$e^{\int H\,\mathrm{d}B - \frac{1}{2}\int |H_s|^2\,\mathrm{d}s}$$

is a true martingale.

Remark. $[M]_t = \int_0^t |H_s|^2 \,\mathrm{d}s < K_t$.

Corollary 1.7.3. Let $M \in c\mathcal{M}_{0,loc}$. Then $\forall\ t > 0,\ y > 0,\ K > 0$,

$$\mathsf{P}\left(\max_{s\le t} M_s \ge y, [M]_t \le K\right) \le e^{-\frac{y^2}{2K}}.$$

$$\implies \mathsf{P}\left(\max_{s\le t} |M_s| \ge y, [M]_t \le K\right) \le 2e^{-\frac{y^2}{2K}}.$$

Proof. Let $T = \inf\{u \ge 0\colon [M]_u \ge K\}$. Then

$$\mathsf{P}\left(\max_{s\le t} M_s \ge y, [M]_t \le K\right) \le \mathsf{P}\left(\max_{s\le t} M_{s\wedge T} \ge y\right) \le e^{-\frac{y^2}{2K}}$$

since $[M^T]_t \le K$. □

Theorem 1.7.12 (Novikov's Thm)**.** If $\forall\ t > 0$, $\mathbb{E}(e^{\frac{1}{2}[M]_t}) < \infty$ then $\mathcal{E}(X)_t$ is a true martingale.

Proof. See P198 of [KS]. □

1.7.3 Burkholder-Davis-Gundy inequalities

Definition 1.7.10. We say a function $F : \mathbb{R}^+ \to \mathbb{R}^+$ is moderate if continuous and increasing, $F(x) = 0 \Leftrightarrow x = 0$, and $\forall\ \alpha > 1$,

$$\sup_{x>0} \frac{F(\alpha x)}{F(x)} < \infty.$$

Example . $F(x) = x^p$, $\forall\ p > 0$. $F(x) = x^p (\log x \vee 1)$, $\forall\ p \ge 0$.

Theorem 1.7.13 (BDG inequality)**.** Let F be moderate. Then $\exists$ universal constants $c_F \le C_F < \infty$ s.t. $\forall\ M \in c\mathcal{M}_{0,loc}$,

$$c_F \mathbb{E}\left(F([M]_\infty^{\frac{1}{2}})\right) \le \mathbb{E}F(M_\infty^*) \le C_F \mathbb{E}\left(F([M]_\infty^{\frac{1}{2}})\right).$$

$M_t^* = \sup_{s\le t} |M_s|$. Fix $t > 0$. Let $\tilde{M}_s^t = M_{s\wedge t}$. Then

$$c_F \mathbb{E}\left(F([M]_t^{\frac{1}{2}})\right) \le \mathbb{E}\left(F(M_t^*)\right) \le C_F \mathbb{E}F([M]_t^{\frac{1}{2}}).$$

Example . If $F(x) = x^p$, then

$$c_p \mathbb{E}[M]_t^{\frac{p}{2}} \le \mathbb{E}\,|M_t^*|^p \le C_p \mathbb{E}[M]_t^{\frac{p}{2}}.$$

Lemma 1.7.9 (“good” λ inequality)**.** Let X and Y be nonegative R.V.s. Suppose $\exists\ \beta > 1$ s.t. $\forall\ \lambda > 0$, $\delta > 0$,

$$\mathsf{P}(X > \beta\lambda, Y \le \delta\lambda) \le \psi(\delta)\mathsf{P}(X > \lambda)$$

where $\psi(\delta) \downarrow 0$ as $\delta \downarrow 0$.

Then for each moderate F, $\exists\ C = C(\beta, \psi, F) > 0$ s.t.

$$\mathbb{E}[F(X)] \le c\mathbb{E}[F(Y)].$$

Proof for Lemma.

■. Assume good λ inequality. Then it holds by replacing X by $X \wedge a$ for any $a > 0$.

If $a > \beta\lambda > \lambda$, then

$$X \wedge a > \beta\lambda \Leftrightarrow X > \beta\lambda$$

$$X \wedge a > \lambda \Leftrightarrow X > \lambda.$$

Hence we may assume $\mathbb{E}F(X) < \infty$.

[$\mathbb{E}F(X \wedge a) \le \mathbb{E}F(Y)$ for all $a > 0$. Then let $a \uparrow \infty$.]

Notice that $\sup_{x>0} \frac{F(x)}{F(\frac{x}{\beta})}$.

Let $\gamma > 0$ s.t. $F(x) \le \frac{1}{\gamma}F(\frac{x}{\beta})$ for all $x > 0$.

Now

$$\int F(\mathrm{d}\lambda)\psi(\delta)\mathsf{P}(X > \lambda) \ge \int F(\mathrm{d}\lambda)\mathsf{P}(X > \beta\lambda, Y \le \delta\lambda).$$

$$LHS = \psi(\delta)\mathbb{E}\left(\int_0^X F(\mathrm{d}\lambda)\right) = \psi(\delta)\mathbb{E}(F(X)).$$

$$\begin{aligned} RHS &= \int F(\mathrm{d}\lambda)\mathbb{E}(1_{\frac{Y}{\delta} \le \lambda < \frac{X}{\beta}}) \\ &= \mathbb{E}\left(\int_{\frac{Y}{\delta}}^{\frac{X}{\beta}} F(\mathrm{d}\lambda)\right) \\ &= \mathbb{E}\left(F\left(\frac{X}{\beta}\right) - F\left(\frac{Y}{\delta}\right)\right) \\ &\ge \mathbb{E}\left(\gamma F(X)\right) - \mathbb{E}\left(F\left(\frac{Y}{\delta}\right)\right). \end{aligned}$$

Let $\delta > 0$ be small sucht that

$$\psi(\delta) < \frac{\gamma}{2} \implies \gamma - \psi(\delta) \ge \frac{\gamma}{2}.$$

Hence

$$\mathbb{E}\left(F(X)\right)\left(\psi(\delta) - \gamma\right) \ge -\mathbb{E}\left(F\left(\frac{Y}{\delta}\right)\right).$$

$$\implies \mathbb{E}\left(F\left(\frac{Y}{\delta}\right)\right) \ge \left(\gamma - \psi(\delta)\right)\mathbb{E}F(X) \ge \frac{\lambda}{2}\mathbb{E}F(X).$$

Using $\sup_{y>0} \dfrac{F(y/\delta)}{F(y)} < \infty$, $\implies \exists\ \mu > 0$ s.t. $F(y/\delta) \le \mu F(y)$ for all $y > 0$

$$\implies \mathbb{E}(F(Y/\delta)) \le \mu \mathbb{E}F(Y)$$

$$\implies \frac{\gamma}{2}\mathbb{E}F(X) \le \mu \mathbb{E}F(Y).$$

$\square$

Proof. of BDF inequality.

$$\mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda\right) \le e^{-\frac{(\beta\lambda)^2}{2(\delta\lambda)^2}} = e^{-\frac{\beta^2}{2\delta^2}}.$$

Define $\tau = \inf\{u \ge 0\colon\ |M_u| > \lambda\}$.
Then

$$\begin{aligned}\mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda\right) &= \mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda, \tau < \infty\right)\\ &= \mathbb{E}\left[1_{\tau<\infty}\mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda \,\middle|\, \mathcal{F}_\tau\right)\right]\end{aligned}$$

Define $N_t \equiv (M_{t+\tau} - M_\tau)^2 - ([M]_{t+\tau} - [M]_\tau)$.
Then $N_t \in \mathcal{F}_{t+\tau}$ is a continuous local martingale

$$\mathbb{E}\left[M_{t+\tau}^2 - [M]_{t+\tau} \mid \mathcal{F}_{s+\tau}\right] = M_{s+\tau}^2 - [M]_{s+\tau}.$$

$\implies N_0 = 0$ and on $\{M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda\}$ we have $\exists\ \tilde{t} > 0$ s.t.

$$M_{\tilde{t}+\tau} \ge \beta\lambda \implies N_{\tilde{t}} = \left(M_{\tilde{t}+\tau} \pm \lambda\right)^2 - \left([M]_{\tilde{t}+\tau} - [M]_\tau\right) \ge [(\beta-1)\lambda]^2 - (\delta\lambda)^2.$$

Moreover $N_t \ge -\delta^2\lambda^2$, $\forall\ t \ge 0$.
Hence N_t reaches $[(\beta-1)\lambda]^2 - (\delta\lambda)^2$ before $(-\delta^2\lambda^2)$.
By (Gambler's Ruin $\mathsf{P}_k(\tau_0 < \tau_N) = \frac{N-k}{N}$)

$$\mathsf{P}(\tau_b < \tau_a) \le \frac{-a}{b-a}$$

$$\implies \mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda \mid \mathcal{F}_\tau\right) \le \mathsf{P}\left(\tau_{[(\beta-1)\lambda]^2-(\delta\lambda)^2} < \tau_{-(\delta\lambda)^2}\right) \le \frac{(\delta\lambda)^2}{[(\beta-1)\lambda]^2} = \frac{\delta^2}{(\beta-1)^2}.$$

Here we take $0 < \delta < \beta - 1$. Therefore

$$\mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda\right) \le \delta^2\frac{1}{(\beta-1)^2}\mathsf{P}(\tau < \infty) = \delta^2\frac{1}{(\beta-1)^2}\mathsf{P}\left(M_\infty^* > \lambda\right).$$

Similarly,

$$\mathsf{P}\left([M]_\infty^{\frac{1}{2}} > \beta\lambda, M_\infty^* \le \delta\lambda\right).$$

$$\tilde{N}_t = [M]_{t+\tilde{\tau}} - [M]_{\tilde{\tau}} - (M_{\tilde{\tau}+t} - M_{\tilde{\tau}})^2$$

with $\tilde{\tau} = \inf\{u \geq 0 \colon [M]_u^{\frac{1}{2}} > \lambda\}$.

$$\implies \mathsf{P}\left([M]_\infty^{\frac{1}{2}} > \beta\lambda, M_\infty^* \leq \delta\lambda, \tilde{\tau} < \infty\right) = \mathbb{E}\left[\mathsf{P}(\cdots \mid \mathcal{F}_{\tilde{\tau}}) 1_{\tilde{\tau}<\infty}\right].$$

Then $\tilde{N}_t$ reaches $(\beta\lambda)^2 - \lambda^2 - (\delta\lambda)^2$ before $-(\delta\lambda)^2$.

□

1.8 Semimartingale local time

For a Brownian motion $(B_t)_{t\geq 0}$. We consider

$$l_t^0 = \int_0^t \delta_0(B_s)\,\mathrm{d}s,$$

where δ_0 is the Dirac delta function such that $\forall$ function f,

$$\int f(x)\delta_0(x)\,\mathrm{d}x = f(0).$$

$$\mu([a,b]) = l_b^0 - l_a^0,$$

supp $(\mu) = \{t \geq 0 \colon B_t = 0\}$, level set $= \{t \geq 0 \colon B_t = x\}$, using local time

$$l_t^x = \int_0^t \delta_x(B_s)\,\mathrm{d}s.$$

Theorem 1.8.1 (Tanaka's formula)**.** Let X be a continuous semimartingale. Then $\exists$ continuous increasing adapted process $\{l_t, t \geq 0\}$

$$|X_t| - |X_0| = \int_0^t \mathrm{sgn}(X_s)\,\mathrm{d}X_s + l_t,$$

where

$$\mathrm{sgn}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0. \end{cases}$$

Then $\{l_t, t \geq 0\}$ is called the local time of X at 0, and

$$\int_0^t 1_{\{X_s \neq 0\}}\,\mathrm{d}l_s = 0.$$

Remark.

(1) $f(x) = |x| \implies f'(x) = \mathrm{sgn}(x),\ \forall\, x \neq 0 \implies f''(x) = 2\delta(x)$. (distributional derivative)

(2) $|X_t - x| - |X_0 - x| = \int_0^t \mathrm{sgn}(X_0 - x)\,\mathrm{d}X_s + l_t^x$.

(3) Note that $|X_t| = X_t^+ + X_t^-$, $X_t = X_t^+ - X_t^-$

$$\implies 2X_t^+ = |X_t| + X_t = \int_0^t 1_{\{X_s>0\}}\,\mathrm{d}X_s - \int_0^t 1_{\{X_s\le 0\}}\,\mathrm{d}X_s + l_t + \int_0^t 1\,\mathrm{d}X_s + X_0 + |X_0|$$

$$\implies 2X_t^+ = 2\int_0^t 1_{\{X_s>0\}}\,\mathrm{d}X_s + l_t + 2X_0^+.$$

$$\implies X_t^+ - X_0^+ = \int_0^t 1_{\{X_s>0\}}\,\mathrm{d}X_s + \frac{1}{2}l_t.$$

Similarly, $2X_t^- = |X_t| - X_t$

$$\implies X_t^- - X_0^- = -\int_0^t 1_{\{X_s\le 0\}}\,\mathrm{d}X_s + \frac{1}{2}l_t.$$

Proof. Apply Itô's formula to the function $f(x) = |x|$.

$$\begin{aligned}\implies |X_t| &= |X_0| + \int_0^t f'(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t f''(X_s)\,\mathrm{d}[X]_s \\ &= |X_0| + \int_0^t \mathrm{sgn}(X_s)\,\mathrm{d}X_s + \frac{1}{2}\underbrace{\int_0^t 2\delta(X_s)\,\mathrm{d}[X]_s}_{=l_t}.\end{aligned}$$

$\exists\ \varphi(x) \in C^\infty(\mathbb{R})$, define $\varphi_n(x) = \begin{cases} -1, & x \le 0 \\ \varphi(nx), & 0 < x < \frac{1}{n} \\ 1, & x \ge \frac{1}{n}, \end{cases}$ then $\varphi_n(x) \to \mathrm{sgn}(x),\ \forall\, x \neq 0$.

Let

$$f_n(x) = \begin{cases} -x, & x \le 0 \\ \int_0^x \varphi_n(t)\,\mathrm{d}t, & x > 0. \end{cases}$$

$$\implies f_n'(x) = \varphi_n(x.)$$

Because $f(x) = |x| = \int_0^{|x|} 1\,\mathrm{d}t$.

$$\implies |f(x) - f_n(x)| \le \int_0^{|x|} |\varphi_n(t) - 1|\,\mathrm{d}t \le \int_0^{\frac{1}{n}} |\varphi_n(t) - 1|\,\mathrm{d}t \le 2\frac{1}{n}.$$

Apply Itô formula to f_n to get that

$$f_n(X_t) = f_n(X_0) + \int_0^t f_n'(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t f_n''(X_s)\,\mathrm{d}[X]_s.$$

Note

$$\int_0^t f_n'(X_s)\,\mathrm{d}X_s = \int_0^t f_n'(X_s)\,\mathrm{d}M_s + \int_0^t f_n'(X_s)\,\mathrm{d}A_s,$$

and

$$\int_0^t f_n'(X_s)\,\mathrm{d}A_s = \int_0^t \varphi_n(X_s)\,\mathrm{d}A_s.$$

We have $\varphi_n(x) \uparrow \operatorname{sgn}(x)$, $\forall\, x$. By Monotone convergence,

$$\left|\int_0^t \varphi_n(X_s) - \operatorname{sgn}(X_s)\,\mathrm{d}A_s\right| \le \int_0^t |\varphi_n(X_s) - \operatorname{sgn}(X_s)|\,|\mathrm{d}A_s| \to 0,$$

which implies $\int_0^t \varphi_n(X_s)\,\mathrm{d}A_s \to \int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}A_s$ uniform in t.

Next, $T_k = \inf\{t \ge 0\colon\ |M_t| \ge k\}$. Then M^{T_k} is bounded.

We may assume M is bounded. Then

$$\left\|\int_0^\infty \big(\operatorname{sgn}(X_s) - f_n'(X_s)\big)\,\mathrm{d}M_s\right\|_2^2 = \mathbb{E}\int_0^\infty \big(\operatorname{sgn}(X_s) - f_n'(X_s)\big)^2\,\mathrm{d}[M]_s \to 0.$$

So

$$\mathbb{E}\left[\sup_{t\le T}\left|\int_0^t \big(\operatorname{sgn}(X_s) - f_n'(X_s)\big)\,\mathrm{d}M_s\right|^2\right] \xrightarrow{n\to\infty} 0.$$

Hence

$$\sup_{t\ge 0}\int_0^t \big(\operatorname{sgn}(X_s) - f_n'(X_s)\big)\,\mathrm{d}M_s \xrightarrow{a.s.} 0.$$

$\implies \int_0^t f_n'(X_s)\,\mathrm{d}X_s \xrightarrow{a.s.} \int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}X_s$, $f_n(X_t) \xrightarrow{a.s.} |X_t|$, $f_n(X_0) \xrightarrow{a.s.} |X_0|$.

$\implies C_t^n \triangleq \int_0^t \frac{1}{2} f_n''(X_s)\,\mathrm{d}[X_s]$, $C_t^n \xrightarrow{a.s.} l_t \triangleq |X_t| - |X_0| - \int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}X_s$.

$\mathrm{d}C_s^n = \frac{1}{2} f_n''(X_s)\,\mathrm{d}[X]_s \implies \forall\, \delta > 0$, if $\frac{1}{n} < \delta$, then

$$|X_s| > \delta > \frac{1}{n} \implies f_n''(X_s) = 0.$$

$$\implies \int_0^t 1_{\{|X_s|>\delta\}}\,\mathrm{d}C_s^n = 0.$$

Hence

$$\int_0^t 1_{\{|X_s|>\delta\}}\,\mathrm{d}l_s \le \lim_{n\to\infty}\int_0^t 1_{\{|X_s|>\frac{\delta}{2}\}}\,\mathrm{d}C_s^n = 0.$$

$$\int_0^t \varphi_\delta(X_s)\,\mathrm{d}l_s = \lim_{n\to\infty}\int_0^t \varphi_\delta(X_s)\,\mathrm{d}C_s^n \le \liminf_{n\to\infty}\int_0^t 1_{\{|X_s|>\frac{\delta}{2}\}}\,\mathrm{d}C_s^n.$$

We prove $\int_0^t 1_{\{|X_s|>\delta\}}\,\mathrm{d}l_s = 0$, $\forall\, \delta > 0 \implies \int_0^t 1_{\{|X_s|\neq 0\}}\,\mathrm{d}l_s = 0.$ $\square$

Let X, Y be continuous semimartingale. Then

$$\int_0^t Y_t \, \mathrm{d}X_t = \text{Itô Integral}.$$

Stratonovich Integral

$$\int_0^t Y_t \partial X_t \triangleq \int_0^t Y_t \, \mathrm{d}X_t + \frac{1}{2}[X, Y]_t.$$

Recall that

$$\begin{aligned} X_t Y_t - X_0 Y_0 &= \int_0^t X_s \, \mathrm{d}Y_s + \int_0^t Y_s \, \mathrm{d}X_s + [X, Y]_t \\ &= \int_0^t X_s \, \mathrm{d}Y_s + \frac{1}{2}[X, Y]_t + \int_0^t Y_s \, \mathrm{d}X_s + \frac{1}{2}[X, Y]_t \\ &= \int_0^t X_s \partial Y_s + \int_0^t Y_s \partial X_s. \end{aligned}$$

Theorem 1.8.2. $f(X_t) = f(X_0) + \int_0^t f'(X_s) \partial X_s$.

Proof. By Itô formula,

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) \, \mathrm{d}X_s + \frac{1}{2} \int_0^t f''(X_s) \, \mathrm{d}[X_s].$$

By definition,

$$\int_0^t f'(X_s) \partial X_s = \int_0^t f'(X_s) \, \mathrm{d}X_s + \frac{1}{2} \left[f'(X), X \right]_t.$$

$$\mathrm{d}f(X_t) = f'(X_t) \, \mathrm{d}X_t + \frac{1}{2} f''(X_t) \, \mathrm{d}[X]_t$$

$$\Longrightarrow \mathrm{d}f'(X_t) = f''(X_t) \, \mathrm{d}X_t + \frac{1}{2} f'''(X_t) \, \mathrm{d}[X]_t.$$

So

$$f'(X_t) = f'(X_0) + \int_0^t f''(X_s) \, \mathrm{d}X_s + \frac{1}{2} \int_0^t f'''(X_s) \, \mathrm{d}[X]_s.$$

$$\Longrightarrow \left[f'(X), X \right]_t = \left[\int_0^t f''(X_s) \, \mathrm{d}X_s, X_t \right] = \int_0^t f''(X_s) \, \mathrm{d}[X]_s.$$

($[H \cdot M, K \cdot N]_t = \int_0^t HK \, \mathrm{d}[M, N]_s$).

So

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) \, \mathrm{d}X_s + \frac{1}{2} \left[f'(X), X \right]_t = f(x_0) + \int_0^t f'(X_s) \partial X_s.$$

□

Theorem 1.8.3 (Riemam-Sum Approximation)**.** Let C be a locally bounded adapted L-process.

Let X be a continuous semimartingale. Then

$$\sum_{i=1}^{\infty} c(t_{i-1}^n)(X_{t_i^n} - X_{t_{i-1}^n}) \to \int_0^t c(s)\,\mathrm{d}X_s.$$

with $t_i^n = t \wedge \frac{i}{2^n}$. The convergence is uniform in probability on compacts, i.e. $\forall\ K > 0$,

$$\sup_{t \le K} |I^n(t) - I(t)| \xrightarrow{P} 0.$$

Proof. Let $c^n(t) = c(t_{i-1}^n)$ for $t \in (t_{i-1}^n, t_i^n]$. Then $c^n(t) \to c(t)$ for all $t \ge 0$.

Also,

$$\begin{aligned} I^n(t) &= \sum_{i=1}^{\infty} c(t_{i-1}^n)(X(t_i^n) - X(t_{i-1}^n)) \\ &= \int_0^t c^n(s)\,\mathrm{d}X_s \\ &= (c^n, X)_t = \int_0^t c^n(s)\,\mathrm{d}M_s + \int_0^t c^n(s)\,\mathrm{d}A_s. \end{aligned}$$

$\forall\ t \le k$, $|c^n(t)| \le c_k$

$$\implies \int_0^t c^n(s)\,\mathrm{d}A_s \to \int_0^t c(s)\,\mathrm{d}A_s.$$

By localization, we may assume M is bounded. If c is bounded and $M \in \mathcal{M}_0^2$, then

$$\int_0^t c^n(s)\,\mathrm{d}M_s \to \int_0^t c(s)\,\mathrm{d}M_s.$$

$$\mathbb{E}\left[\sup_{t \le k}\left|\int_0^t (c^n(s) - c(s))\,\mathrm{d}M_s\right|^2\right] \le c_k \mathbb{E}\int_0^k (c^n(s) - c(s))^2\,\mathrm{d}[M]_s \to 0.$$

□

1.8.1 Stratonovich Calculus

Let X, Y be continuous semimartingale. Define

$$\begin{aligned} S^n(t) &\equiv \sum_{i=1}^{n} \frac{1}{2}\left(Y(t_i^n) + Y(t_{i-1}^n)\right)\left(X(t_i^n) - X(t_{i-1}^n)\right) \\ S(t) &= \int_0^t Y_s \partial X_s. \end{aligned}$$

Then

$$\sup_{t \le k} |S^n(t) - S(t)| \xrightarrow{P} 0.$$

Proof.

$$\begin{aligned}
S^n(t) &= \sum_{i=1}^{\infty}\left[Y(t_{i-1}^n) + \frac{1}{2}(Y(t_i^n) - Y(t_{i-1}^n))\right](X(t_i^n) - X(t_{i-1}^n)) \\
&= \sum_{i=1}^{\infty} Y(t_{i-1}^n)(X(t_i^n) - X(t_{i-1}^n)) + \frac{1}{2}\sum_{i=1}^{\infty}\Delta Y(t_i^n)\Delta X(t_i^n).
\end{aligned}$$

It remains to prove

$$\begin{aligned}
&\sum_{i=1}^{\infty}\Delta Y(t_i^n)\Delta X(t_i^n) \to [X,Y]_t = \frac{1}{4}([X+Y]_t - [X-Y]_t) \\
&\Leftrightarrow \sum_{i=1}^{\infty}\Delta X(t_i^n)^2 \to [X]_t \\
&\Leftrightarrow \sum_{i=1}^{\infty}(\Delta M(t_i^n) + \Delta A(t_i^n))^2 \to [X]_t = [M]_t.
\end{aligned}$$

We have $\sum_{i=1}^{\infty}(\Delta M(t_i^n))^2 \to [M]_t$,

$$\left|\sum_{i=1}^{\infty}\Delta M(t_i^n)\Delta A(t_i^n)\right| \le \sup_{i\ge 1}|\Delta M(t_i^n)|\sum_{i=1}^{\infty}|\Delta A(t_i^n)| \le \sup_{\substack{t\le k\\ i\ge 1}}|\Delta M(t_i^n)|\operatorname{Var}_A(t) \xrightarrow{a.s.} 0.$$

□

1.9 Stochastic Differenctial Equation

$$\begin{aligned}
\mathrm{d}X_t &= b(t,X_t)\,\mathrm{d}t + \sigma(t,X_t)\,\mathrm{d}W_t, \\
X_t &= X_0 + \int_0^t b(s,X_s)\,\mathrm{d}s + \int_0^t \sigma(s,X_s)\,\mathrm{d}W_s.
\end{aligned}$$

d-dimension SDE ...

Definition 1.9.1 (Strong solution)**.** A strong solution of the SDE, on a given probability space $(\Omega,\mathcal{F},\mathsf{P})$ and a given B.M. $(W_t,\mathcal{F}_t^W)_{t\ge 0}$, with initial condition ξ is a process $\{X_t, t\ge 0\}$ with continuous sample path and

(i) $X\in\mathcal{F}_t$ with $(\mathcal{F}_t)_{t\ge 0}$ being the augmented filtration of the B.M. $(\mathcal{F}_t^W)_{t\ge 0}$.

(ii) $\mathsf{P}(X_0=\xi)=1$.

(iii)
$$\mathsf{P}\left(\int_0^t |b_i(s,X_s)| + \sigma_{ij}^2(s,X_s)\,\mathrm{d}s < \infty\right) = 1$$
for all i,j,t.

(iv) $X_t = X_0 + \int_0^t b(s,X_s)\,\mathrm{d}s + \int_0^t \sigma(s,X_s)\,\mathrm{d}W_s$.

Definition 1.9.2 (Strong Uniqueness)**.** Given $b(t,x)$ and $\sigma(t,x)$. Suppose $\forall$ B.M. $\{W_t\}_{t\geq 0}$ on some $(\Omega, \mathcal{F}, \mathsf{P})$ and ξ is the initial condition, and $\{\mathcal{F}_t\}$ is the augmented filtration, if X and $\tilde{X}$ are two strong solutions with respect to the sam W, then

$$\mathsf{P}\left(X_t = \tilde{X}_t, \forall\, t \geq 0\right) = 1.$$

Example . (i) $\mathrm{d}X_t = \mathrm{d}W_t \implies X_t = W_t + \xi_0$.

(ii) $\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + \mathrm{d}W_t$ where $b : [0,\infty) \times \mathbb{R} \to \mathbb{R}$ is bounded Borel-mble and $\forall\, x \leq y$, we have $b(t,x) \geq b(t,y)$, $\forall\, t \geq 0$. Fot this SDE, strong uniqueness holds.

Proof. Eg.(ii) $\forall$ B.M. on $(\Omega, \mathcal{F}, \mathsf{P})$, $\forall\, \xi$. If $X^{(1)}$ & $X^{(2)}$ are strong solutions, then

$$X_t^{(j)} = \xi + \int_0^t b(s, X_s^{(j)})\,\mathrm{d}s + W_t, \quad \forall\, j = 1, 2.$$

$$X_t^{(1)} - X_t^{(2)} = \int_0^t \left[b(s, X_s^{(1)}) - b(s, X_s^{(2)})\right] \mathrm{d}s.$$

$$\implies \mathrm{d}\Delta_t = \left[b(t, X_t^{(1)}) - b(t, X_t^{(2)})\right] \mathrm{d}t.$$

By Itô's formula,

$$\Delta_t^2 = \int_0^t 2\Delta_s\,\mathrm{d}\Delta_s + \frac{1}{2}\int_0^t 2\,\mathrm{d}[\Delta]_t = 2\int_0^t \left(X_s^{(1)} - X_s^{(2)}\right)\left[b(s, X_s^{(1)}) - b(s, X_s^{(2)})\right] \mathrm{d}s.$$

If $x \leq y$, then $b(s,x) - b(s,y) \geq 0$.
If $x > y$, then $b(s,x) - b(s,y) \leq 0$.
$\implies (x-y)(b(s,x) - b(s,y)) \leq 0$. So

$$\Delta_t^2 = 2\int_0^t \left(X_s^{(1)} - X_s^{(2)}\right)\left[b(s, X_s^{(1)}) - b(s, X_s^{(2)})\right] \mathrm{d}s \leq 0, \quad \text{a.s.}$$

$\implies \mathsf{P}\left(\Delta_t = 0, \forall\, t \geq 0\right) = 1.$
$\implies$ strong uniqueness. □

Theorem 1.9.1 (Itô Uniqueness Theorem)**.** Suppose $b(t,x)$ and $\sigma(t,x)$ are locally Lipschitz in the space variable: $\forall\, n \geq 1$, $\exists\, K_n > 0$ s.t. $\forall\, t \geq 0$, $\|x\|, \|y\| \leq n$, we have

$$\|b(t,x) - b(t,y)\| + \|\sigma(t,x) - \sigma(t,y)\| \leq K_n \|x - y\|.$$

Then strong uniqueness holds.

Lemma 1.9.1 (Gronwall's Inequality)**.** Suppose a continuous function $g(t)$ satisfies $0 \leq g(t) \leq \alpha(t) + \beta\int_0^t g(s)\,\mathrm{d}s$ for all $t \in [0,T]$ with $\beta \geq 0$ and $\alpha : [0,T] \to \mathbb{R}$ integrable. Then

$$g(t) \leq \alpha(t) + \beta\int_0^t \alpha(s) e^{\beta(t-s)}\,\mathrm{d}s, \quad \forall\, t \leq T.$$

Proof. $I(t) := \int_0^t g(s)\,\mathrm{d}s$.

$\implies I'(t) = g(t) \le \alpha(t) + \beta I(t)$.

$\implies I'(t) - \beta I(t) - \alpha(t) \le 0$.

Notice that

$$\left[e^{-\beta t} I(t)\right]' = \left(I'(t) - \beta I(t)\right) e^{-\beta t} \le \alpha(t) e^{-\beta t}.$$

$e^{-\beta t} I(t) - 0 = \int_0^t \left[e^{-\beta s} I(s)\right]' \mathrm{d}s \le \int_0^t \alpha(s) e^{-\beta s}\,\mathrm{d}s$.

$\implies I(t) \le \int_0^t \alpha(s) e^{\beta(t-s)}\,\mathrm{d}s$.

$\implies g(t) \le \alpha(t) + \beta I(t) = \alpha(t) + \beta \int_0^t \alpha(s) e^{\beta(t-s)}\,\mathrm{d}s$. □

Proof. of Itô Uniqueness Theorem.

Given W and $(\Omega, \mathcal{F}, \mathsf{P})$ and ξ. If X and $\tilde{X}$ are two strong solutions with respect to the same W. Define

$$\tau_n = \inf\{t \ge 0\colon \ \|X_t\| \ge n\}$$
$$\tilde{\tau}_n = \inf\{t \ge 0\colon \ \left\|\tilde{X}_t\right\| \ge n\}.$$

Set $\tau_n \wedge \tilde{\tau}_n$. Then $\lim_{n\to\infty} S_n = \infty$, P-a.s.

Note that

$$X_{t\wedge S_n} - \tilde{X}_{t\wedge S_n} = \int_0^{t\wedge S_n} \left[b(s, X_s) - b(s, \tilde{X}_s)\right] \mathrm{d}s + \int_0^{t\wedge S_n} \left[\sigma(s, X_s) - \sigma(s, \tilde{X}_s)\right]$$

$$\implies \mathbb{E}\left\|X_{t\wedge S_n} - \tilde{X}_{t\wedge S_n}\right\|^2 \le 4\mathbb{E}\left[\int_0^{t\wedge S_n} (b - b)\,\mathrm{d}s\right]^2 + 4\mathbb{E}\left[\int_0^{t\wedge S_n} (\sigma - \sigma)\,\mathrm{d}W_s\right]^2$$

.

$(X_t^1, X_t^2, \ldots, X_t^d) = \left(\int_0^t b^1\,\mathrm{d}s + \int_0^t \sigma^1\,\mathrm{d}W_s, \ldots, \int_0^t b^d\,\mathrm{d}s + \int_0^t \sigma^d\,\mathrm{d}W_s\right)$.

$$\begin{aligned} \implies \|X\|^2 &= \sum_{i=1}^d (X_t^j)^2 \\ &= \sum_{i=1}^d \left(\int_0^t b^j\,\mathrm{d}s + \int_0^t \sigma^j\,\mathrm{d}W_s\right)^2 \\ &\le 2\sum_{i=1}^d \left(\int_0^t b^j\,\mathrm{d}s\right)^2 + \left(\int_0^t \sigma^j\,\mathrm{d}W_s\right)^2 \end{aligned}$$

$$
\begin{aligned}
\mathbb{E}\left\|X_{t\wedge S_n}-\tilde{X}_{t\wedge S_n}\right\|^2 \leq & 2\sum_{i=1}^{d}\mathbb{E}\left(\int_0^{t\wedge S_n}\left(b_i(s,X_s)-b(s,\tilde{X}_s)\right)\mathrm{d}s\right)^2 \\
& +2\sum_{i=1}^{d}\mathbb{E}\left(\sum_{j=1}^{r}\int_0^{t\wedge S_n}\left(\sigma_{ij}(s,X_s)-\sigma_{ij}(s,\tilde{X}_s)\right)\mathrm{d}W_s^{(j)}\right)^2 \\
\leq & 2\sum_{i=1}^{d}\mathbb{E}\left[t\cdot\int_0^{t\wedge S_n}\left(b_i(s,X_s)-b_i(s,\tilde{X}_s)\right)^2\mathrm{d}s\right] \\
& +2\sum_{i=1}^{d}\sum_{j=1}^{r}\mathbb{E}\left[\int_0^{t\wedge S_n}(\sigma_{ij}-\sigma_{ij})\,\mathrm{d}W_s^{(j)}\right]^2
\end{aligned}
$$

And $\mathbb{E}\left[\int_0^{t\wedge S_n}(\sigma_{ij}-\sigma_{ij})\,\mathrm{d}W_s^{(j)}\right]^2=\mathbb{E}\left[\int_0^{t\wedge S_n}(\sigma_{ij}-\sigma_{ij})^2\,\mathrm{d}s\right]$.

We conclude that

$$
\begin{aligned}
\mathbb{E}\left\|X_{t\wedge S_n}-\tilde{X}_{t\wedge S_n}\right\|^2 \leq & 2t\mathbb{E}\int_0^{t\wedge S_n}\left\|b(s,X_s)-b(s,\tilde{X}_s)\right\|^2\mathrm{d}s+2\mathbb{E}\int_0^{t\wedge S_n}\left\|\sigma(s,X_s)-\sigma(s,\tilde{X}_s)\right\|^2\mathrm{d}s \\
\leq & 2T\mathbb{E}\int_0^{t\wedge S_n}K_n^2\left\|X_s-\tilde{X}_s\right\|^2\mathrm{d}s+2\mathbb{E}\int_0^{t\wedge S_n}K_n^2\left\|X_s-\tilde{X}_s\right\|^2\mathrm{d}s \\
\leq & 2(T+1)K_n^2\int_0^t\mathbb{E}\left\|X_{s\wedge S_n}-\tilde{X}_{s\wedge S_n}\right\|^2\mathrm{d}s,
\end{aligned}
$$

since

$$\|\sigma\|^2=\sum_{i=1}^{d}\sum_{j=1}^{r}\sigma_{ij}^2.$$

If we let $g(t) =\triangleq \mathbb{E}\left\|X_{t\wedge S_n}-\tilde{X}_{t\wedge S_n}\right\|^2$, then $g(t)\leq 2(T+1)K_n^2\int_0^t g(s)\,\mathrm{d}s$.

By Gronwall's inequality, we get $g(t)\leq 0$.

$\implies X_{t\wedge S_n}=\tilde{X}_{t\wedge S_n}$ a,s, $\forall\, t\geq 0$.

$\implies X_{t\wedge S_n}=\tilde{X}_{t\wedge S_n}$, $\forall\, t\geq 0$, a.s.

Let $S_n\uparrow\infty$ to conclude $X_t=\tilde{X}_t$, $\forall\, t\geq 0$, a.s. □

Theorem 1.9.2 (Existence)**.** Suppose $\exists\, K>0$ s.t. $\forall\, t\geq 0$, $\forall\, x,y\in\mathbb{R}^d$,

$$
\begin{cases}
\|b(t,x)-b(t,y)\|+\|\sigma(t,x)-\sigma(t,y)\|\leq K\|x-y\| \\
\|b(t,x)\|+\|\sigma(t,x)\|\leq K(1+\|x\|).
\end{cases}
$$

On $(\Omega,\mathcal{F},\mathsf{P})$, let ξ be R.V. and r-dimensional $B.M.$ $W=\{W_t\}_{t\geq 0}$ independent of ξ, and $\mathbb{E}\|\xi\|^2<\infty$. Let $(\mathcal{F}_t)_{t\geq 0}$ be the augmented filtration. Then $\exists$ continuous adapted process $X=\{X_t\}_{t\geq 0}$ which is a strong solution with respect to W and $X_0=\xi$ a.s.

Moreover, $\forall\, T>0$, $\exists\, C=C(K,T)>0$ s.t. $\forall\, 0\leq t\leq T$,

$$\mathbb{E}\|X_t\|^2\leq C(1+\mathbb{E}\|\xi\|^2)e^{Ct}.$$

Proof. Define $X_t^{(0)} = \xi$, $\forall\ t \geq 0$.

For any $k \geq 1$, define

$$X_t^{(k)} \triangleq \xi + \int_0^t b(s, X_s^{(k-1)})\,\mathrm{d}s + \int_0^t \sigma(s, X_s^{(k-1)})\,\mathrm{d}W_s.$$

Then $\forall\ T > 0$, $\exists\ C = C(K,T) > 0$ s.t.

$$\mathbb{E}\left\|X_t^{(k)}\right\|^2 \leq C(1 + \mathbb{E}\left\|\xi\right\|^2)e^{Ct}, \quad \forall\ t \leq T, \forall\ k \geq 0.$$

(Exercise)

For any $k \geq 0$,

$$X_t^{(k+1)} - X_t^{(k)} = \int_0^t \left(b(s, X_s^{(k)}) - b(s, X_s^{(k-1)})\right)\mathrm{d}s + \int_0^t \left(\sigma(s, X_s^{(k)}) - \sigma(s, X_s^{(k-1)})\right)\mathrm{d}W_s.$$

$$\begin{aligned}
\mathbb{E}\left[\sup_{s\leq t}\left\|B_s\right\|^2\right] &= \mathbb{E}\left\|\sup_{s\leq t}\int_0^s \left(b(u, X_u^{(k)}) - b(u, X_u^{(k-1)})\right)\mathrm{d}u\right\|^2 \\
&\leq \mathbb{E}\left[t \cdot \int_0^t \left\|b(u, X_u^{(k)}) - b(u, X_u^{(k-1)})\right\|^2 \mathrm{d}u\right] \\
&\leq K^2 + \int_0^t \mathbb{E}\left\|X_u^{(k)} - X_u^{(k-1)}\right\|^2 \mathrm{d}u.
\end{aligned}$$

$M_t = \int_0^t \sigma(s, X_s^{(k)}) - \sigma(s, X_s^{(k-1)})\,\mathrm{d}W_s$.

$[M]_t = \int_0^t \left\|\sigma(s, X_s^{(k)}) - \sigma(s, X_s^{(k-1)})\right\|^2 \mathrm{d}s$.

$\implies \mathbb{E}[M]_t \leq \int_0^t K^2\left(1 + \left\|X_s^{(k)}\right\| + \left\|X_s^{(k-1)}\right\|\right)^2 \mathrm{d}s < \infty$ by $\mathbb{E}\left\|X_t^{(k)}\right\|^2 \leq Ce^{Ct}$.

$$\begin{aligned}
\mathbb{E}\left[\sup_{s\leq t}\left\|M_s\right\|^2\right] &\leq C\mathbb{E}[M_t] \\
&= C\mathbb{E}\int_0^t \left\|\sigma(s, X_s^{(k)}) - \sigma(s, X_s^{k-1})\right\|^2 \mathrm{d}s \\
&\leq C\mathbb{E}\int_0^t K\left\|X_s^{(k)} - X_s^{(k-1)}\right\|^2 \mathrm{d}s.
\end{aligned}$$

Combine the two inequalities to get

$$\begin{aligned}
\mathbb{E}\left[\max_{s\leq t}\left\|X_s^{(k+1)} - X_s^{(k)}\right\|^2\right] &\leq 2\mathbb{E}\left[\max_{s\leq t}\left\|B_s\right\|^2\right] + 2\mathbb{E}\left[\max_{s\leq t}\left\|M_s\right\|^2\right] \\
&\leq C \cdot 4K^2(1+T)\int_0^t \mathbb{E}\left\|X_s^{(k)} - X_s^{(k-1)}\right\|^2 \mathrm{d}s \\
&\leq L\int_0^t \mathbb{E}\left(\left\|X_u^{(k)} - X_u^{(k-1)}\right\|^2\right)\mathrm{d}s.
\end{aligned}$$

When $k=1$, we have

$$\mathbb{E}\left(\left\|X_s^{(1)}-X_s^{(0)}\right\|^2\right)\le 2\mathbb{E}\left\|X_t^{(1)}\right\|^2+2\mathbb{E}\left\|X_t^{(0)}\right\|^2\le C^*$$

for all $0\le t\le T$.

Hence

$$\mathbb{E}\left(\max_{0\le s\le t}\left\|X_s^{(2)}-X_s^{(1)}\right\|^2\right)\le L\int_0^t C^*\,\mathrm{d}s=C^*(Lt).$$

$$\Longrightarrow\ \mathbb{E}\left(\max_{0\le s\le t}\left\|X_s^{(3)}-X_s^{(2)}\right\|^2\right)\le L\int_0^t C^*(Ls)\,\mathrm{d}s=C^*\frac{1}{2}(Lt)^2.$$

By induction, we can prove $\forall\ k\ge 1$,

$$\mathbb{E}\left(\max_{s\le t}\left\|X_s^{(k+1)}-X_s^{(k)}\right\|^2\right)\le C^*\frac{1}{k!}(Lt)^k.$$

By chebyshev's Inequality,

$$\mathsf{P}\left(\max_{0\le t\le T}\left\|X_t^{(k+1)}-X_t^{(k)}\right\|>\frac{1}{2^{k+1}}\right)\le 4^{k+1}C^*\frac{1}{k!}(LT)^k.$$

$\sum_{k=1}^{\infty}\mathsf{P}(\max\|\ldots\|>\frac{1}{2^{k+1}})<\infty$.

$\Longrightarrow$ B.C. Lemma gives

$$\mathsf{P}\left(\max_{t\le T}\left\|X_t^{(k+1)}-X_t^{(k)}\right\|>\frac{1}{2^{k+1}}\text{ i.o.}\right)=0.$$

$\Longrightarrow$ P-a.s. $\exists\ N(\omega)<\infty$ s.t. $\forall\ k\ge N$,

$$\max_{t\le T}\left\|X_t^{(k+1)}-X_t^{(k)}\right\|\le\frac{1}{2^{k+1}}.$$

$\Longrightarrow$ $\forall\ k\ge N$, $\forall\ m\ge 1$, we have

$$\max_{t\le T}\left\|X_t^{(k+m)}-X_t^{(k)}\right\|\le\sum_{j=k}^{k+m-1}\frac{1}{2^{j+1}}\le\frac{1}{2^k}.$$

Hence $\left\{X_t^{(k)}(\omega),0\le t\le T\right\}$ is a Cauchy sequence in $\|\ldots\|_{sup}$.
Hence $\exists$ limit $\{X_t(\omega),0\le t\le T\}$ s.t.

$$\lim_{k\to\infty}\max_{t\le T}\left\|X_t^{(k)}-X_t\right\|=0.$$

Since $T>0$ is arbitrary, $\exists\ X=\{X_t,0\le t<\infty\}$ s.t. $\forall\ T>0$, $\sup_{t\le T}\left\|X_t^{(k)}-X_t\right\|\to 0$ as $k\to\infty$.

$$\begin{aligned}\mathbb{E}\left\|X_t\right\|^2 &= \mathbb{E}\lim_{k\to\infty}\left\|X_t^{(k)}\right\|^2 \\ &\le \lim_{k\to\infty}\mathbb{E}\left\|X_t^{(k)}\right\|^2 \le C(1+\mathbb{E}\left\|\xi\right\|^2)e^{Ct}.\end{aligned}$$

$\implies \mathsf{P}\left(\int_0^t |b(s,X_s)|\,\mathrm{d}s + \int_0^t \sigma_{ij}^2(s,X_s)\,\mathrm{d}s < \infty\right) = 1.$

$$X_t^{(k+1)} = \xi + \int_0^t b(s, X_s^{(k)})\,\mathrm{d}s + \int_0^t \sigma(s, X_s^{(k)})\,\mathrm{d}W_s.$$

$$X_t = \xi + \int_0^t b(s, X_s)\,\mathrm{d}s + \int_0^t \sigma(s, X_s)\,\mathrm{d}W_s.$$

Need to prove $X_t^{(k+1)} \xrightarrow{k\to\infty} X_t$ a.s. □

Theorem 1.9.3 (Yamada-Watanabe)**.** Let $\mathrm{d}X_t = \sigma(X_t)\,\mathrm{d}W_t + b(X_t)\,\mathrm{d}t$, where

$$\sigma(x) = \begin{pmatrix} \sigma_1(x_1) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_n(x_) \end{pmatrix}$$

$$b(x) = (b^1(x), b^2(x), \ldots, b^n(x))$$

such that

(i) $\exists$ positive increasing function $\rho(u)$, $u \in (0,\infty)$ s.t.

$$|\sigma_i(\xi) - \sigma_i(\eta)| \le \rho\left(|\xi - \eta|\right)$$

for all $\xi, \eta \in \mathbb{R}^1$, $i = 1, \ldots, n$ and

$$\int_{0^+} \rho^{-2}(u)\,\mathrm{d}u = \infty.$$

i.e. $\forall\, \varepsilon > 0$, $\int_0^\varepsilon \rho^{-2}(u)\,\mathrm{d}u = \infty$.

(ii) $\exists$ positive increasing concave function $k(u)$, $u \in (0,\infty)$ s.t.

$$|b_i(x) - b_j(y)| \le k\left(|x - y|\right)$$

for all $x, y \in \mathbb{R}^n$, $i = 1, \ldots, n$ and $\int_{0^+} k^{-1}(u)\,\mathrm{d}u = \infty$.

Then strong uniqueness holds.

Remark. If $\rho(u) = u^\alpha$, then strong uniqueness holds if $\alpha \ge \frac{1}{2}$; fails if $0 < \alpha < \frac{1}{2}$.

Proposition 1.9.1. (Yamada & Watanabe 1971)

"$d = 1$" Consider

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + \sigma(t, X_t)\,\mathrm{d}W_t.$$

Assume that $\exists$ constant $K > 0$ and $h : [0, \infty) \to [0, \infty)$ is a strictly increasing function with $h(0) = 0$ such that

$$\begin{cases} |b(t,x) - b(t,y)| \leq K\,|x - y| \\ |\sigma(t,x) - \sigma(t,y)| \leq h(|x - y|) \end{cases}$$

hold for all $t \geq 0$ and $x, y \in \mathbb{R}$ with

$$\int_0^\varepsilon h(u)^{-2}\,\mathrm{d}u = \infty, \quad \forall\, \varepsilon > 0.$$

Then strong uniqueness holds.

Proof. Idea: Constant a C^2 function $\psi_n(x)$ s.t. $\psi_n(x) \uparrow |x|$. Let $a_0 = 1 > a_1 > \cdots > a_n \downarrow 0$ such that

$$\begin{aligned} &\int_{a_1}^{a_0} h(u)^{-2}\,\mathrm{d}u = 1, \\ &\int_{a_2}^{a_1} h(u)^{-2}\,\mathrm{d}u = 2, \\ &\cdots \\ &\int_{a_n}^{a_{n-1}} h(u)^{-2}\,\mathrm{d}u = n, \end{aligned}$$

So

$$\int_{a_n}^{a_{n-1}} \frac{2}{n} h(x)^{-2}\,\mathrm{d}x = 2.$$

Then $\exists\ \rho_n(x)$ s.t.

$$\int_{a_n}^{a_{n-1}} \rho_n(x)\,\mathrm{d}x = 1$$

and $\rho_n(x) = 0$, $\forall\, x \notin (a_n, a_{n-1})$, and $x \mapsto \rho_n(x)$ is continuous.

$$\implies \int_0^\infty \rho_n(x)\,\mathrm{d}x = 1.$$

Let $\psi_n'(x) = \int_0^x \rho_n(y)\,\mathrm{d}y$, $\forall\, x \geq 0$.

Let $\psi_n(x) = \int_0^x \psi_n'(y)\,\mathrm{d}y$ for $x \geq 0$ and

$$\psi_n(x) = \psi_n(-x), \quad \forall\, x < 0.$$

One can check that $\psi_n(x) \uparrow |x|$, $\forall\, x \in \mathbb{R}$.

Summary:

(i) $\psi_n(x) \to |x|$, $\forall\, x \in \mathbb{R}$,

(ii) $|\psi_n'(x)| \le 1$, $\forall\, x \in \mathbb{R}$,

(iii) $|\psi_n''(x)| = |\rho_n(x)| \le \frac{2}{n} h(x)^{-2}$.

Suppose $X^{(1)}$ & $X^{(2)}$ are two strong solutions with $X_0^{(1)} = X_0^{(2)}$ a.s.

Define

$$\tau_N = \inf\left\{t \ge 0\colon\ \int_0^t \sigma(s, X_s^{(1)})^2 \,\mathrm{d}s \ge N \text{ or } \int_0^t \sigma(s, X_s^{(2)})^2 \,\mathrm{d}s \ge N\right\}.$$

$$\implies \mathbb{E}\int_0^{t\wedge\tau_N} \sigma(s, X_s^{(i)})^2 \,\mathrm{d}s \le N$$

for $i = 1, 2$.

We may assume that

$$\mathbb{E}\int_0^t \sigma(s, X_s^{(i)})^2 \,\mathrm{d}s < \infty,$$

for $\forall\, t \ge 0$, $i = 1, 2$, due to $\tau_N \uparrow \infty$ a.s.

Otherwise we may prove

$$X_s^{(1)} = X_s^{(2)}, \quad \forall\, 0 \le s \le \tau_N \text{ a.s.} \implies X_s^{(1)} = X_s^{(2)}, \quad \forall\, s \ge 0.$$

Define $\Delta_t = X_t^{(1)} - X_t^{(2)}$.

$$\begin{aligned} \implies \mathrm{d}\Delta_t &= \mathrm{d}X_t^{(1)} - \mathrm{d}X_t^{(2)} \\ &= \left[\sigma(t, X_t^{(1)}) - \sigma(t, X_t^{(2)})\right] \mathrm{d}W_t + \left[b(t, X_t^{(1)}) - b(t, X_t^{(2)})\right] \mathrm{d}t. \end{aligned}$$

Apply Itô's formula to get

$$\begin{aligned} \psi_n(\Delta_t) &= \psi_n(\Delta_0) + \int_0^t \psi_n'(\Delta_s)\,\mathrm{d}\Delta_s + \frac{1}{2}\int_0^t \psi_n''(\Delta_s)\,\mathrm{d}[\Delta]_s \\ &= \psi_n(0) + \int_0^t \psi_n'(\Delta_s)[\sigma - \sigma]\,\mathrm{d}W_s + \int_0^t \psi_n'(\Delta_s)[b - b]\,\mathrm{d}s + \frac{1}{2}\int_0^t \psi_n''(\Delta_s)[\sigma - \sigma]^2\,\mathrm{d}s. \end{aligned}$$

Take expectation to see that

$$\begin{aligned} \mathbb{E}\psi_n(\Delta_t) &= \mathbb{E}\int_0^t \psi_n'(\Delta_s)\left[b(t, X_t^{(1)}) - b(t, X_t^{(2)})\right] \mathrm{d}s + \mathbb{E}\int_0^t \frac{1}{2}\psi_n''(\Delta_s)[\sigma - \sigma]^2\,\mathrm{d}s \\ &\le \mathbb{E}\int_0^t K\left|X_s^{(1)} - X_s^{(2)}\right| \mathrm{d}s + \mathbb{E}\int_0^t \frac{1}{2}\frac{2}{n} h(|\Delta_s|)^{-2} \cdot h(\left|X_s^{(1)} - X_s^{(2)}\right|)^2\,\mathrm{d}s \\ &= K\int_0^t \mathbb{E}\left|X_s^{(1)} - X_s^{(2)}\right| \mathrm{d}s + \frac{1}{n}t. \end{aligned}$$

Let $n \to \infty$ to get

$$\mathbb{E}\,|\Delta_t| \le K\int_0^t \mathbb{E}\,|\Delta_s|\,\mathrm{d}s.$$

By Gronwall's Inequality,

$$\mathbb{E}\,|\Delta_t| = 0 \implies X_t^{(1)} = X_t^{(2)} \text{ a.s..}$$

By continuity,

$$X_t^{(1)} = X_t^{(2)}, \quad \forall\, t \geq 0 \text{ a.s..}$$

□

Proposition 1.9.2. (R & W)

If $Y = X - X'$, and $h\colon [0,\infty) \to [0,\infty)$ is increasing s.t.

$$\int_0^\varepsilon h(x)^{-2}\,\mathrm{d}x = \infty, \quad \forall\, \varepsilon > 0,$$

then $l_\cdot^0(Y) = 0$ if

$$\int_0^t h(Y_s)^{-2} 1_{\{Y_s>0\}}\,\mathrm{d}[Y]_s < \infty$$

a.s. $\forall\, t > 0$.

Here $l_t^0(Y)$ is the local time of Y at 0

(Recall $|Y_t| = |Y_0| + \int_0^t \operatorname{sgn}(Y_s)\,\mathrm{d}Y_s + l_t^0(Y)$)

Proof.

$$l_t^a \text{``}=\text{''} \int_0^t \delta_a(Y_s)\,\mathrm{d}[Y]_s.$$

Then $\forall$ function $f\colon\ :\mathbb{R} \to \mathbb{R}$,

$$\begin{aligned}\int_0^t f(Y_s)\,\mathrm{d}[Y]_s &= \int_0^t \mathrm{d}[Y]_s \int_{-\infty}^\infty f(a)\delta_a(Y_s)\,\mathrm{d}a \\ &= \int_{-\infty}^\infty f(a)\,\mathrm{d}a \int_0^t \delta_a(Y_s)\,\mathrm{d}[Y]_s \\ &= \int_{-\infty}^\infty f(a) l_t^a(Y)\,\mathrm{d}a.\end{aligned}$$

So

$$\int_0^t h(Y_s)^{-2} 1_{\{Y_s>0\}}\,\mathrm{d}[Y]_s = \int_{-\infty}^\infty h(a)^{-2} 1_{\{a>0\}} l_t^a\,\mathrm{d}a = \int_0^\infty h(a)^{-2} l_t^a\,\mathrm{d}a.$$

If $l_t^a(Y) > 0$, then by the right-continuity of $a \mapsto l_t^a(Y)$, we get $\int_t^a(Y) \geq \varepsilon_0$ for some $\varepsilon_0 > 0$ and $0 < a < \varepsilon_0$.

$$\implies \int_0^\infty h(a)^{-2} l_t^a\,\mathrm{d}a \geq \int_0^{\varepsilon_0} h(a)^{-2}\varepsilon_0\,\mathrm{d}a = \infty.$$

Contradiction.

Hence we conclude that $l_t^0(Y) = 0$, $\forall\, t \geq 0$, a.s. □

$$\textbf{Assumption: } \begin{cases} |b(t,x)-b(t,y)| \le K\,|x-y| \\ |\sigma(t,x)-\sigma(t,y)| \le h(|x-y|) \end{cases} \quad \text{and } \int_0^\varepsilon h(x)^{-2}\,\mathrm{d}x = \infty,\ \forall\, \varepsilon > 0.$$

Proof. Second proof of (Y-W 1971) (Le Gall)

Since (Recall $\mathrm{d}Y_s = \mathrm{d}X_s^{(1)} - \mathrm{d}X_s^{(2)}$)

$$\begin{aligned} \int_0^t h(Y_s)^{-2}\mathbf{1}_{\{Y_s>0\}}\,\mathrm{d}[Y]_s &= \int_0^t h(Y_s)^{-2}\mathbf{1}_{\{Y_s>0\}}(\sigma-\sigma)^2\,\mathrm{d}s \\ &\le \int_0^t h(Y_s)^{-2}\mathbf{1}_{\{Y_s>0\}}h(\left|X_s^{(1)}-X_s^{(2)}\right|)^2\,\mathrm{d}s \le t < \infty \text{ a.s..} \end{aligned}$$

$\Longrightarrow\ l_t^0(Y) = 0$ a.s.

Thus

$$|Y_t| = |Y_0| + \int_0^t \operatorname{sgn}(Y_s)\,\mathrm{d}Y_s + l_t^0(Y).$$

$$\Longrightarrow\ \left|X_t^{(1)}-X_t^{(2)}\right| = \int_0^t \operatorname{sgn}(X_s^{(1)}-X_s^{(2)})(b-b)\,\mathrm{d}s + \int_0^t \operatorname{sgn}(X_s^{(1)}-X_s^{(2)})(\sigma-\sigma)\,\mathrm{d}W_s.$$

$$\Longrightarrow\ \mathbb{E}\left|X_t^{(1)}-X_t^{(2)}\right| \le \int_0^t \mathbb{E}K\left|X_s^{(1)}-X_s^{(2)}\right|\mathrm{d}s.$$

By Gronwall's, the proof is done. □

Theorem 1.9.4 (Comparision Principle)**.** Given $(\Omega,\mathcal{F},\mathsf{P})$ and a B.M. $\{W_t,\mathcal{F}_t,t\ge 0\}$. If two continuous adapted process $X^{(1)},X^{(2)}$ satisfy that

$$\begin{cases} X_t^{(1)} = X_0^{(1)} + \int_0^t b_1(s,X_s^{(1)})\,\mathrm{d}s + \int_0^t \sigma(s,X_s^{(1)})\,\mathrm{d}W_s, \\ X_t^{(2)} = X_0^{(2)} + \int_0^t b_2(s,X_s^{(2)})\,\mathrm{d}s + \int_0^t \sigma(s,X_s^{(2)})\,\mathrm{d}W_s. \end{cases}$$

Assume

(1) $\sigma(t,x)$, $b_j(t,x)$ are continuous, real-values functions on $[0,\infty)\times\mathbb{R}$.

(2) $|\sigma(t,x)-\sigma(t,y)| \le h(|x-y|)$ with

$$\int_0^\varepsilon h(x)^{-2}\,\mathrm{d}x = \infty.$$

(3) $X_0^{(1)} \le X_0^{(2)}$.

(4) $b_1(t,x) \le b_2(t,x)$.

(5) either $b_1(t,x)$ or $b_2(t,x)$ satisfy $|b(t,x)-b(t,y)| \le K\,|x-y|$.

Then $\mathsf{P}\left(X_t^{(1)} \le X_t^{(2)}, \forall\, t\ge 0\right) = 1$.

Proof. Assume $|b_1(t,x) - b_1(t,y)| \le K\,|x-y|$. Again we assume

$$\mathbb{E}\int_0^t \sigma(s, X_s^{(j)})^2 \, \mathrm{d}s < \infty, \quad \forall\, t \ge 0.$$

Let $\psi_n(x)$ be as before. Define

$$\varphi_n(x) = \psi_n(x) 1_{(0,\infty)}(x).$$

$$\mathbb{E}[\varphi_n(\Delta_t)] \le \mathbb{E}\int_0^t \varphi_n'(\Delta_s)\left[b_1(s, X_s^{(1)}) - b_2(s, X_s^{(2)})\right] \mathrm{d}s + \frac{t}{n}$$

where $\Delta_t = X_t^{(1)} - X_t^{(2)}$.

So

$$\begin{aligned}
\mathbb{E}\varphi_n(\Delta_t) - \frac{t}{n} &\le \mathbb{E}\int_0^t \varphi_n'(\Delta_s)\left[b_1(s, X_s^{(1)}) - b_2(s, X_s^{(2)})\right] \mathrm{d}s \\
&\le \mathbb{E}\int_0^t K\left|X_s^{(1)} - X_s^{(2)}\right| \varphi_n'(\Delta_s)\,\mathrm{d}s \\
&= \mathbb{E}\int_0^t K\Delta_s^+ \varphi_n'(\Delta_s^+)\,\mathrm{d}s \\
&\le \mathbb{E}\int_0^t K\Delta_s^+\,\mathrm{d}s,
\end{aligned}$$

since $\varphi_n'(\Delta_s) = 0$ if $\Delta_s \le 0$. $\varphi_n'(\Delta_s) > 0$ if $\Delta_s > 0$.

By $\psi_n(x) \to |x|$, $\varphi_n(x) = \psi_n(x)1_{(0,\infty)}(x) \to x^+$.

Let $n \to \infty$ to get

$$\mathbb{E}\Delta_t^+ \le K\int_0^t \mathbb{E}\Delta_s^+\,\mathrm{d}s.$$

By Gronwall, $\mathbb{E}\Delta_t^+ = 0 \implies \mathsf{P}\left(X_t^{(1)} \le X_t^{(2)}\right) = 1$, $\forall\, t > 0$. □

Definition 1.9.3 (Weak Solution)**.** A weak solution of SDE is a triple (X, W), $(\Omega, \mathcal{F}, \mathsf{P})$, $\{\mathcal{F}_t\}$ where

(1) $(\Omega, \mathcal{F}, \mathsf{P})$ probability space, $\{\mathcal{F}_t\}$ is filtration.

(2) $(X_t, \mathcal{F}_t)$ is continuous and adapted $(W_t, \mathcal{F}_t)$ is an r-dim B.M.

(3) $\mathsf{P}\left(\int_0^t b_j(s, X_s) + \sigma_{ij}(s, X_s)^2\,\mathrm{d}s < \infty, \quad \forall\, t \ge 0\right) = 1.$

(4) $X_t = X_0 + \int_0^t b(s, X_s)\,\mathrm{d}s + \int_0^t \sigma(s, X_s)\,\mathrm{d}W_s.$

Definition 1.9.4 (Pathwise Uniqueness)**.** Suppose whenever (X, W), $(\Omega, \mathcal{F}, \mathsf{P})$, $\{\mathcal{F}_t\}$ and $(\tilde{X}, W)$, $(\Omega, \mathcal{F}, \mathsf{P})$, $\{\tilde{\mathcal{F}}_t\}$ are two weak solutions with $X_0 = \tilde{X}_0$ a.s., if

$$\mathsf{P}\left(X_t = \tilde{X}_t, \quad \forall\, t \ge 0\right) = 1,$$

then pathwise uniqueness holds.

Definition 1.9.5 (Weak Uniqueness (Uniqueness in law))**.** If for any two weak solutions (X, W), $(\Omega, \mathcal{F}, \mathsf{P})$, $\{\mathcal{F}_t\}$ and $(\tilde{X}, \tilde{W})$, $(\tilde{\Omega}, \tilde{\mathcal{F}}_t, \tilde{\mathsf{P}})$, $\{\tilde{\mathcal{F}}_t\}$ with same initial distribution $\forall\ B \in B(\mathbb{R}^d)$,

$$\mathsf{P}(X_0 \in B) = \tilde{\mathsf{P}}(\tilde{X}_0 \in B).$$

We get X and $\tilde{X}$ have the same law, i.e. $\forall\ t_0 \le t_1 \le \cdots \le t_n$,

$$\mathsf{P}\left(X_{t_0} \in B_0, \ldots, X_{t_n} \in B_n\right) = \tilde{\mathsf{P}}\left(\tilde{X}_{t_0} \in B_0, \ldots, X_{t_n} \in B_n\right)$$

for all $B_i \in B(\mathbb{R}^d)$, then weak uniqueness holds.

Remark. (Tanaka SDE)

$$X_t = \int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}W_s$$

with $\operatorname{sgn}(x) = \begin{cases} 1, & x > 0 \\ -1, & x \le 0 \end{cases}$.

Weak Uniqueness: If (X, W), $(\Omega, \mathcal{F}, \mathsf{P})$, $\{\mathcal{F}_t\}$ is a weak solution, then the process (X_t) is a continuous, square-integrable martingale with

$$[X]_t = \int_0^t \operatorname{sgn}(X_s)^2\,\mathrm{d}s = t.$$

So X is a B.M. $\implies$ Weak Uniqueness holds.

However, $(-X, W)$, $(\Omega, \mathcal{F}, \mathsf{P})$, $\{\mathcal{F}_t\}$ is also a weak solution.

Because

$$X_t = \int_0^t \operatorname{sgn}(X_t)\,\mathrm{d}W_s \implies -X_t = \int_0^t -\operatorname{sgn}(X_s)\,\mathrm{d}W_s = \int_0^t \operatorname{sgn}(-X_s)\,\mathrm{d}W_s.$$

$$\int_0^t (-\operatorname{sgn}(X_t) - \operatorname{sgn}(-X_s))\,\mathrm{d}W_s \sim \int_0^t (-\operatorname{sgn}(X_s) - \operatorname{sgn}(-X_s))^2\,\mathrm{d}s.$$

Notice that $-\operatorname{sgn}(x) - \operatorname{sgn}(-x) = 0$ for all $x \neq 0$ and

$$\mathsf{P}\left(X_s \neq 0,\ \text{for a.e. } s \ge 0\right) = 1.$$

W.P.1

$$\int_0^t (-\operatorname{sgn}(X_s) - \operatorname{sgn}(-X_s))^2\,\mathrm{d}s = 0 \implies \int_0^t -\operatorname{sgn}(X_s)\,\mathrm{d}W_s = \int_0^t \operatorname{sgn}(-X_s)\,\mathrm{d}W_s \text{ a.s.}$$

$\implies -X_t = \int_0^t \operatorname{sgn}(-X_s)\,\mathrm{d}W_s$.

$\implies (-X, W)$ is also a weak solution.

$$\implies \mathsf{P}(X_t \neq -X_t,\ \text{for a.e. } t \ge 0) = 1.$$

Hence pathwise uniqueness fails.

Weak Existence: (X,W), $(\Omega,\mathcal{F},\mathsf{P})$, $\{\mathcal{F}_t\}$.

Construction: Given $(\Omega,\mathcal{F},\mathsf{P})$ and a B.M. $X=\{X_t,\mathcal{F}_t^X,t\geq 0\}$. Let $\mathring{\mathcal{F}}_t^X$ =augmention of $\mathcal{F}_t^X$.

Assume $\mathsf{P}(X_0=0)=1$.

Define $W_t \triangleq \int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}X_s$.

Then W_t is a B.M. with respect to $\mathring{\mathcal{F}}_t^X$.

Also
$$\int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}W_s=\int_0^t \operatorname{sgn}(X_s)\operatorname{sgn}(X_s)\,\mathrm{d}X_s=\int_0^t 1\,\mathrm{d}X_s=X_t.$$

Hence (X,W), $(\Omega,\mathcal{F},\mathsf{P})$, $\mathring{\mathcal{F}}_t^X$ is a weak solution.

No strong solution.

Given $(\Omega,\mathcal{F},\mathsf{P})$ and a B.M. W, if X is a strong solution, then

$$X_t\in\mathcal{F}_t^W\subseteq\mathring{\mathcal{F}}_t^W \implies \mathcal{F}_t^X\subseteq\mathring{\mathcal{F}}_t^W.$$

By X is a B.M., by Tanaka's formula,

$$|X_t|=|X_0|+\int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}X_s+L_t^X(0).$$

By SDE, we get
$$\int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}X_s=\int_0^t \operatorname{sgn}(X_s)^2\,\mathrm{d}W_s=W_t.$$
$$\implies W_t=|X_t|-L_t^X(0).$$
$$L_t^X(0)=\int_0^t \delta_0(X_s)\,\mathrm{d}s=\lim_{\varepsilon\downarrow 0}\frac{1}{2\varepsilon}\int_0^t \mathsf{1}_{|X_s|\leq\varepsilon}\,\mathrm{d}s=\lim_{\varepsilon\downarrow 0}\frac{1}{2\varepsilon}\left|\{0\leq s\leq y\colon\ |X_s|\leq\varepsilon\}\right|.$$

Hence $W_t=|X_t|-\lim_{\varepsilon\downarrow 0}\frac{1}{2\varepsilon}\left|\{0\leq s\leq t\colon\ |X_s|\leq\varepsilon\}\right|\in\mathcal{F}_t^{|X|}$.

$\implies \mathring{\mathcal{F}}_t^W\subseteq\mathring{\mathcal{F}}_t^{|X|}$.

$\implies \mathring{\mathcal{F}}_t^X\subseteq\mathring{\mathcal{F}}_t^{|X|}$. Contradiction.

$$\begin{cases}\text{strong uniqueness} \implies \text{weak uniqueness}\\ \text{weak existence}+\text{pathwise uniqueness} \implies \text{strong existence.}\end{cases}$$

1.9.1 Girsanov Theorem

Given $(\Omega,\mathcal{F},\mathsf{P})$ and a d-dim B.M. $W=\{W_t,\mathcal{F}_t,t\geq 0\}$ with $\mathsf{P}(W_0=0)=1$. Let $X=\{X_t,\mathcal{F}_t,t\geq 0\}$ be a d-dim measurable, adapted process such that $\forall\, T\geq 0$, $\forall\, 1\leq i\leq d$, $\mathsf{P}\left(\int_0^T (X_t^{(i)})^2\,\mathrm{d}t<\infty\right)=1$. Then
$$\int_0^t X_s^{(i)}\,\mathrm{d}W_s^{(i)}$$

is well-defined. Set

$$Z_t(X) = \exp\left(\int_0^t X_s \,\mathrm{d}W_s - \frac{1}{2}\int_0^t \|X_s\|^2 \,\mathrm{d}s\right).$$

Assume $Z(X)$ is a martingale i.e. $\mathbb{E}Z_t(X) = 1$, $\forall\, t \geq 0$.

Define a probability measure $\tilde{\mathsf{P}}_T$ for any $T > 0$ by

$$\tilde{\mathsf{P}}_T(A) = \mathbb{E}\left(Z_T(X) \cdot 1_A\right)$$

for all $A \in \mathcal{F}_T$, $\frac{\mathrm{d}\tilde{\mathsf{P}}_T}{\mathrm{d}\mathsf{P}} = Z_T(X)$.

Remark. If $A \in \mathcal{F}_t \subseteq \mathcal{F}_T$ for $0 \leq t \leq T$, we must

$$\tilde{\mathsf{P}}_T(A) = \mathbb{E}\left(1_A Z_T(X)\right),$$
$$\tilde{\mathsf{P}}_t(A) = \mathbb{E}\left(1_A Z_t(X)\right).$$

Note that

$$\mathbb{E}(1_A Z_T(X)) = \mathbb{E}\left[\mathbb{E}\left(1_A Z_T(X) \mid \mathcal{F}_t\right)\right] = \mathbb{E}\left[1_A \mathbb{E}\left(Z_T(X) \mid \mathcal{F}_t\right)\right] = \mathbb{E}\left[1_A Z_t(X)\right].$$

Hence $\tilde{\mathsf{P}}_T|_{\mathcal{F}_t} = \tilde{\mathsf{P}}_t$.

Theorem 1.9.5 (Girsanov Theorem)**.** Assume $Z(X)$ is a martingale. Define $\tilde{W} = \{\tilde{W}_t, t \geq 0\}$ by $\tilde{W}_t^{(i)} = W_t^{(i)} - \int_0^t X_s^{(i)} \,\mathrm{d}s$ for $1 \leq i \leq d$, and $t \geq 0$. Then for each fixed $T > 0$, $\{\tilde{W}_t, \mathcal{F}_t, 0 \leq t \leq T\}$ is a d-dim B.M. on $(\Omega, \mathcal{F}_T, \tilde{\mathsf{P}}_T)$.

Lemma 1.9.2. Fix $T > 0$. If $0 \leq s \leq t \leq T$ and $Y \in \mathcal{F}_t$ with $\tilde{E}_T|Y| < \infty$. Then

$$\tilde{E}_T\left(Y \mid \mathcal{F}_s\right) = \frac{1}{Z_s}\mathbb{E}\left(Y Z_t \mid \mathcal{F}_s\right)$$

a.s. $\tilde{\mathsf{P}}_T$ and P.

$\tilde{\mathsf{P}}_T << \mathsf{P}$ and $\mathsf{P} << \tilde{\mathsf{P}}_T$. $\tilde{\mathsf{P}}_T(A) = \mathbb{E}\left(1_A Z_T(X)\right)$.

If $\mathsf{P}(A) = 0 \implies \tilde{\mathsf{P}}_T(A) = 0$.

If $\tilde{\mathsf{P}}_T(A) = 0 \implies \mathsf{P}(A) = 0.(Z_T(X) > 0$ a.s.).

Proof. It suffices to prove $\forall\, A \in \mathcal{F}_s$,

$$\tilde{E}_T(1_A Y) = \tilde{E}_T(1_A \frac{1}{Z_s}\mathbb{E}(Y Z_t \mid \mathcal{F}_s)).$$

$$\begin{aligned} RHS &= \tilde{E}_s(1_A \frac{1}{Z_s}\mathbb{E}(YZ_t \mid \mathcal{F}_s)) \\ &= \mathbb{E}(1_A \mathbb{E}(YZ_t \mid \mathcal{F}_s)) \\ &= \mathbb{E}(1_A Y Z_t) \\ &= \tilde{E}_t(1_A Y) \\ &= \tilde{E}_T(1_A Y). \end{aligned}$$

□

Proposition 1.9.3. Define $M_T^{c,loc} = \{M \colon M \text{ is a continuous local martingale on } (\Omega, \mathcal{F}_T, \mathsf{P})\}$.
$\tilde{M}_T^{c,loc} = \{\text{continuous local martingale on } (\Omega, \mathcal{F}_T, \tilde{\mathsf{P}}_T)\}$.

(1) If $M \in M_T^{c,loc}$, then

$$\tilde{M}_t \triangleq M_t - \sum_{i=1}^{d} \int_0^t X_s^{(i)} \,\mathrm{d}[M, W^{(i)}]_s$$

for $0 \le t \le T$ is in $\tilde{M}_T^{c,loc}$.

(2) If $N \in M_T^{c,loc}$ and

$$\tilde{N}_t = N_t - \sum_{i=1}^{d} \int_0^t X_s^{(i)} \,\mathrm{d}[N, W^{(i)}]_s,$$

then

$$[\tilde{M}, \tilde{N}]_t = [M, N]_t$$

for all $0 \le t \le T$, a.s. P and $\tilde{\mathsf{P}}_T$.

Proof. of Girsanov Theorem:
If $M = W^{(j)}$ with $1 \le j \le d$, then

$$\tilde{M}_t = W_t^{(j)} - \int_0^t X_s^{(j)} \,\mathrm{d}W_s^{(j)} \in \tilde{M}_T^{c,loc}$$

$$\implies \tilde{W}_t^{(j)} \in \tilde{M}_T^{c,loc}.$$

Set $N = W^{(k)} \implies \tilde{N}_t = \tilde{W}_t^{(k)} \in \tilde{M}_T^{c,loc}$ and $[\tilde{M}, \tilde{N}]_t = [M, N]_t = \delta_{jk} t$.
$\implies [\tilde{W}^{(j)}, \tilde{W}^{(k)}]_t = \delta_{jk} t$
$\implies \tilde{W}$ is a d-dim B.M.. □

Proof. of Prop. Assume M, N are bounded martingale with bounded quadrastic variation.

$$\left[\tau_n^M = \inf\{t \ge 0 \colon |M_t| \ge n \text{ or } [M]_t \ge n\}\right]$$

Consdier $t \le \tau_n^M \wedge \tau_n^N$.
Assume also that $Z_t(X)$ and $\sum_{i=1}^d \int_0^t (X_s^{(j)})^2 \,\mathrm{d}s$ are bounded for all $t \ge 0$ a.s.

By Kunita & Watanabe inequality

$$\begin{aligned}\left|\int_0^t X_s^{(j)}\,\mathrm{d}[M,W^{(j)}]_s\right|^2 &\le \int_0^t (X_s^{(j)})^2\,\mathrm{d}[W^{(j)}]_s\cdot\int_0^t 1^2\,\mathrm{d}[M]_s\\ &= \int_0^t (X_s^{(j)})^2\,\mathrm{d}s\cdot[M]_t\\ &\le C \quad \text{a.s..}\end{aligned}$$

Hence $\tilde{M}_t$ is also bounded for all $t\ge 0$. To prove $\tilde{M}\in\tilde{M}_T^{c,loc}$, we only need to show that

$$\tilde{E}_T(\tilde{M}_t\mid\mathcal{F}_s)=\tilde{M}_s \quad \text{a.s.}$$

$$LHS = (\text{by Lemma}) = \frac{1}{Z_s(X)}\mathbb{E}(\tilde{M}_t Z_t(X)\mid\mathcal{F}_s).$$

By IBP, we get

$$\tilde{M}_t Z_t(X)=\tilde{M}_0 Z_0(X)+\int_0^t \tilde{M}_s\,\mathrm{d}Z_s(X)+\int_0^t Z_s(X)\,\mathrm{d}\tilde{M}_s+[\tilde{M},Z]_t.$$

Recall $Z_t(X)=e^{\int_0^t X_s\,\mathrm{d}W_s-\frac{1}{2}\int_0^t\|X_s\|^2\,\mathrm{d}s}$,
we have

$$\mathrm{d}Z_t(X)=Z_t(X)X_t\cdot\mathrm{d}W_t=\sum_{i=1}^d Z_t(X)X_t^{(i)}\,\mathrm{d}W_t^{(i)}.$$

Recall $\mathrm{d}\tilde{M}_t=\mathrm{d}M_t-\sum_{i=1}^d X_t^{((i))}\,\mathrm{d}[M,W^{(i)}]_t$.
So

$$\begin{aligned}&Z_t(X)\tilde{M}_t\\ &=\int_0^t \tilde{M}_s\sum_{i=1}^d Z_s(X)X_s^{(i)}\,\mathrm{d}W_s^{(i)}\\ &\quad+\int_0^t Z_s(X)\left[\mathrm{d}M_s-\sum_{i=1}^d X_s^{(i)}\,\mathrm{d}[M,W^{(i)}]_s\right]+\int_0^t Z_s(X)X_s^{(i)}\,\mathrm{d}[M,W^{(i)}]_s\\ &=\int_0^t \tilde{M}_s\sum_{i=1}^d Z_s(X)X_s^{(i)}\,\mathrm{d}W_s^{(i)}+\int_0^t Z_s(X)\,\mathrm{d}M_s.\end{aligned}$$

Hence $\tilde{M}_t Z_t(X)$ is a martingale.

$$\Longrightarrow\ LHS=\frac{1}{Z_s(X)}\tilde{M}_s Z_s(X)=RHS.$$

(2) $\mathrm{d}\tilde{N}_t=\mathrm{d}N_t=\sum_{i=1}^d X_t^{(i)}\,\mathrm{d}[N,W^{(i)}]_t$.

$\mathrm{d}\tilde{M}_t = \mathrm{d}M_t - \sum_{i=1}^d X_t^{(i)}\,\mathrm{d}[M, W^{(i)}]_t$.

$$\begin{aligned}\tilde{M}_t\tilde{N}_t =& \tilde{M}_0\tilde{N}_0 + \int_0^t \tilde{M}_s\,\mathrm{d}\tilde{N}_s + \int_0^t \tilde{N}_s\,\mathrm{d}\tilde{M}_s + [\tilde{M},\tilde{N}]_t \\ =& 0 + \int_0^t \tilde{M}_s\left(\mathrm{d}W_s - \sum_{i=1}^d X_s^{(i)}\,\mathrm{d}[N,W^{(i)}]_s\right) \\ &+ \int_0^t \tilde{N}_s\left(\mathrm{d}M_s - \sum_{i=1}^d X_s^{(i)}\,\mathrm{d}[M,W^{(i)}]_s\right) + [M,N]_t.\end{aligned}$$

Then

$$\begin{aligned}\tilde{M}_t\tilde{N}_t - [M,N]_t = \int_0^t \tilde{M}_s\,\mathrm{d}N_s + \int_0^t \tilde{N}_s\,\mathrm{d}M_s - \sum_{i=1}^d \int_0^t X_s^{(i)}\tilde{M}_s\,\mathrm{d}[N,W^{(i)}]_s \\ - \sum_{i=1}^d \int_0^t X_s^{(i)}\tilde{N}_s\,\mathrm{d}[M,W^{(i)}]_s.\end{aligned}$$

Next

$$\begin{aligned}&Z_t(X)\left[\tilde{M}_t - \tilde{N}_t - [M,N]_t\right] \\ =& \int_0^t Z_s(X)\left[\tilde{M}_s\,\mathrm{d}N_s + \tilde{N}_s\,\mathrm{d}M_s - \sum_{i=1}^d X_s^{(i)}\left(\tilde{M}_s\,\mathrm{d}[N,W^{(i)}]_s + \tilde{N}_s\,\mathrm{d}[M,W^{(i)}]_s\right)\right] \\ &+ \int_0^t (\tilde{M}_s\tilde{N}_s - [M,N]_s)\sum_{i=1}^d Z_s(X)X_s^{(i)}\,\mathrm{d}W_s^{(i)} \\ &+ \int_0^t \sum_{i=1}^d Z_s(X)X_s^{(i)}\left(\tilde{M}_s\,\mathrm{d}[N,W^{(i)}]_s + \tilde{N}_s\,\mathrm{d}[M,W^{(i)}]_s\right).\end{aligned}$$

Therefore

$$\tilde{E}_T(\tilde{M}_t\tilde{N}_t - [M,N]_t \mid \mathcal{F}_s) = \frac{1}{Z_s(X)}\mathbb{E}(Z_t(X)(\tilde{M}_t\tilde{N}_t - [M,N]_t) \mid \mathcal{F}_s) = (\tilde{M}_s\tilde{N}_s - [M,N]_s).$$

This proves $[\tilde{M},\tilde{N}]_t = [M,N]_t$. □

Weak solutions constructed by means of the Girsanov Theorem.

Proposition 1.9.4. Consider the SDE

$$\mathrm{d}X_t = \mathrm{d}W_t + b(t,X_t)\,\mathrm{d}t$$

for $0 \le t \le T$, W is B.M. and $b(t,x)$ is Borel-measurable with

$$\|b(t,x)\| \le K(1+\|x\|)$$

for all $0 \le t \le T$, $x \in \mathbb{R}^d$.

Then for any probability measure μ on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ the SDE has a weak solution (X, W), $(\Omega, \mathcal{F}, \mathsf{P})$, $\{\mathcal{F}_t\}$ such that

$$\mathsf{P}(X_0 \in \cdot) = \mu.$$

Proof. Let $X = \{X_t, \mathcal{F}_t, t \ge 0\}$ be a B.M. on $(\Omega, \mathcal{F})$ with law $(\mathsf{P}^X)_{X \in \mathbb{R}^d}$ s.t.

$$\mathsf{P}^X(X_0 = x) = 1.$$

By corollary 3.5.16, by assuming

$$\|b(t,x)\| \le K(1 + \|x\|)$$

we get

$$Z_t \triangleq \exp\left(\sum_{i=1}^{d} \int_0^t b_j(s, X_s)\,\mathrm{d}X_s^{(j)} - \frac{1}{2}\int_0^t \|b(s, X_s)\|^2\,\mathrm{d}s\right)$$

is a martingale.

By Girsanov Theorem, if we define

$$Q^X(A) = \mathsf{P}^X(A \cdot Z_T)$$

for all $A \in \mathcal{F}_T$, then

$$W_t \triangleq X_t - X_0 - \int_0^t b(s, X_s)\,\mathrm{d}s$$

is a B.M. with $Q^X(W_0 = 0) = 1$.

Then $X_t = X_0 + W_t + \int_0^t b(s, X_s)\,\mathrm{d}s$ for all $0 \le s \le T$.

Note that $\mathsf{P}^X(X_0 = x) = 1$.

Then $Q^X(X_0 = x) = 1$.

Define $Q^\mu(A) = \int_{\mathbb{R}^d} Q^X(A)\mu(\mathrm{d}x)$.

Then $Q^\mu(W_0 = 0) = 1$ and

$$\begin{aligned} Q^\mu(X_0 \in A) &= \int_{\mathbb{R}^d} Q^X(X_0 \in A)\mu(\mathrm{d}x) \\ &= \int_{\mathbb{R}^d} 1_{\mathsf{x}\in\mathsf{A}}\mu(\mathrm{d}x) = \mu(A). \end{aligned}$$

So under Q^μ we get $X_0 \sim \mu$ and (X, W), $(\Omega, \mathcal{F}, Q^\mu)$, $\{\mathcal{F}_t\}$.

$\square$

(1) pathwise uniqueness $\implies$ weak uniqueness.

Proof. If

$$\begin{cases}(X^{(1)},W^{(1)})\\(\Omega_1,\mathcal{F}_1,v_1)\\(\mathcal{F}_t^{(1)})_{t\geq 0},\end{cases}\qquad\begin{cases}(X^{(2)},W^{(2)})\\(\Omega_2,\mathcal{F}_2,v_2)\\(\mathcal{F}_t^{(2)})_{t\geq 0},\end{cases}$$

are two weak solutions.

If $v_1(X_0^{(1)}\in\cdot)=v_2(X_0^{(2)}\in\cdot)$, then we will show that

$$v_1(X^{(1)}\in\cdot)=v_2(X^{(2)}\in\cdot).$$

In fact, we will prove that if

$$Y^{(j)}=X^{(j)}-X_0^{(j)},$$

then

$$v_1((X_0^{(1)},W^{(1)},Y^{(1)})\in A)=v_2((X_0^{(2)},W^{(2)},Y^{(2)})\in A)$$

for all $A\in\mathcal{B}(\Theta)$ with $\Theta=\mathbb{R}^d\times C[0,\infty)^r\times C[0,\infty)^d$ and $\mathcal{B}(\Theta)=\mathcal{B}(\mathbb{R}^d)\otimes\mathcal{B}(C[0,\infty)^r)\otimes\mathcal{B}(C[0,\infty)^d)$.

Let $\mu(\cdot)=v_1(X_0^{(1)}\in\cdot)=v_2(X_0^{(2)}\in\cdot)$.

Let P_* denote the wiener measure on $C[0,\infty)^r$.

Define $(\Omega,\mathcal{F})$ by $\Omega=\Theta\times C[0,\infty)^d$, $\mathcal{F}=\mathcal{B}(\Theta)\otimes\mathcal{B}(C[0,\infty)^d)$.

For each $\omega\in\Omega$, we write $\omega=(x,w,y_1,y_2)$.

Define

$$\mathsf{P}(\mathrm{d}\omega)=\mu(\mathrm{d}x)P_*(\mathrm{d}w)Q_1(x,w,\mathrm{d}y_1)Q_2(x,w,\mathrm{d}y_2),$$

where $Q_j(x,w,y_i)$ is the regular conditional probability s.t.

(1) $\forall\ x\in\mathbb{R}^d$, $w\in C[0,\infty)^r$, $Q_j(x,w,\cdot)$ is a probability measure on $(C[0,\infty)^d,\mathcal{B}(C[0,\infty)^d))$.

(2) $\forall\ F\in\mathcal{B}(C[0,\infty)^d)$, the map $(x,w)\to Q_j(x,w;F)$ is measurable w.r.t $\mathcal{B}(\mathbb{R}^d)\otimes\mathcal{B}(C[0,\infty)^r)$.

(3) $\forall\ G\in\mathcal{B}(\mathbb{R}^d)\otimes\mathcal{B}(C[0,\infty)^r)$, $\forall\ F\in\mathcal{B}(C[0,\infty)^d)$,

$$v_j(X_0^{(j)},W^{(j)}\in G,Y^{(j)}\in F)=\int_G\mu(\mathrm{d}x)P_*(\mathrm{d}w)Q_j(x,w;F)$$

i.e. $Q_j(x,w;F)=v_j(Y^{(j)}\in F\mid X_0^{(j)}=x,W^{(j)}=w)$.

Let $G_t\triangleq\sigma((x,w(s),y_1(s),y_2(s))\colon x\in\mathbb{R}^d,0\leq s\leq t)$.

Let $F_t\triangleq$ augmention of σ_t.

Now we conclude that

$$\begin{aligned}\mathsf{P}\left(\{\omega\in\Omega\colon(x,w,y_1)\in A\}\right)&=\int\mathsf{P}(\mathrm{d}\omega)1_{(x,w,y_1)\in A}\\&=\int_A\mu(\mathrm{d}x)P_*(\mathrm{d}w)Q_1(x,w;\mathrm{d}y_1)\\&=v_1((X_0^{(1)},W^{(1)},Y^{(1)})\in A).\end{aligned}$$

So

$$(x + y_1, w) \stackrel{d}{=} (X^{(1)}, X^{(1)})$$

and

$$(x + y_2, w) \stackrel{d}{=} (X^{(2)}, W^{(2)}).$$

pathwise uniqueness implies

$$\mathsf{P}(x + y_1(t) = x + y_2(t), \forall\ t \geq 0) = 1.$$

$$\begin{aligned} v_1((X_0^{(1)}, W^{(1)}, Y^{(1)}) \in A) &= \mathsf{P}(\{\omega \in \Omega \colon (x, w, y_1) \in A\}) \\ &= \mathsf{P}(\{\omega \in \Omega \colon (x, w, y_2) \in A\}) \\ &= v_2((X_0^{(2)}, W^{(2)}, Y^{(2)}) \in A). \end{aligned}$$

□

1.10 Harmonic Function

$\forall$ smooth function $f : \mathbb{R}^d \to \mathbb{R}$,

$$\frac{\Delta}{2} f(x) = \lim_{t \downarrow 0} \frac{E^x[f(W_t) - f(x)]}{t}.$$

We call $\frac{\Delta}{2}$ the generator of Brownian mothion.

$$\Delta u \triangleq \sum_{i=1}^{d} \frac{\partial^2 u}{\partial x_i^2}.$$

Proof.

$$f(W_t) = f(W_0) + \int_0^t \sum_{i=1}^{d} \frac{\partial f}{\partial x_i}(W_s)\,\mathrm{d}W_s^{(i)} + \frac{1}{2}\int_0^t \sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_i^2}(W_s)\,\mathrm{d}s.$$

So

$$\frac{1}{t} E^x[f(W_t) - f(W_0)] = \frac{1}{2}\frac{1}{t}\int_0^t E[\Delta f(W_s)]\,\mathrm{d}s.$$

Let $t \downarrow 0$, we get the limit is $\frac{1}{2}\Delta f(x)$. □

A function u defined on some open subset $D \subseteq \mathbb{R}^d$ is called harmonic if $u \in C^2(D)$ and $\Delta u = 0$.
Let $\{W_t, \mathcal{F}_t, 0 \leq t < \infty\}$, $(\Omega, \mathcal{F})$ and $\{\mathsf{P}^x\}_{x \in \mathbb{R}^d}$ is a d-dim B.M. such that

$$\mathsf{P}^x(W_0 = x) = 1.$$

For any $D \subseteq \mathbb{R}^d$ open, we set $\tau_D = \inf\{t \geq 0 : W_t \notin D\}$.
If D is bounded, then

$$\mathsf{P}^x(\tau_D < \infty) = 1, \quad \forall\ x \in D,$$

and $B_{\tau_D} \in \partial D$ with $\partial D =$ topological boundary, $\overline{D} = D \cup \partial D$ is the closure of D.

Let $B_r = \{x \in \mathbb{R}^d : \|x\| < r\}$, $B_r(x) = \{y \in \mathbb{R}^d : \|y - x\| < r\}$ be the open ball centered at $0/x$ with radius $r > 0$. Volume

$$V_r = \frac{2r^d \pi^{\frac{d}{2}}}{d\Gamma(\frac{d}{2})}.$$

Surface area

$$S_r \triangleq \frac{2r^{d-1}\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})}.$$

Define $\mu_r(\mathrm{d}x) = \mathsf{P}^0(W_{\tau_{B_r}} \in \mathrm{d}x)$. Then μ_r is a measure on ∂B_r.

By rotational invariance μ_r is a uniform measure on ∂B_r. In particular, we have

$$\mu_r = \frac{\sigma_r}{S_r}$$

where $\sigma_r(\mathrm{d}x)$ =surface measure on ∂B_r.

Recall that for any function f,

$$\int_{B_r} f(x)\,\mathrm{d}x = \int_0^t \mathrm{d}\rho \int_{\partial B_\rho} f(x)\sigma_\rho(\mathrm{d}x) = \int_0^r \mathrm{d}\rho \int_{\partial B_\rho} f(x) S_\rho \mu_\rho(\mathrm{d}x).$$

Definition 1.10.1 (Mean Value Property (MVP))**.** A function $u : D \to \mathbb{R}$ has MVP if $\forall\ a \in D$, $\forall\ r > 0$ with $B_r(a) \subseteq D$, we have

$$u(a) = \int_{\partial B_r} \mu(a + x) u_r(\mathrm{d}x) = \int_{\partial B_r(a)} u(x)\mu_r(\mathrm{d}x).$$

Assume MVP, we have

$$\int_{B_r(a)} u(x)\,\mathrm{d}x = \int_0^t \mathrm{d}\rho \int_{\partial B_\rho(a)} u(x) S_\rho \mu_\rho(\mathrm{d}x) = \int_0^r \mathrm{d}\rho \cdot u(a) S_\rho = V_r \cdot u(a).$$

Then

$$u(a) = \frac{1}{V(r)} \int_{B_r(a)} u(x)\,\mathrm{d}x.$$

Proposition 1.10.1. If u is harmonic on D, then u has MVP. And if u has MVP on D, then u is harmonic & $u \in C^\infty$.

Proof. (1). Assume $\Delta u = 0$, $\forall\ x \in D$. Then $\forall\ r > 0$ with $\overline{B_r(a)} \subseteq D$, $\forall\ a \in D$.

By Itô's formula

$$u(W_{t\wedge\tau_{B_r(a)}}) = u(W_0) + \int_0^{t\wedge\tau_{B_r(a)}} \nabla u(W_s)\,\mathrm{d}W_s + \frac{1}{2}\int_0^{t\wedge\tau_{B_r(a)}} \Delta u(W_s)\,\mathrm{d}s, \quad \forall\ t \geq 0.$$

By $\nabla u(x)$ is bounded on $\overline{B_r(a)}$, we get

$$E\left[\int_0^{t\wedge\tau_{B_r(a)}} \nabla u(W_0)\,\mathrm{d}W_s\right].$$

Hence

$$E^a\left[u(W_{t\wedge\tau_{B_r(a)}})\right] = u(a).$$

Let $t\to\infty$ to get

$$\begin{aligned} u(a) = \lim_{t\to\infty} E^a\left[u(W_{t\wedge\tau_{B_r(a)}})\right] &= E^a\left[u(W_{\tau_{B_r(a)}})\right] \\ &= E^0\left[u(W_{\tau_{B_r}}+a)\right] \\ &= \int_{\partial B_r} u(x+a)\mu_r(\mathrm{d}x). \end{aligned}$$

(2) If u has MVP, then $\forall\ a\in D$, $\forall\ r>0$ with $\overline{B_r(a)}\subseteq D$,

$$u(a) = \int_{\partial B_r(a)} u(x)\mu_r(\mathrm{d}x).$$

For any $\varepsilon>0$, define $g_\varepsilon:\mathbb{R}^d\to\mathbb{R}$ by

$$g_\varepsilon(x) = \begin{cases} c(\varepsilon)e^{-\frac{1}{\varepsilon^2-\|x\|^2}}, & \|x\|<\varepsilon \\ 0, & \|x\|\ge\varepsilon. \end{cases}$$

where $c(\varepsilon)>0$ is a constant s.t.

$$\int_{\mathbb{R}^d} g_\varepsilon(x)\,\mathrm{d}x = \int_{B_\varepsilon} g_\varepsilon(x)\,\mathrm{d}x = 1.$$

Mollifier. For any $a\in D$ and $\varepsilon>0$ with $\overline{B_\varepsilon(a)}\subseteq D$, define

$$u_\varepsilon(a) \triangleq \int_{B_\varepsilon} u(a+x)g_\varepsilon(x)\,\mathrm{d}x = \int_{\mathbb{R}^d} u(y)g_\varepsilon(y-a)\,\mathrm{d}y.$$

Then $u_\varepsilon(a)\in C^\infty$.

$$\begin{aligned} \lim_{\delta\downarrow 0}\frac{u_\varepsilon(a+\delta)-u_\varepsilon(a)}{\delta} &= \lim_{\delta\downarrow 0}\int_{\mathbb{R}^d} u(y)\frac{g_\varepsilon(y-a-\delta)-g_\varepsilon(y-a)}{\delta}\,\mathrm{d}y \\ &= -\int_{\mathbb{R}^d} u(y)\left(\frac{\mathrm{d}}{\mathrm{d}x}g_\varepsilon(x)\Big|_{x=y-a}\right)\mathrm{d}y. \end{aligned}$$

And

$$
\begin{aligned}
u_\varepsilon(a) &= c(\varepsilon)\int_{B_\varepsilon} u(a+x)e^{-\frac{1}{\varepsilon^2-\|x\|^2}}\,\mathrm{d}x \\
&= c(\varepsilon)\int_0^\varepsilon S_\rho\,\mathrm{d}\rho\int_{\partial B_\rho} u(a+x)e^{-\frac{1}{\varepsilon^2-\rho^2}}\mu_\rho(\mathrm{d}x) \\
&= c(\varepsilon)\int_0^\varepsilon S_\rho\,\mathrm{d}\rho\cdot e^{-\frac{1}{\varepsilon^2-\rho^2}}u(a) \\
&= u(a)\int_0^\varepsilon S_\rho c(\varepsilon)e^{-\frac{1}{\varepsilon^2-\rho^2}}\,\mathrm{d}\rho \\
&= u(a)\int_{B_\varepsilon} c(\varepsilon)e^{-\frac{1}{\varepsilon^2-\|x\|^2}}\,\mathrm{d}x = u(0).
\end{aligned}
$$

So $u(a) = u_\varepsilon(0)$ for all $a \in D$ with $\overline{B_\varepsilon(a)} \subseteq D$.
Then $u \in C^\infty(D)$.
Next, we will prove $\Delta u = 0$. $\forall\ a \in D$, $\exists\ \varepsilon > 0$ s.t. $\overline{B_\varepsilon(a)} \subseteq D$. By Taylor's expansion at a,

$$u(a+y) = u(a) + \sum_{i=1}^d y_i\frac{\partial u}{\partial x_i}(a) + \frac{1}{2}\sum_{i=1}^d\sum_{j=1}^d y_iy_j\frac{\partial^2 u}{\partial x_i\partial x_j}(a) + o(\|u\|^2), \quad y \in \overline{B}_\varepsilon.$$

By symmetry,

$$\int_{\partial B_\varepsilon} y_i\mu_\varepsilon(\mathrm{d}y) = 0, \quad \int_{\partial B_\varepsilon} y_iy_j\mu_\varepsilon(\mathrm{d}y) = 0, \quad \forall\ i \neq j.$$

Now integrate $u(a+y)$ over ∂B_ε to obtain

$$\int_{\partial B_\varepsilon} u(a+y)\mu_\varepsilon(\mathrm{d}y) = u(a) + 0 + \int_{\partial B_\varepsilon}\frac{1}{2}\Delta u(a)\sum_{i=1}^d y_i^2\mu_\varepsilon(\mathrm{d}y) + o(\varepsilon^2).$$

For

$$\frac{1}{2}\sum_{i=1}^d\int_{\partial B_\varepsilon}\frac{\partial^2 u}{\partial x_i^2}(a)\cdot y_i^2\mu_\varepsilon(\mathrm{d}y).$$

Notice that

$$\int_{\partial B_\varepsilon} y_i^2\mu_\varepsilon(\mathrm{d}y) = \frac{1}{d}\sum_{i=1}^d\int_{\partial B_\varepsilon} y_i^2\mu_\varepsilon(\mathrm{d}y) = \frac{1}{d}\int_{\partial B_\varepsilon}\|y\|^2\mu_\varepsilon(\mathrm{d}y) = \frac{1}{d}\varepsilon^2\cdot 1 = \frac{1}{d}\varepsilon^2.$$

Hence

$$\frac{1}{2}\sum_{i=1}^d\int_{\partial B_\varepsilon}\frac{\partial^2 u}{\partial x_i^2}(a)\cdot y_i^2\mu_\varepsilon(\mathrm{d}y) = \frac{1}{2}\sum_{i=1}^d\frac{\partial^2 u}{\partial x_i^2}(a)\cdot\frac{\varepsilon^2}{d} = \frac{1}{2}\Delta u(a)\cdot\frac{\varepsilon^2}{d}.$$

Now we conclude

$$u(a) = \int_{\partial B_\varepsilon} u(a+y)\mu_\varepsilon(\mathrm{d}y) = u(a) + \frac{\Delta}{2}u(a)\cdot\frac{\varepsilon^k}{d} + o(\varepsilon^2).$$

So $0 = \frac{\Delta}{2}u(a)\cdot\frac{1}{d} + o(1)$, be letting $\varepsilon \downarrow 0$ to get $\Delta u(a) = 0$.

□

1.10.1 Dirichlet Problem

Given an open set $D \subseteq \mathbb{R}^d$, and a continuous function f on ∂D, the Dirichlet probability (D.f) is to find $u \in C(\overline{D})$ s.t.

$$\begin{cases} \Delta u = 0, & \text{in } D \\ u = f, & \text{on } \partial D. \end{cases}$$

$$\frac{\partial u(t,x)}{\partial t} = \frac{\Delta}{2} u(t,x).$$

Heat Equation $u(t,x)$ =temperature at $x \in D$ at time t.

Existence.

Let $u(x) = E^x[f(W_{\tau_D})]$ for all $x \in \overline{D}$, provided that $E^x\,|f(W_{\tau_D})| < \infty$, $\forall\ x \in D$. Then for $x \in \partial D$, we get

$$\mathsf{P}^x(\tau_D = 0) = 1$$

by $\tau_D = \inf\{t \geq 0 : W_t \notin D\}$. Then

$$\mathsf{P}^x(W_{\tau_D} = x) = 1,$$

which implies $E^x[f(W_{\tau_D})] = f(x)$.

So $u(x) = f(x)$, $\forall\ x \in \partial D$.

Next, we will prove $\Delta u = 0$, $\forall\ x \in D \Leftrightarrow u$ has MVP.

For all $a \in D$ and $r > 0$ s.t. $\overline{B_r(a)} \subseteq D$, we have

$$\begin{aligned} u(a) = E^a[f(W_{\tau_D})] &= E^a\left[E^a\left[f(W_{\tau_D}) \,\middle|\, \mathcal{F}_{\tau_{B_r(a)}}\right]\right] \\ (\tau_{B_r(a)} \leq \tau_D) \ &= E^a\left[E^{W_{\tau_{B_r(a)}}}[f(W_{\tau_D})]\right] \\ &= E^a\left[u(W_{\tau_{B_r(a)}})\right] \\ &= \int_{\partial B_r} u(a+y)\mu_r(\mathrm{d}y). \end{aligned}$$

Therefore u has MVP. It remains to prove $u \in C(\overline{D})$ i.e. $\forall\ a \in \partial D$

$$\lim_{\substack{x \to a \\ x \in D}} u(x) = u(a) = f(a).$$

$\forall\ a \in \partial D$, we need that

$$\lim_{\substack{x \to a \\ x \in D}} E^x[f(W_{\tau_D})] = f(a).$$

Uniqueness.

If f is bdd and $\mathsf{P}^a(\tau_D < \infty) = 1$, $\forall\ a \in D$. Then any bounded solution u to Dirichlet Problem (D,f) is given by $u(a) = E^a[f(W_{\tau_D})]$.

Proof. Let $u \in C(\overline{D})$ be bounded and

$$\begin{cases} \Delta u = 0, & \text{in } D \\ u = f, & \text{on } \partial D. \end{cases}$$

For each $n \geq 1$, we let

$$\begin{aligned} D_n &= \{x \in D : \inf_{y \in \partial D} \|x - y\| > \frac{1}{n}\} \\ &= \{x \in D : d(x, \partial D) > \frac{1}{n}\}. \end{aligned}$$

By Itô's formula

$$u(W_{t\wedge\tau_{D_n}\wedge\tau_{B_n}}) = u(W_0) + \sum_{i=1}^{d} \int_0^{t\wedge\tau_{D_n}\wedge\tau_{B_n}} \frac{\partial u}{\partial x_i}(W_s)\, \mathrm{d}W_s^{(i)}.$$

Since $\frac{\partial u}{\partial x_i}(W_s)$ is bounded for all $0 \leq s \leq t \wedge \tau_{B_n} \wedge \tau_{D_n}$, we get

$$E\left[\int_0^{t\wedge\tau_{D_n}\wedge\tau_{B_n}} \frac{\partial u}{\partial x_i}(W_s)\, \mathrm{d}W_s^{(i)}\right] = 0.$$

Then

$$u(a) = E^a\left[u(W_{t\wedge\tau_{B_n}\wedge\tau_{D_n}})\right].$$

By $\mathsf{P}^a(\tau_D < \infty) = 1$, we have $(t \to \infty, n \to \infty)$

$$u(W_{t\wedge\tau_{B_n}\wedge\tau_{D_n}}) \xrightarrow{a.s.} u(W_{\tau_D}) = f(W_{\tau_D}).$$

So by bounded convergence theorem,

$$u(a) = \lim_{n\to\infty} \lim_{t\to\infty} E^a\left[u(W_{t\wedge\tau_{B_n}\wedge\tau_{D_n}})\right] = E^a(f(W_{\tau_D})).$$

□

To prove $\forall\ a \in \partial D$,

$$\lim_{\substack{x\to a \\ x\in D}} E^x\left[f(W_{\tau_D})\right] = f(a.)$$

Define $\sigma_D = \inf\{t > 0\colon W_t \notin D\}$. It is obvious that $\tau_D \leq \sigma_D$.

Definition 1.10.2 (regular point)**.** We say a point $a \in \partial D$ is regular for D if $\mathsf{P}^a(\sigma_D = 0) = 1$.

$\sigma_D = \frac{1}{2} \Leftrightarrow \forall\ 0 < t < \frac{1}{2}$, $W_t \in D$.
$\sigma_D = 0 \Leftrightarrow \exists\ t_n \downarrow 0$ s.t. $W_{t_n} \notin D$.

When $d = 1$, $D = (a, b)$. Then $\partial D = \{a, b\}$. By LIL, a and b are both regular points.

$$\limsup_{t\downarrow 0} \frac{B_t - B_0^n}{\sqrt{2t\log\log\frac{1}{t}}} = 1 \quad \text{a.s.}$$

$\implies \exists\, t_n \downarrow 0$ s.t.

$$B_t - b \geq (\sqrt{2} - \varepsilon)\sqrt{t_n \log\log\frac{1}{t_n}} > 0.$$

Definition 1.10.3 (irrgular point)**.** A point $a \in \partial D$ is irrgular if $\mathsf{P}^a(\sigma_D = 0) < 1 \Leftrightarrow \mathsf{P}^a(\sigma_D > 0) > 0$.

By Blumenthal 0-1 law, $\mathsf{P}^a(\sigma_D > 0) \in \{0, 1\}$. So if $a \in \partial D$ is irrgular, then $\mathsf{P}^a(\sigma_D > 0) = 1$.

Theorem 1.10.1. $(d \geq 2)$ and fix $a \in \partial D$. The following are equivalent

(i)

$$\lim_{\substack{x\to a\\ x\in D}} E^x\left[f(W_{\tau_D})\right] = f(a)$$

holds for every bounded, measurable function f on ∂D s.t. f is continuous at a.

(ii) a is regular.

(iii) $\forall\, \varepsilon > 0$,

$$\lim_{\substack{x\to a\\ x\in D}} \mathsf{P}^x(\tau_D > \varepsilon) = 0.$$

$(\tau_D \xrightarrow{\mathsf{P}^x} 0$ as $x \to a)$

Proof. (iii) $\implies$ (i) By continuity of Brownian path,

$$\mathsf{P}^x\left(\max_{0\leq t\leq\varepsilon} \|W_t - W_0\| < r\right) \to 1$$

as $\varepsilon \downarrow 0$, and

$$\mathsf{P}^0\left(\max_{0\leq t\leq\varepsilon} \|W_t\| < r\right) \to 1.$$

Then

$$\begin{aligned}\mathsf{P}^x\left(\|W_{\tau_D} - W_0\| < r\right) &\geq \mathsf{P}^x\left(\max_{0\leq t\leq\varepsilon} \|W_t - W_0\| < r,\ \tau_D \leq \varepsilon\right)\\ &\geq \mathsf{P}^x\left(\max_{0\leq t\leq\varepsilon} \|W_t - W_0\| < r\right) - \mathsf{P}^x(\tau_D > \varepsilon).\end{aligned}$$

Let $x \in D \to a$ to get

$$\lim_{\substack{x\to a\\ x\in D}} \mathsf{P}^x\left(\|W_{\tau_D} - W_0\| < r\right) \geq \mathsf{P}^0(\max_{0\leq t\leq\varepsilon} \|W_t\| < r) - 0.$$

$\forall\, \varepsilon > 0$, $\exists\, r > 0$ s.t. $|f(x) - f(a)| < \varepsilon$ for all $|x - a| < r$,

$$E^x\left|f(W_{\tau_D}) - f(a)\right| \leq \|f\|_\infty \cdot E^x\left(1_{\|W_{\tau_D} - x\| > \frac{r}{2}}\right) + E^x\left(|f(W_{\tau_D}) - f(a)| \cdot 1_{\|W_{\tau_D} - x\| < \frac{r}{2}}\right).$$

If we let $|x-a|<\frac{r}{2}$, then

$$\|W_{\tau_D}-a\| \leq \|W_{\tau_D}-x\|+\|x-a\|<r.$$

$$\implies |f(W_{\tau_D})-f(a)|<\varepsilon.$$

(ii) $\implies$ (iii) Assume $a=0$.

$$g(x)=\mathsf{P}^x(\sigma_D>\varepsilon).$$

Because $\tau_D \leq \sigma_D$, we get

$$\limsup_{\substack{x\to 0\\ x\in D}} \mathsf{P}^x(\tau_D>\varepsilon) \leq \limsup_{\substack{x\to 0\\ x\in D}} \mathsf{P}^x(\sigma_D>\varepsilon) \leq \limsup_{\substack{x\to 0\\ x\in D}} g(x).$$

Because $g(0)=0$, it suffices to prove g is upper semicontinuos, i.e. $\forall\ x_0\in\mathbb{R}$,

$$\limsup_{x\to x_0} g(x) \leq g(x_0).$$

Note that

$$g(x)=\mathsf{P}^x(\sigma_D>\varepsilon)=\mathsf{P}^x\left(W_s\in D,\ \forall\ 0<s\leq\varepsilon\right).$$

$\forall\ 0<\delta<\varepsilon$,

$$g_\delta(x)=\mathsf{P}^x(W_s\in D,\ \forall\ \delta\leq s\leq\varepsilon).$$

Then (1)$\forall\ x\in\mathbb{R}$, $g_{\delta(x)}\downarrow g(x)$. Next, we will show that (2) $x\mapsto g_\delta(x)$ is continuous.
Then (1)+(2) $\implies$ $g(x)$ is upper-semicontinuos.
Fix $\delta\in(0,\varepsilon)$, we have

$$\begin{aligned}
g_\delta(x) &= \mathsf{P}^x\left(W_s\in D,\ \delta\leq s\leq\varepsilon\right)\\
&= E^x\left[\mathsf{P}^{W_\delta}\left(W_s\in D,\ 0\leq s\leq\varepsilon-\delta\right)\right]\\
&= E^x\left[\mathsf{P}^{W_\delta}\left(\tau_D>\varepsilon-\delta\right)\right]\\
&= \int_{\mathbb{R}^d}\mathsf{P}_\delta(x,y)\mathsf{P}^y\left(\tau_D>\varepsilon-\delta\right)\mathrm{d}y.
\end{aligned}$$

So $x\mapsto\int_{\mathbb{R}^d}\mathsf{P}_\delta(x,y)\mathsf{P}^y(\tau_D>\varepsilon-\delta)\,\mathrm{d}y$ is continuous.
(i) $\implies$ (ii) Assume 0 is not regular, i.e.

$$\mathsf{P}^0(\sigma_D=0)=0 \implies \mathsf{P}^0(\sigma_D>0)=1.$$

Because starting from 0,

$$\mathsf{P}^0\left(W_t\neq 0,\ \forall\ t>0\right)=1 \implies \mathsf{P}^0\left(W_{\sigma_D}=0\right)=0.$$

Hence

$$\lim_{r\downarrow 0} \mathsf{P}^0\left(W_{\sigma_D} \in B_r\right) = 0.$$

Fix $r > 0$ small s.t.

$$\mathsf{P}^0\left(W_{\sigma_D} \in B_r\right) < \frac{1}{4}.$$

Let $\delta_n \downarrow 0$, $0 < \delta_n < r$.

Define

$$\tau_n = \inf\{t \geq 0\colon\ \|W_t\| \geq \delta_n\}.$$

Then

$$\mathsf{P}^0\left(\tau_n \downarrow 0\right) = 1.$$

By

$$\mathsf{P}^0(\sigma_D > 0) = 1,$$

we have

$$\lim_{n\to\infty} \mathsf{P}^0(\tau_n < \sigma_D) = 1.$$

Since $\tau_n \downarrow 0$, $\sigma_D > 0$, we have $\{\tau_n < \sigma_D\} \subseteq \{\tau_{n+1} < \sigma_D\}$.

$$\mathsf{P}\left(\bigcup_{n=1}^{\infty}\{\tau_n < \sigma_D\}\right) = \lim_{n\to\infty} \mathsf{P}(\tau_n < \sigma_D).$$

$$\mathsf{P}^0\left(\bigcup_{n=1}^{\infty}\{\tau_n < \sigma_D\}\right) = 1 = \lim_{n\to\infty} \mathsf{P}^0(\tau_n < \sigma_D) = 1.$$

So $\exists\ N \geq 1$ s.t. $\forall\ n \geq N$,

$$\mathsf{P}^0(\tau_n < \sigma_D) \geq \frac{1}{2}.$$

$\square$

Definition 1.10.4 (defn example)**.** Example

Theorem 1.10.2 (thm example)**.** Example

Proof. proof □

Lemma 1.10.1 (lemma example)**.** Example

Proof for Lemma.

■. proof for lemma example □

Proposition 1.10.2. pro example

Proof for Proposition.

■. Example □

Claim 1.10.1. claim example Example

Proof for Claim.

■. proof for claim example □

Corollary 1.10.1 (cor example)**.** example

Proof for Corollary.

■. proof for corollary example □

Fact 1.10.1. Example fact

Remark. Example remark

✉ zhengzh2025@mail.sustech.edu.cn